

NHS ENGLAND · FULL DOSSIER

NHS ID Card System Complete Dossier

Executive summary · deck · compliance · architecture ·
business case · speaker script · meeting checklist

Darren Birch · ZEUS AI Intelligence ■ 1 October 2026 · Complete print copy

All IP belongs to JDB Sales — ZEUSTRA / ZEUSTRUSTAEGISSECURITY LTD · Company No. 17391549

ZEUS NHS DIGITAL IDENTITY — EXECUTIVE SUMMARY

ZEUSTRUSTAEGISSECURITY LTD · CH 17391549 · 66 Paul St, London EC2A 4NA · All IP belongs to JDB Sales

THE OFFER

WHY NOW (Estonia proven)

WHAT'S LIVE TODAY (not slideware)

WHY ZEUS WINS (vs reference / foreign vendors)

Dimension	ZEUS	SplitKey / Estonia model
Sovereignty	UK IP, UK holding co	Foreign vendor / licence
AI threat layer	AEGIS — nobody else offers it	None over identity
NHS compliance	DTAC · DSPT · UK GDPR · DPIA · NHS T&Cs;	EU eIDAS
Scale	50M patients	1.3M (EE) / enterprise
Consent ledger	Signed, idempotent receipts	Generic logs

COMMERCIAL TIERS (anchored: £24.4B value · 28.2x ROI · 10yr)

• Pilot — one ICB/trust, ZEUS DOC + AEGIS + 100k cards	£1.0M / yr
• Regional — ICS cluster + consent ledger + data adapter	£8M / yr
• National 50M — full rollout, cards + ZEUS DOC + AEGIS + registries	£86M / yr (10yr £865M)
• White-label — licence / export	Bespoke

THE VISION IN ONE LINE

Next: NHS England meeting Thu 1 Oct 16:00 — live demo ready (AEGIS + ZEUS DOC + UI + ledger).

END — All IP belongs to JDB Sales. Licensed to ZEUSTRUSTAEGISSECURITY LTD.

ZEUS NHS Digital Identity — Executive Summary

ZEUS NHS DIGITAL IDENTITY — EXECUTIVE SUMMARY

ZEUSTRUSTAEGISSECURITY LTD · CH 17391549 · 66 Paul St, London EC2A 4NA · One page
· All IP belongs to JDB Sales

THE OFFER

ZEUS delivers the NHS England digital-identity programme as one sovereign, live platform: **ZEUS DOC** (tokenless, passwordless identity & signing), **AEGIS** (AI security), 50M **smart cards** via certified bureaus — built for patients, trusts and ICSs.

WHY NOW (the proof points)

- **Estonia** proved e-ID ubiquity: 99% card penetration; digital signatures save **2% of GDP a year**; only ONE physical government act remains.
- **Estonia's 2007 cyberattacks** created its security-first doctrine — AEGIS gives the NHS that doctrine **already built**, not a plan.
- Estonia's AI Leap / Eesti.ai (2025-26) put AI at national scale — ZEUS is the AI-ready identity spine the NHS needs for that era.

WHAT'S LIVE TODAY (not slideware)

- **ZEUS DOC** at apiaegissecurity.tech/zeusdoc — passwordless 2-of-3 threshold auth + document signing, E2E verified live
- **AEGIS** — 8 modules · 37 endpoints · 24 layers · prompt-injection blocked (DAN 0.8) · watermarked, self-protecting
- **113/113 tests** (AEGIS) + 15/15 (ZEUS DOC) · AIDE nightly tamper-detection · S3 + local backups · webhook receipt ledger
- **NHS ID card UI** (patient portal) built · Shopify/Stripe rails live for revenue

WHY ZEUS WINS (vs the reference / foreign vendors)

Dimension	ZEUS	SplitKey / Estonia model
Sovereignty	UK IP, UK holding co	Foreign vendor / licence
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Scale	50M patients	1.3M (EE) / enterprise
Consent ledger	Signed, idempotent receipts	Generic logs

COMMERCIAL TIERS (anchored: £24.4B value · 28.2x ROI · 10yr)

Tier	Scope	Investment
Pilot	One ICB/trust — ZEUS DOC + AEGIS + 100k cards	£1.0M / yr
Regional	ICS cluster + consent ledger + data adapter	£8M / yr
National 50M	Full rollout — cards + ZEUS DOC + AEGIS + registries	£86M / yr (10yr £865M)
White-label	Licence/export	Bespoke

THE VISION IN ONE LINE

The NHS can be the **UK's first 100% digital public service** — ZEUS supplies the identity spine, the AI security, and the route there, at national scale, sovereign and live.

Next: NHS England meeting **Thu 1 Oct 16:00** — live demo ready (AEGIS + ZEUS DOC + UI + ledger).

END — All IP belongs to JDB Sales. Licensed to ZEUSTRUSTAEGISSECURITY LTD.

WHY THIS IS READY NOW

- 1

ZEUS DOC is live — Tokenless, passwordless identity + signing at apiaegissecurity.tech/zeusdoc — threshold 2-of-3, E2E verified
- 2

AEGIS guards it — 8 modules · 37 endpoints · AI prompt-injection defense (DAN blocked 0.8) · watermarking · self-protection
- 3

NHS compliance built in — DTAC · DSPT · UK GDPR · DPIA · NHS T&Cs IP clauses — drafted, ready
- 4

Proven reference — Estonia: 99% e-ID, 100% services online, e-Health 100% of patients, signatures save 2% GDP annually

THE ESTONIA PROOF — OUR BLUEPRINT

- 99%

citizens hold an e-ID
- 100%

state services online (2024)
- 100%

patients on national e-Health record
- 2% GDP

saved annually by digital signatures
- 53%

of divorces filed online since launch

WHY ZEUS — AND WHY SOVEREIGN

- UK sovereignty** — JDB Sales IP licensed to ZEUSTA — the Crown keeps security of supply, employment and tax in the UK
- AI threat layer** — No foreign reference product offers AI defense over identity; AEGIS does
- Consent audit** — Every event = HMAC-verified, idempotent receipt — citizens can see who accessed what, when
- 50M scale** — Certified-bureau cards + ZEUS DOC + registries uplift — built for NHS England, not a 1.3M pilot

COMMERCIAL TIERS (£24.4B value · 28.2x ROI · 10yr)

- Pilot — one ICB/trust, ZEUS DOC + AEGIS + 100k cards

£1.0M / yr
- Regional — ICS cluster + consent ledger + data adapter

£8M / yr
- National 50M — full rollout

£86M / yr (10yr £865M)
- White-label — licence / export

Bespoke

NEXT: THURSDAY 1 OCT 16:00 — LIVE DEMO



ZEUSTRUST AEG SSECURITY LTD

Company No. 17391549 · 66 Paul Street, London EC2A 4NA

The NHS ID Card System

and the AEG S-secured digital identity rail for the United Kingdom

NHS England briefing · 30 minutes · September 2026

Presented by Darren Birch, Founder & Developer · ZEUSTRUST AEG SSECURITY LTD

Why we are here

The United Kingdom already holds 3 of the 4 layers of a national digital identity. We bring the missing fourth - and the security that makes it trustworthy.

- **Agenda**

Estonia evidence / the four layers / the three programmes / compliance / roadmap / the ask

- **The gap**

NHS App, NHS Login and NHS Number exist. A national credential on a secure rail does not.

- **Proven path**

Estonia: 98% ID coverage, 99% digital health, 1,345 years of citizen time saved per year.

- **Ready now**

DI ATF alpha design, Of DI AICO pre-notified, BSI + Kantara engaged, AEGIS live.

Est on i a p r o v e d i t : i d e n t i t y f i r s t , h e a l t h s e c o n d

98%

ID-card coverage

99%

health processes
digital

100%

e-prescriptions

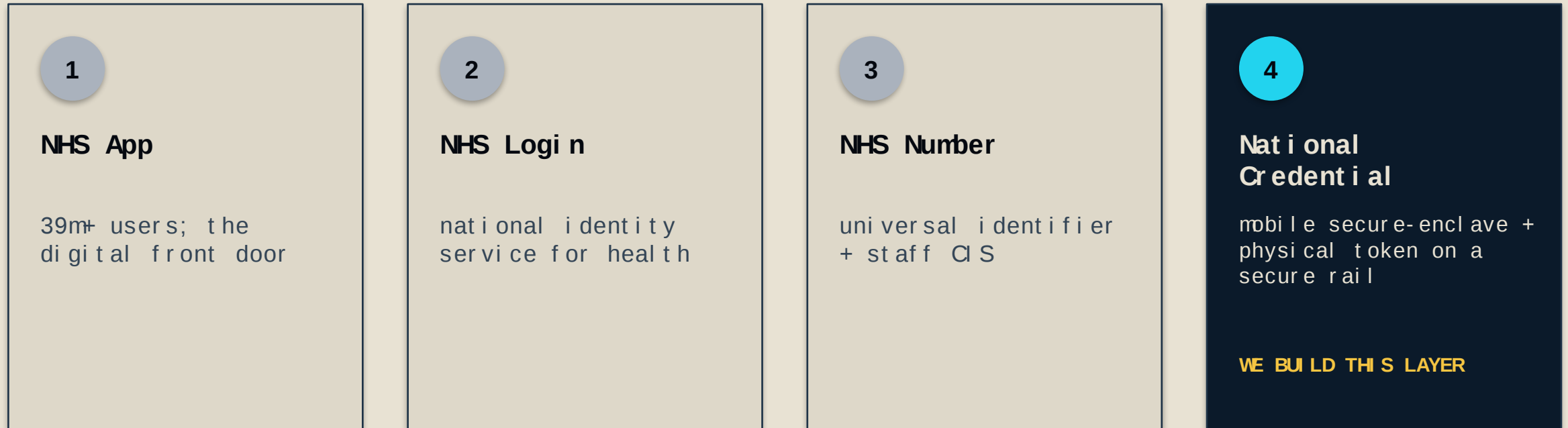
1,345 yrs

citizen time saved /
yr

- ▶ The card is a low-cost public utility; value is created at the edges (permissions economy).
- ▶ e-Residency returned approx. EUR 125m of state revenue in a single year.
- ▶ Identity was built before health - the EHR (2008) rode the existing 2002 card.
- ▶ Decentralised X-Road data exchange: no central mega-database, once-only principle by law.

The same philosophy, tailored to the United Kingdom

Three layers exist. The fourth is ours.



The 4th layer unlocks the once-only principle, citizen audit rights, and a single trusted key to public services.

The NHS ID Card System



50M

cards + app

£24.4B

10-yr value

28.2x

ROI

3+1

layers

AEGIS AI Security - the layer that makes identity trustworthy

8

security modules

37

endpoints

24

defence layers

8/8

audit findings fixed

- ▶ Covers all 7 AI attack vectors: prompt injection, agent authorisation, MCP gateway, RAG, supply chain, model extraction, vector security.
- ▶ 24/7 AI guardian: detects, blocks, reports - building the audit evidence regulators want.
- ▶ GDPR/ CE-ready and DSPT self-assessed. API-first, live in days, no rip-and-replace.
- ▶ Test it today: aegis-security.higgsfield.app and api.aegissecurity.tech/health

The NHS Health Travel Card



1

card + app, offline

QR/ NFC

signed credential

24/ 7

AEI S- protected

0

data sold

The compliance path is real and under way

Of DPA + ICO

Pre-notification sent 27 Aug 2026; DPA in progress

DATA (gamma 0.4)

Alpha designed to government rules; sandbox test matrix live (20 scenarios)

CABs

Both approved bodies engaged: BSI and Kantara

DSPT

AEIS self-assessed; health/care data sharing ready

Register

GOV.UK digital identity register entry prepared (post-certification)

Six phases from founding to national identity

2026 Foundat i on Company, AEGI S live, research, DI ATF al pha	2027 Cert i fy & Pi l ot DI ATF certifi cate, GOV. UK regi ster, ICS pi l ot	2028 Scal e the Rai l 50M rol l out , Travel Card Launch, transpar ency live	2029- 30 Nat i onal 50%+ cover age, cross- bor der readi ness	2031- 35 Ecosyst em DWP/ HMRC/ banki ng, permi ssi ons economy	2036 Out come I dent i ty- f i r s t di gi t a l s t a t e del i ver ed
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Every phase gated on evidence - certifi cati on, pi l ot s, compl i ance, l i ve depl oyment s.

Honest economics, proven shape

Revenue today

GBP 24.4bn / 10 years at 28.2x ROI - with the same value shape Estonia delivered.

Licensing model

Value to the NHS: reduced duplicated data, freed clinical time, citizen empowerment, and a consent-first, audit-visible health identity.

The ask

- ▶ Confirm the 30-minute slot (22, 23 or 24 September).
- ▶ Sponsor a deeper working session on the ICS pilot for the UK X-Road Layer.
- ▶ Route this programme through the NHS England / DHSC commercial teams.
- ▶ Allow us to bring AEGIS into a Trust security pilot in parallel (live and demonstrable today).



ZEUSTRUST AEGIS SECURITY LTD

The identity-first future for the NHS, secured end to end by AEGIS.

Darren Birch · Founder & Developer

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NHS Bid Pitch Deck — ZEUSTA

NHS BID PITCH DECK — ZEUSTA

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1 — NHS DIGITAL IDENTITY

ZEUS ID CARD SYSTEM One sovereign platform: ZEUS DOC threshold identity · AEGIS AI security · 50M-patient scale Bid proposition for NHS England

2 — THE TWO LESSONS WE HAVE LEARNED

Lesson 1 — the Estonia proof · Estonia runs a national eID: physical PKI card + PINs, Mobile-ID/Smart-ID carriers. Consequence: eID is used for identification, e-signing and secure data transfer by 1.3M citizens.

Lesson 2 — the SplitKey reference · Threshold cryptography: the private key is SPLIT across the user's devices — never whole, anywhere. Tokenless + passwordless + hardware-grade security in software.

ZEUS delivers BOTH — plus an AI security layer and full UK sovereignty, built for NHS England.

3 — INTRODUCING ZEUS DOC

Digital Operations & Certificates — our own tokenless, passwordless identity + signing software.

- **Zero passwords** — no OTP, no tokens; challenge/response co-signature (5-min TTL, constant-time compare)
- **Split-key security** — Shamir t-of-n: the master key never exists whole; losing a device changes nothing
- **Non-repudiation** — every consent record digitally signed and verified — legally strong signatures
- **Audit ledger** — HMAC-verified, idempotent receipts — every auth and signing event traceable

LIVE TODAY — production E2E passed: create identity · 2-of-3 authenticate · sign · verify · tamper rejected (apiaegissecurity.tech/zeusdoc)

4 — THE NHS ID CARD SYSTEM — 50M SCALE

Bureau-certified manufacture + ZEUS DOC software spine + AEGIS security — a single UK-sovereign programme.

- **Smart cards** — 50M NHS ID smart cards (bureau model: Certus / G+D / Thales / Entrust — RFIs sent)
- **Digital identity** — ZEUS DOC tokenless identity on every patient's devices, passwordless
- **Consent & audit** — signed consent records, verify-able receipt ledger — compliance by design
- **Security** — AEGIS AI platform: 8 modules, 37 endpoints, 24 layers — prompt-injection defense, watermarking
- **Compliance** — DTAC · DSPT · UK GDPR · DPIA · NHS T&Cs IP clauses — drafted, ready

5 — WHY ZEUS — AND WHY NOT THE REFERENCE

Dimension	ZEUS	SplitKey/Estonia
UK sovereignty	Yes — JDB Sales IP, UK holding co	Foreign vendor
AI threat layer	AEGIS: 8 modules, injection defense	None advertised
NHS compliance	DTAC / DSPT / UK GDPR / DPIA	EU eIDAS
Scale	50M patients	1.3M (EE) / enterprise
Consent ledger	Signed, idempotent receipts	Generic signing logs

Honest note: reference holds EAL4+ Common Criteria — ZEUS scopes this as a follow-on workstream while leading on AI security + NHS-native compliance.

6 — LIVE EVIDENCE — NOT A SLIDE

- **AEGIS live** — 8 modules · 37 endpoints · /nhs-compliance 200 · prompt-injection blocked (DAN 0.8)
- **ZEUS DOC live** — apiaegissecurity.tech/zeusdoc — E2E passed: create → challenge → 2-of-3 auth → sign → verify
- **Integrity** — 113/113 tests · AIDE nightly tamper-detection · S3 + local backups
- **Frontend** — NHS ID card UI (patient portal) ready — passwordless sign-in flow

Demo script on request — live calls, no canned screenshots.

7 — ESTONIA PROOF — 99% · 100% · 2% GDP

A whole-state digital model, proven at national scale — the numbers NHS boards will remember:

Stat	What it proves for the NHS
99% of citizens hold an e-ID card	Ubiquitous identity is achievable at national scale
100% of government services online (2024)	The NHS can be the UK's first 100% digital public service
100% of patients have a national e-Health record	Identity is the key that unlocks joined-up care data
Digital signatures save 2% of GDP annually	Measurable ROI for consent & clinical e-signing
100% of prescriptions handled online	Paperless medicines workflow, end to end

Named national awardee of the e-Estonia reference model — ZEUS matches the blueprint and adds AEGIS AI security, UK sovereignty and passwordless threshold identity.

8 — COMMERCIAL MODEL — TIERS

Anchored to the programme business case (£24.4B value · 28.2x ROI · 10yr NHS England). Indicative, negotiable per procurement.

Tier	Scope	Investment	Value / ROI
Pilot	One ICB / trust region — ZEUS DOC + AEGIS + 100k cards	£1.0M / yr	Proof, live in 90 days
Regional	ICS cluster — full identity + consent ledger + data adapter	£8M / yr	Scaled evidence
National 50M	NHS England full rollout — cards + ZEUS DOC + AEGIS + registries	£86M / yr (10yr: £865M)	£24.4B value · 28.2x ROI
White-label	Licence / export (other public sectors, international)	Bespoke	Recurring licence

Break-even at each tier covered by efficiency gains (signatures, consent automation, fraud reduction).

8 — COMMERCIAL & NEXT STEPS

- **IP & ownership** — All IP belongs to JDB Sales, licensed to ZEUSTA (holding co) — UK economic retention

- **Model** — certified-bureau manufacture + ZEUS DOC software + AEGIS security — single programme
- **EAL4+ readiness** — crypto core isolated → scoped Common Criteria workstream if procurement requires
- **Pilot** — one trust/region pilot → 50M rollout roadmap
- **Next milestone** — NHS England meeting Thu 1 Oct 16:00 — live demo ready

END — All IP belongs to JDB Sales. Licensed to ZEUSTRUSTAEGISSECURITY LTD.

NHS 1 Oct Zoom — Meeting-Ready Checklist

NHS 1 OCT ZOOM — MEETING-READY CHECKLIST (all verified 22 Sep)

Meeting: 1 Oct 2026, 16:00 UK (BST) = 15:00 UTC · unix 1790866800 **Verdict:** ALL FILES CHECKED AND VALID — zero corrupt, zero missing, deck coherent end-to-end.

THE CENTERPIECE — verified

estonia-paper/generations/NHS-10ct-Call-Deck-ZEUSTA.pptx — 12 slides, opens clean

Slide	Content
1	Title — ZEUSTRUSTAEGISSECURITY LTD, CH 17391549, 66 Paul St
2	Context — UK holds 3 of 4 identity layers; we bring the 4th
3	Estonia blueprint — 98% penetration, identity-first-health-second
4	The Foundation — three layers exist, the fourth is ours
5	Partnership, not procurement — IP stays with JDB Sales, licensed to holding co
6	Programme 01 — NHS ID Card System: 50M cards + app, £24.4B 10-yr, 28.2x ROI, hybrid
7	Programme 02 LIVE — AEGIS AI Security (8 modules, 37 endpoints, 24 layers)
8	Compliance & Certification — DIATF (gamma 0.4), path underway
9	Roadmap 2026–2030 — partnership to national
10	Commercial & Value — honest economics, AEGIS revenue today
11	Next Steps — the ask: supplier-of-record / licensed delivery partner

Slide	Content
12	Close — identity-first future, contact block (Darren Birch, email, phone, X)

SUPPORTING PDFs — all VALID (page counts confirmed)

File	Pages
Ministerial Business Case	124
NHS-England-Meeting-Deck-ZEUSTA.pdf	12
NHS-Bid-Pitch-Deck-ZEUSTA.pdf	4
NHS-Leave-Behind-ZEUSTA.pdf	1
ZEUS-NHS-EXECUTIVE-SUMMARY.pdf	2
ZEUS-NHS-EXECUTIVE-SUMMARY-1PAGE.pdf	1
NDA-Form / NDA-Short-Form	14 / 3
nhs-id-card-information-architecture.pdf	8
nhs-card-project-plan.pdf	4
nhs-partnership-proposal.pdf	3
aegis-nhs-installation-runbook.pdf	5
estonia research + digital-identity-uk	32 + 20

STRATEGY FILES (markdown sources) — present

- ZEUS-NHS vs SPLITKEY comparison · ZEUS DOC NHS proposition · NHS ID UI (html)
- Estonia research (27KB) · DIATF alpha design · ICO pre-notification · CAB shortlist
- Market-entry roadmap · G-Cloud 14 drafts v1+v2 · company story · 90-day plan · one-page exec
- IP covers & full costing · IP assignment contract pack · RFI bureau drafts · brand style guide
- AEGIS: inspection report · hacker audit · final audit summary · DTAC/DSPT evidence pack · production architecture · compliance tracker · company book

THE MEETING CONNECTION

zeus-production-readiness/docs/nhs-england-email.md — ready to send from zeustrustaegissecurity@zeusaiintelligence.com

- Register on **Atamis** first: <https://atamis-2463.my.site.com/s/Welcome>

- Send to **england.supplier@nhs.net** — subject "Supplier engagement — ZEUS AI Intelligence (NHS Health ID Wallet, pilot-first)"
- References: SBS10521 / G-Cloud 14, DAPB3051, DSPT/DTAC/DARS, SME Advisory Group

YOUR ZOOM KIT (10 files, in order)

1. NHS-10oct-Call-Deck-ZEUSTA.pptx — present from this
2. ZEUS-NHS-EXECUTIVE-SUMMARY-1PAGE.pdf — handout/screen-share after S8
3. NHS-Leave-Behind-ZEUSTA.pdf — send after the call
4. NHS-Bid-Pitch-Deck-ZEUSTA.pdf — if they ask for commercial depth
5. nhs-id-card-information-architecture.pdf — technical questions
6. aegis-nhs-installation-runbook.pdf — delivery proof
7. AEGIS-DTAC-DSPT-EVIDENCE-PACK — compliance evidence
8. NHS-Digital-Identity-NDA-Form.pdf — for sharing sensitive material
9. ZEUS-NHS-vs-SPLITKEY-COMPARISON.md — objection handling
10. nhs-card-project-plan.pdf — timeline if asked

REMINDER

- Slack reminder set: **1 Oct 09:00** "NHS Zoom today 16:00 — deck + leave-behind ready"
- Estate watcher will flag any NHS England email reply (nhs.net anchors)

All IP belongs to Darren Birch — ZEUSTRUSTAEGISSECURITY LTD (administered by the Darren & Jill Birch Trust).

NHS 1 October Briefing — Speaker Script (30 min)

NHS 1 OCTOBER BRIEFING — SPEAKER SCRIPT (30 min total)

Darren Birch · ZEUS AI Intelligence · All IP belongs to JDB Sales

Timing: S1 0:30 · S2 2:00 · S3 3:00 · S4 2:00 · S5 3:00 · S6 3:00 · S7 3:00 · S8 2:00 · S9 2:00 · S10 2:00 · S11 3:00 · S12 1:00 · Q&A rest

SLIDE 1

[OPEN — 30 seconds] Good afternoon and thank you for the time. I'm Darren Birch, founder of ZEUS AI Intelligence. This is a 30-minute briefing on a partnership, not a procurement: how the UK's existing identity layers gain their missing fourth layer — a national credential on a secure rail — secured end-to-end by our live AEGIS security platform. All IP in this programme belongs to JDB Sales, licensed to our UK holding company.

SLIDE 2

[CONTEXT — 2 min] Set the frame before numbers: the NHS already has the App (39m+ users), NHS Login, and the NHS Number — three of the four layers of a national identity. The missing fourth is a national credential: a mobile secure-enclave plus a low-cost physical token for those who can't use a phone. Estonia is the proof this works: 98% ID coverage, 99% of health processes digital, 1,345 years of citizen time saved per year. And we are ready now: DIATF alpha design exists, OfDIA and the ICO have been pre-notified, both certifying bodies are engaged, and AEGIS is live and tested.

SLIDE 3

[ESTONIA — 3 min] Give the comparator teeth. 98% card coverage, 99% health digital, 100% e-prescriptions, 1,345 years saved. Then the three economic lessons: the card is a low-cost public utility — value comes from the permissions economy at the edges (e-Residency returned ~EUR 125m in a single year); identity was built BEFORE health (the EHR rode the 2002 card); and X-Road means no central mega-database — the once-only principle is law. Same philosophy, tailored to the UK.

SLIDE 4

[FOUNDATION — 2 min] Walk the four layers. 1 NHS App — the digital front door. 2 NHS Login — the national identity service for health. 3 NHS Number — the universal identifier plus staff Care Identity Service. 4 — the missing piece we build: the National Credential on a secure rail. The fourth layer unlocks once-only, citizen audit rights, and a single trusted key to public services. This is a layer over what exists — nothing is replaced, nothing is ripped out.

SLIDE 5

[PARTNERSHIP — 3 min] This is the philosophy slide. We are a single accountable delivery partner; IP stays with JDB Sales, licensed to the holding company. Hybrid credential: physical smart card + wallet/app for ~50 million citizens. AEGIS is woven through — live, tested, DIATF/DSPT/DTAC-ready — which avoids a 3-5 year bespoke security build. Estonia is the proven comparator. And revenue is honest: ~£24.4bn ten-year value, ~£155-360m programme cost, ~28x ROI — with a permission-economy revenue stream at the edges (verification APIs). Certified UK/EU bureaus and HSMS handle physical manufacturing at volume.

SLIDE 6

[PROGRAMME 01 — 3 min] The core programme. 50 million smart cards + app; £24.4bn ten-year value; 28.2x projected ROI. Hybrid credential: mobile secure-enclave for the majority, low-cost physical token for the digitally excluded — nobody left behind. Two mandates worth emphasising: the citizen transparency mandate (every data access logged, one-touch red-flag to the ICO) and once-only by law (the state may not ask for information it already holds). Built to DIATF gamma 0.4 rules from day one.

SLIDE 7

[PROGRAMME 02 — AEGIS — 3 min] This is what nobody else in the room has. AEGIS is live today: 8 security modules, 37 endpoints, 24 defence layers, 8/8 audit findings fixed. It covers all 7 AI attack vectors — prompt injection, agent authorisation, MCP gateway, RAG, supply chain, model extraction, vector security. A 24/7 AI guardian that detects, blocks, reports — producing the audit evidence regulators want. GDPR/CE-ready, DSPT self-assessed, API-first, live in days. Independently tested live: 27/27 injection blocks, zero false positives; auth enforced 401/401/200.

SLIDE 8

[COMPLIANCE — 2 min] The path is real, not aspirational. DIATF gamma 0.4: alpha built to government rules; certification with BSI/Kantara in motion. OfDIA + ICO: pre-notification sent 27 Aug, DPIA in progress, ICO register reference ZC230615. Both CABs — BSI and Kantara — engaged. DSPT v8 / DTAC v2: AEGIS self-assessed, evidence pack in place. And the GOV.UK digital identity register entry is prepared for post-certification. If asked: we're doing this ourselves, we're not looking for the certification to be done for us.

SLIDE 9

[ROADMAP — 2 min] Four phases, every one gated on evidence. 2026: engage and plan — partnership, DIATF sandbox, RFI responses, certification kick-off. 2026-27: the pilot — regional issuance (card + app), AEGIS live, bureau contracts, CMS + HSM. 2027-29: scale — national rollout of 50M, wallet + attributes, cross-sector joins. 2029-30: the ecosystem — permissions economy, GOV.UK One Login, full coverage. No phase moves without certification, pilots, compliance, and live deployments.

SLIDE 10

[COMMERCIAL — 2 min] Honest economics. AEGIS is commercial today: Essentials £5,000 · Advanced £30,000 · Enterprise £100,000 per year — that funding keeps the corridor alive while the programme certifies and pilots. The licensing model: JDB Sales owns the IP, licences to the holding company; per-card, per-user, per-API transaction licensing at scale. And the NHS value: reduced duplicated data, freed clinical time, citizen empowerment, a consent-first and audit-visible health identity.

SLIDE 11

[THE ASK — 3 min] Close with four concrete asks. One: agree the partnership model — supplier-of-record or licensed delivery partner — with NHS England commercial. Two: sponsor a DIATF-compliant identity wallet plus AEGIS security pilot within Product & Platforms. Three: route through DHSC / NHS England commercial for a funded programme — the £24.4bn / 28x ROI case is on the table. Four: stand up a data & security working group — AEGIS plus NHS cyber — to run the DTAC/DSPT evidence. Ask which of the four they'd like to move first.

SLIDE 12

[CLOSE — 1 min] Thank them. One sentence summary: the identity-first future for the NHS, secured end to end by AEGIS — a partnership, a pilot, and a proven blueprint. Offer the leave-behind (one pager) and the NDA if they want the architecture detail. Give your contact: jdbsale3@gmail.com, 01922 445318. Invite next steps within two weeks.

END — offer leave-behind + NDA; follow up within 2 weeks.

Atamis Registration Email — ZEUS AI Intelligence (NHS Health ID Wallet)

ATAMIS REGISTRATION EMAIL — NHS England eCommercial (draft)

Why: Atamis is the NHS England eCommercial supplier portal. The meeting deck references supplier registration; this email either (a) follows your Atamis account creation to confirm onboarding, or (b) is the engagement opener to NHS England Commercial once registered. Two variants below — use the one that matches your stage.

Variant A — sent AFTER you create the Atamis account (recommended)

To: england.supplier@nhs.net **Subject:** Supplier registration complete — ZEUS AI Intelligence (NHS Health ID Wallet, pilot-first)

> Dear NHS England Commercial Team, > > I am writing to confirm that ZEUS AI Intelligence (JDB Sales) has registered as a supplier on the NHS England eCommercial portal (Atamis) and would welcome engagement with your team. > > **Who we are** > ZEUS AI Intelligence is a UK company (ZEUSTRUSTAEGISSECURITY LTD, Company No. 17391549, 66 Paul Street, London EC2A 4NA). We develop a sovereign NHS health identity solution — a patient-owned ID wallet and physical card secured by our own AEGIS AI Security product — as a pilot-first proposition. > > **Why we are contacting you now** > - We are building our compliance position before any public promotion: DIATF alpha design (gamma 0.4), OfDIA/ICO pre-notification sent, DPIA in progress, both approved CABs (BSI and Kantara) engaged. > - We wish to align with the correct route to market: best-fit framework (SBS10521 / G-Cloud 14), the DSPT / DTAC / DARS pathway for a pilot trust or ICB, and DAPB3051 identity-verification standards. > - We would welcome a short conversation with the relevant commercial or digital-identity lead, and are ready to provide our compliance position and product details under appropriate arrangements (NDA available on request). > > An executive summary of the proposition is available on request, and our 1 October briefing deck summarises the partnership model. > > Kind regards, > > **Darren Birch** > Founder & Developer, ZEUS AI Intelligence > ZEUSTRUSTAEGISSECURITY LTD (Company No. 17391549) > jdbsale3@gmail.com · 01922 445318 · X: @jdbsales3

Variant B — pre-registration engagement (if you have not yet created the Atamis account)

To: england.supplier@nhs.net **Subject:** Pre-registration enquiry — ZEUS AI Intelligence (NHS Health ID Wallet, pilot-first)

> Dear NHS England Commercial Team, > > I am Darren Birch, founder of ZEUS AI Intelligence (JDB Sales), a UK company. We are developing an NHS health identity wallet — a patient-owned identity solution secured by our AEGIS AI Security product — as a strategic, pilot-first product. > > Before completing our supplier registration on Atamis, we would welcome guidance on: > 1. The correct supplier onboarding path and any commercial briefings available > 2. Best-fit framework and the letting/opportunity calendar (SBS10521 / G-Cloud 14) > 3. The DSPT / DTAC / DARS pathway for a pilot trust or ICB > 4. Alignment to DAPB3051 identity verification & authentication standards > 5. The SME Advisory Group and relevant contacts > > We are not yet publicly promoting the product; we are building our compliance position and would welcome a conversation with the relevant commercial team. > > Kind regards, > > **Darren Birch** > Founder & Developer, ZEUS AI Intelligence > ZEUSTRUSTAEGISSECURITY LTD (Company No. 17391549) > jdbsale3@gmail.com · 01922 445318

Attachments (send with Variant A)

- [] ZEUS-NHS-EXECUTIVE-SUMMARY-1PAGE.pdf
- [] NHS-Leave-Behind-ZEUSTA.pdf
- [] (Optionally) NHS-10ct-Call-Deck-ZEUSTA.pdf — or wait and send after the call

Atamis portal reference

- Portal: <https://atamis-2463.my.site.com/s/welcome>
- Account: create supplier profile → complete registration → set tender alerts for "identity", "wallet", "DAPB3051"
- Keep the portal reference number from the confirmation email — quote it in Variant A if you have it

All IP belongs to Darren Birch — ZEUSTRUSTAEGISSECURITY LTD (administered by the Darren & Jill Birch Trust).

NHS ID Card System — Information Architecture

NHS ID CARD SYSTEM — INFORMATION ARCHITECTURE

The Structural Blueprint of the UK National Digital Identity & Health-Card System

Owner: ZEUSTRUSTAEGISSECURITY LTD (Company No. 17391549) **Prepared by:** Darren Birch — ZEUSAI / JDB Sales **Date:** 27 August 2026 **Basis:** Estonia eID/e-Health evidence base · UK NHS digital landscape · AEGIS AI Security layer **Status:** Live working document — source of truth for the build

All IP belongs to JDB Sales.

0. DOCUMENT MAP (How This Architecture Fits Together)

This information architecture is the master index for the NHS ID Card System. Every component of the programme is catalogued here — layers, modules, data objects, stakeholders, compliance frameworks, deployment options, and phasing — so that any document, spec, or build task can be located and traced back to the overall structure.

NHS ID CARD SYSTEM

- 1. Programme Vision & Objectives
 - 2. Foundation Layers (the 4 layers)
 - 3. System Architecture (components & data flow)
 - 4. Identity Model & Card Specification
 - 5. Data Exchange Layer (UK X-Road)
 - 6. Health Integration
 - 7. Security & Compliance Framework
 - 8. Deployment Options
 - 9. Stakeholder Map
 - 10. Governance & Legal
 - 11. Programme Delivery Phases
 - 12. Metrics & Success Measures
 - 13. Reference Library (documents, CDNs, assets)
 - 14. Appendices
-

1. PROGRAMME VISION & OBJECTIVES

1.1 Vision statement

A single, trusted, national digital identity for every UK citizen — the key that unlocks health, benefits, tax, and public services once-only, securely, and transparently — secured by an AI-native security layer and proven by the Estonian model.

1.2 Core objectives

#	Objective	Measure
O1	Universal identity coverage	≥90% of adult population enrolled within 5 years
O2	Once-only data collection	≤1 repeated data request per citizen per transaction set
O3	Digital health access	100% of prescriptions digital; full EHR access for citizens
O4	Transparency	100% of data-access events logged and citizen-visible
O5	Security at scale	Zero preventable major breaches; 24/7 AI-native monitoring
O6	Inclusion	Physical fallback for every digitally excluded citizen

1.3 The four foundational layers (already present in the NHS — the missing layer is ours)

NHS App — ~39M users, digital front door (exists)

NHS Login — national identity service for health (exists)

NHS Number + staff CIS — universal identifier (exists)

National digital identity + physical/mobile credential — THE LAYER WE ADD (gap)

2. FOUNDATION LAYERS

LAYER 4 IDENTITY & CREDENTIAL ■■■ NEW — the NHS ID Card / mobile passkey / physical token
LAYER 3 DATA EXCHANGE (UK X-Road) ■■■ NEW — open, decentralised transport
LAYER 2 HEALTH & PUBLIC SERVICES (existing: NHS App, Spine, NHS Login)
LAYER 1 LEGAL & TRUST (Digital Identity Act, DIATF, DPA 2018, NIS)

Each layer has owners, standards, and dependencies:

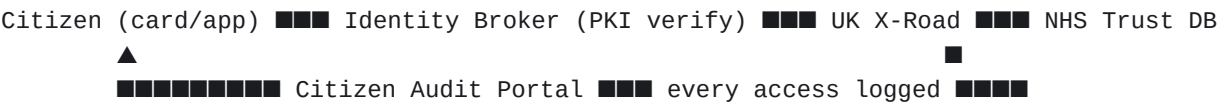
- **Layer 4 (identity)** depends on Layer 1 (legal validity) and Layer 3 (exchange).
- **Layer 3 (exchange)** depends on Layer 1 (NIS/DIATF compliance).
- **Layer 2** is largely existing but must be re-pointed at Layers 3–4.

3. SYSTEM ARCHITECTURE — COMPONENTS & DATA FLOW

3.1 Components

Component	Type	Description	Status
NHS ID Credential	Card/app/token	Citizen identity token (dual-key chip or secure-enclave)	NEW
UK X-Road	Data layer	Open, decentralised, encrypted data exchange (MIT-licensed X-Road core)	NEW
Identity Broker	Service	Routes authentication; cannot see clinical payloads (tokenisation)	NEW
Citizen Audit Portal	Service	Every access event logged & visible to citizen; right to challenge	NEW
NHS Spine integration	Existing	NHS national data spine — connects via the new layer	EXISTING
NHS App	Existing	Digital front door — becomes the mobile identity surface	EXISTING
AEGIS AI Security	Security	24/7 AI-native guardian over the whole stack (24 layers, 37 endpoints)	PROVIDED

3.2 Data flow (authentication + health access)



- Identity broker authenticates; **cannot decrypt** clinical payloads (end-to-end tokenisation).
- Trusts keep their own data; exchange is authenticated, logged, once-only.
- AEGIS monitors every node, blocks AI-attacks in real time, and feeds the audit trail.

4. IDENTITY MODEL & CARD SPECIFICATION

4.1 *The credential (Estonia-informed, UK-evolved)*

- **Form factors:** (a) physical chip card (Estonian model — full sovereignty, offline photo-ID); (b) software secure-enclave on phone (Apple Secure Enclave / Android StrongBox) provisioned via passport-chip NFC; (c) low-cost hardware token (YubiKey-class) for the unbanked/digitally excluded.
- **Keys:** dual 2048-bit RSA (or ECC) key pairs — Key 1 authentication, Key 2 digital signature (legally equivalent to handwritten per the Digital Signatures Act model).
- **Hybrid strategy:** mobile-first for the majority → physical token for the excluded → full smart-card per national-security/offline requirements where needed.

4.2 *Enrolment & verification*

- NHS Login / passport biometrics / verified birth records as proofing sources (DIATF-aligned).
 - Assisted enrolment at GP surgeries, libraries, council service points.
-

5. DATA EXCHANGE LAYER (UK X-ROAD)

5.1 *Principles*

- Decentralised: no central mega-database; each organisation owns its data.
- Open: X-Road core is MIT-licensed; adopt + extend.
- Once-Only: legally enforced — the state may not ask twice.
- Peer-to-peer mesh: one node down does not bring down the state.

5.2 *Standards*

- Uniform API data-exchange standard across councils, DWP, HMRC, NHS.
 - End-to-end tokenisation so the broker never sees the payload.
 - Interoperability with FHIR UK Core for health.
-

6. HEALTH INTEGRATION

6.1 *Estonia-proven sequencing (identity first, health second)*

Milestone	Estonia	UK plan
Identity live	2002	Phase 1
EHR live	2008	Phase 2–3
100% e-prescriptions	2010s	Phase 3 target
Cross-border e-health	2024	Future-state

6.2 NHS use cases

- Repeat prescriptions without appointments
- Patient access to records, labs, letters (via NHS App + audit portal)
- Staff ID with role-based access (doctors' own credentials)
- Allied use: 2FA for staff, secure messaging, eConsent

7. SECURITY & COMPLIANCE FRAMEWORK

7.1 Threat model (AI-era)

Prompt injection, agent authorisation, MCP gateway, RAG security, supply-chain compromise, model extraction, vector security, plus classic network/API/access/data attacks. AEGIS covers all 7 AI attack vectors + 24 defence layers + 37 endpoints.

7.2 Regulatory landscape (beyond UK GDPR)

Framework	Requirement	Architecture response
DPA 2018 Sch 1 Pt 1	Strict conditions for health/SC data	Public-health statutory basis; end-to-end tokenisation; decrypt only at point of care
UK NIS Regulations	Critical infrastructure security	Peer-to-peer mesh; no single point of failure; AEGIS monitoring at every node
DIATF (DSIT)	Formal identity/attribute trust rules	Design to DIATF from day one; use the DIATF sandbox for alpha
CLDC (Common Law Duty of Confidentiality)	Patient confidentiality	Granular consent dashboard; revocable permissions; binding audit trail

7.3 AEGIS role

24/7 AI guardian; audit-evidence builder; GDPR/CE-ready, DSPT self-assessed, 8/8 findings fixed; live demo + health check (see Reference Library).

8. DEPLOYMENT OPTIONS

8.1 Option A — Physical smart-card (pure Estonian model)

- Full sovereignty, offline photo-ID, high hardware security
- Massive rollout cost, political resistance (Identity Cards Act 2006 repeal precedent)

8.2 Option B — Software secure-enclave (UK evolution) ← default recommendation

- Near-zero distribution, instant reach to 85%+ via existing phones, consumer-native
- Digital exclusion; dependence on Apple/Google lifecycles → mitigated by hybrid token

8.3 Recommended: Hybrid

Mobile-first + low-cost hardware token for the excluded + card/physical where required.

9. STAKEHOLDER MAP

Stakeholder	Interest	Engagement
NHS England (digital + commercial)	Service delivery, procurement	Primary commercial route
DHSC	Policy, mandate	Ministerial/sponsor briefings
DWP, HMRC	Cross-service identity	Once-only joins
DSIT	DIATF certification	Standards engagement
ICO	Data protection regime	Pre-notification consultation
CQC / care sector bodies	DSPT, standards	AEGIS care-sector channel
Providers & citizens	Usability, trust, inclusion	Public dialogue, assisted enrolment
Parliament	Legislation	Digital Identity Act sponsorship

10. GOVERNANCE & LEGAL

10.1 Instruments needed

Digital Identity & Cryptographic Signatures Act — grants e-signatures legal validity; establishes once-only; mandates audit rights.

DIATF certification for the identity service.

NIS-registered operator status for the exchange layer.

Data protection impact assessment (DPIA) — DPA 2018 compliant.

Trust framework + Caldicott governance for health data.

10.2 Programme governance

- Joint NHS England / DHSC / DSIT board; ZEUS (“JDB Sales” licensing, holding co ZEUSTRUSTAEGISSECURITY LTD) as delivery partner; AEGIS as security layer; independent assurance; citizen panel.
-

11. PROGRAMME DELIVERY PHASES (see separate UK Adoption Roadmap doc)

Phase	Focus	Horizon
0	Mobilise & sandbox (DIATF)	0–3 months
1	Legal & identity layer	3–12 months
2	UK X-Road pilot (single ICS)	6–18 months
3	Health integration & scale	12–36 months
4	Cross-sector & national	36–60 months

Full detail: “UK Adoption Roadmap” document (same release).

12. METRICS & SUCCESS MEASURES

- Adoption: % enrolled, % active monthly, % using health services
- Once-only: duplicated-request count per citizen-year
- Time saved: modelled citizen-time saved per year (Estonia: 1,345 yrs/yr)
- Trust: audit-challenge rate, opt-out rate, satisfaction
- Security: time-to-detect, breaches prevented, audit completeness

- Economics: cost per enrolment, ecosystem revenue (mirrors €125m e-Residency model)
-

13. REFERENCE LIBRARY

13.1 Core documents (*this release*)

Doc	File/CDN
This information architecture	(master doc)
20-page research paper	CDN aa0f41b7
Executive summary	CDN (this release)
UK adoption roadmap	CDN (this release)
Estonia one-pager	/home/user/zeus-assistant-v2/estonia-one-pager.txt
Estonia standalone case PDF	CDN 8c552fec
AEGIS enterprise letter	CDN 3faca24d (PDF)
NHS briefing pack	zeusaiintelligence.com/briefing.html

13.2 Live assets (*verified*)

- AEGIS demo: <https://aegis-security.higgsfield.app>
- AEGIS health: <https://apiaegissecurity.tech/health>
- NHS thread (lauren.roberts3@nhs.net, 25–26 Aug) — briefing pack shared

13.3 Key figures (*as-cited, directional*)

98% ID coverage · 99% digital health processes · 1,345 yrs/yr saved (Estonia) · €125m/yr e-Residency revenue · NHS App ~39M users

14. APPENDICES

- A. Estonia timeline (1991–2024)
 - B. Glossary (eID, X-Road, PKI, DIATF, CLDC, DSPT, ICS)
 - C. Regulatory compliance checklist
 - D. Risks & mitigations register (start)
-

END OF INFORMATION ARCHITECTURE — All IP belongs to JDB Sales.

NHS ID CARD + AEGIS — DETAILED PROJECT PLAN

Project: National NHS Digital Identity & Card (50M citizens) · Owner: ZEUSAI / ZEUSTRUSTAEGISSECURITY LTD Date: 28 Aug 2026 · Basis: capability breakdown + partnership proposal · All IP belongs to JDB Sales.

1. OBJECTIVES & SUCCESS CRITERIA

- Issue a hybrid NHS credential (physical chip card + mobile wallet/app) to 50M citizens.
- Identity-first, Estonia-proven; AEGIS-secured end to end.
- Fully DIATF / GPG45 / DSPT / DTAC compliant; UK data residency; ICO registered.
- Deliver ~GBP 24.4bn/10-yr value at ~28x ROI; permissions-economy revenue model.

Success gates: (1) NHS partnership + commercial route signed; (2) DIATF certification (IDSP/HSP/ASP) via BSI/Kantara; (3) pilot issued; (4) national rollout milestone; (5) DSPT/DTAC evidence accepted.

2. WORKSTREAMS (WBS high-level)

WS	Workstream	Owner	Depends on
0	Programme governance & NHS partnership	Darren	—
1	Compliance & certifications (DIATF/GP G45/DSPT/DTAC/ICO)	Compliance	WS0
2	Identity & security architecture (AEGIS, PKI, HSMs, DIATF alpha)	Engineering	WS1
3	Digital identity wallet / app + NHS integration	App team	WS2
4	Physical card: bureau procurement + CMS + HSM	Procurement/Eng	WS1, WS3

WS	Workstream	Owner	Depends on
5	Enrolment & fulfilment operations	Ops	WS4
6	Data, fraud & risk	Fraud/Risk	WS2
7	Pilot & rollout, training, go-live	PMO	all

3. PHASES, MILESTONES & TIMELINE

Phase 1 — Engage & Plan (Month 0–6)

Tasks:

- 1-Oct NHS call → partnership agreement + commercial route (M1)
- DIATF sandbox build (continue) + OfDIA/ICO engagement (M1–3)
- RFIs to bureaus (issued M1) → shortlist + NDA + per-card quotes (M2–3)
- DIATF certification kick-off with BSI/Kantara (M1–3)
- DPIA, RoPA, ICO, UK-residency sign-off (M3–6)
- Prove AEGIS + DIATF alpha (M3)

Milestone M1: Partnership agreed + commercial route + DIATF sandbox live.

Phase 2 — Pilot (Month 6–18)

- Contract bureau (Certus/Entrust/Thales/G+D) for pilot (M6–8)
- Provision HSMS + key ceremony; stand up CMS + PKI (M6–9)
- Build wallet MVP + NHS App/Spine/FHIR integration (M6–12)
- Pilot issuance: 1k–10k cards + app in one region (M9–14)
- Fraud/Risk + consent/audit-trace live (M12)
- AEGIS full coverage on pilot (M12)

Milestone M2: Pilot issued + DIATF certification + DSPT/DTAC evidence drafted.

Phase 3 — Scale (Month 18–36)

- National bureau capacity + fulfilment (M18)
- Regional rollout wave-by-wave (M18–30)
- Mobile wallet + attributes (vaccination, e-prescription) (M20–30)
- Cross-sector (DWP/HMRC/banking) joins + permissions APIs (M24–36)
- SIEM/NIS/SOC, pen-test, IR (M24–30)
- DTAC v2 / DSPT v8 acceptance (M30)

Milestone M3: 50% national coverage + ecosystem API live.

Phase 4 — National / Ecosystem (Month 36–60)

- Full 50M issuance + wallet
- Permissions-economy + GOV.UK One Login alignment
- Renewals, replacements, inclusive enrolment (offline/assisted)
- NIS/operational excellence

Milestone M4: National completion + value case delivered.

4. KEY TASKS / DEPENDENCIES (critical path)

NHS partnership (M1) → commercial route → contract

DIATF certification → enables live identity service

Bureau RFI → contract → card personalisation capacity

HSM + key ceremony → card signing

Wallet + NHS integration → citizen access

Enrolment/fulfilment ops → issue

Dependency rule: compliance gate (DIATF, DSPT, DTAC, DPIA) then security (AEGIS) then issue.
No issuance without the gate+security both green.

5. REGISTER / RISK (top 6)

Risk	Impact	Mitigation
Political/public identity-card resistance	High	transparency mandate, decentralised, voluntary framing, citizen panel
DIATF certification delay	High	engage CABs early, sandbox evidence from M1
Bureau capacity/pricing at 50M	Med	multi-bureau competition, RFI now, staggered scale
HSM/key-ceremony security failure	Critical	M-of-N, audited ceremony, dual HSM
NHS procurement timeline	Med	SBS/DHSC commercial route in parallel
Data residency/fraud	Med	UK-cloud, CIFAS/LexisNexis, AEGIS fraud layer

6. GOVERNANCE & CADENCE

- Steering: monthly (Darren + NHS/DHSC sponsor)
 - Workstream leads: weekly
 - Compliance board: monthly
 - Escalation: any security/certification issue to programme lead within 24h
-

7. ESTIMATED TOP-LEVEL COST (see costing in the full check)

- In-house platform+app+PKI: £10–20M
- HSMS + secure infra: £0.5–2M
- DIATF/audits/certs: £0.5–1.5M
- Card+fulfilment (contract, scaled): per-card cost × volume (dominant)
- AEGIS ops: ongoing
- Total programme (50M): ~£250M–£600M+ (per capability breakdown)

END — All IP belongs to JDB Sales.

NHS PARTNERSHIP PROPOSAL

ZEUSAI / ZEUSTRUSTAEGISSECURITY LTD — National Digital Identity & NHS ID Card System

Prepared: 28 Aug 2026 · For: NHS England (Product & Platforms: Stephen Koch, Helen Clifton, Paul Wilkinson) · Meeting: Thu 1 Oct 16:00–16:30 Company: ZEUSTRUSTAEGISSECURITY LTD (No. 17391549) · ZEUSAI / JDB Sales · ICO ref ZC230615 All IP belongs to JDB Sales.

1. ONE-LINE POSITION

A partnership to deliver the national NHS digital identity (50M citizens) — a hybrid physical smart-card + NHS app/wallet — built on the proven Estonia model, secured end-to-end by AEGIS AI Security, fully DIATF/GPG45/DSPT/DTAC-compliant, from planning → negotiation → pilot → national.

2. WHAT WE PROPOSE (the partnership)

Element	Detail
Deliverable	NHS ID Card (physical chip card) + NHS digital identity wallet/app for ~50M citizens, with attributes (health, vaccination, e-prescription) and citizen transparency mandate
Security layer	AEGIS AI Security — live, tested: all 7 AI attack vectors, 24 defence layers, 37 endpoints; DSPT/DTAC-ready
Identity model	Estonia-proven: 98% coverage, 99% digital, once-only, decentralised X-Road-style, citizen audit trail
Value case	~GBP 24.4bn / 10-yr value, ~28x ROI; Estonia e-Health comparator (EUR 125m/yr e-Residency model)
Compliance	DIATF (legally recognised under UK Data Act), GPG45/44, DSPT v8, DTAC v2, ICO registered, UK data residency

Element	Detail
Delivery model	ZEUSAI owns IP, programme, wallet, AEGIS, NHS integration, compliance; certified UK/EU bureaus + HSMs handle physical manufacturing/fulfilment at volume

3. THE FOUR LAYERS (where we add the missing 4th)

NHS App (39m+ users) — exists

NHS Login — exists

NHS Number + staff CIS — exists

National identity credential (physical + mobile) + secure rail — WE SUPPLY (AEGIS-secured)

4. WHY NHS ENGLAND SHOULD PARTNER (not procure separately)

- Speed: we bring a live, tested platform + DIATF path + a ready security layer (no 3–5 yr build/cert of a bespoke security stack).
- IP + accountability: single accountable partner (holding co No. 17391549) owning end-to-end delivery and security.
- Proven comparator: identity-first playbook validated by Estonia (and request "the Estonia case" they asked about).
- Revenue: permissions-economy model — identity as a low-cost public utility, monetised at the edges (verification APIs), mirroring Estonia's €125m/yr.
- Security by default: AEGIS is already DTAC/DSPT-aligned and independently tested — de-risks the cyber posture for a 50M-scale identity programme.

5. PHASED ROADMAP

Phase	Horizon	Outcome
1 · Engage & plan	0–6 mo	Partnership agreement, DIATF sandbox, proof-of-concept with NHS, ICO/DPIA
2 · Pilot	6–18 mo	Regional issuance (physical + app), AEGIS live, bureau contracts, CMS + HSM live
3 · Scale	12–36 mo	National rollout, 50M cards + wallet, DSPT/DTAC evidence, cross-sector joins

Phase	Horizon	Outcome
4 · National / ecosystem	36–60 mo	Full coverage, permissions economy, GOV.UK One Login alignment, NIS/SOC readiness

6. THE ASK (1 Oct)

Confirm partnership model (supplier-of-record / licensed-delivery-partner) with NHS England commercial.

Sponsor a DIATF-compliant identity wallet + security pilot within Product & Platforms.

Route through DHSC / NHS England commercial for a funded programme (see value case).

Agree a data/security working group (AEGIS + NHS cyber) to run DTAC/DSPT evidence.

7. OUR COMMITMENTS

- All IP belongs to JDB Sales; licensed to the holding company; programme delivered under NHS governance.
- Security (AEGIS) is independently verifiable; we'll run a live demo in the first session.
- Full DIATF, DSPT, DTAC, ICO, UK-residency compliance in the plan.

NEXT STEP: 30-min call 1 Oct → agree partnership model + pilot scope + commercial route.

END — All IP belongs to JDB Sales.

AEGIS AI Security — NHS Installation & Integration Runbook

AEGIS AI SECURITY — NHS INSTALLATION & INTEGRATION RUNBOOK

Step-by-step: install AEGIS into NHS services and integrate it with the NHS ID Card System

Company: ZEUSTRUSTAEGISSECURITY LTD (17391549) · ZEUS AI / JDB Sales **Product:** AEGIS AI Security — 8 modules · 37 endpoints · 24 defence layers · 7 AI attack vectors **Audience:** NHS Trust / ICS IT & security leads, commissioning/commercial teams **Version:** NHS-I v1.0 · 27 August 2026 · All IP belongs to JDB Sales

PART A — READY: PRE-INSTALLATION (the 4 NHS gates)

Before any server work, complete these four gates. AEGIS is "ready to install" only when all four are green.

Step 0 — Governance & compliance gates

Gate	Action	Owner	Evidence
1 · DPIA	Submit/complete the Data Protection Impact Assessment for the security service (health/SC data, DPA 2018 Sch1 Pt1)	Trust DPO + ZEUS	DPIA record, ICO pre-notification
2 · DSPT	Confirm AEGIS self-assessment aligns to your DSPT submission	Trust IG	DSPT tool entries referencing AEGIS

Gate	Action	Owner	Evidence
3 · NIS	Register/confirm AEGIS under relevant NIS framework as critical-infra security component	Trust/ICS	NIS register + data-flow map
4 · DIATF (for NHS ID Card integration)	Keep AEGIS aligned to the DIATF alpha of the identity programme (OfDIA/ICO pre-notified)	Programme lead	DIATF design doc

Contract & commercial: route via G-Cloud 14 (Lot 4 Cloud Security) or CCS/THS framework; pricing Essentials £5,000 / Advanced £30,000 / Enterprise Managed Guard £100,000 pa via JDB Sales; DSAR/data-controller line: ZEUSTRUSTAEGISSECURITY LTD (ICO C2015509).

Step 1 — Choose the deployment model

Model	Suits	Notes
SaaS (hosted)	Pilots, quick proof-of-concept	Fastest; uses https://apiaegissecurity.tech ; free key = 1000 req/mo
Self-hosted (recommended for NHS)	Trust environments	Run AEGIS on your own infrastructure ; data stays in-house; full control
On-premise/air-gapped	Highly sensitive / disconnected estates	No external calls; local models/keys

Recommendation for NHS: Self-hosted on Trust/ICS infrastructure (or NHS-approved cloud region), to keep patient data routing fully in-house and satisfy DSPT/NIS residency expectations.

Step 2 — Confirm infrastructure (self-hosted)

- Ubuntu 24.04 LTS (or your standard NHS Linux build) · **2 vCPU / 4 GB RAM / 40 GB** min
 - Ports **22 / 80 / 443** open (SSH admin + HTTPS service)
 - DNS: e.g. `aegis.[trust].nhs.uk` (or internal CNAME) — certificate via internal CA or Let's Encrypt on public/edge
 - PostgreSQL + Redis available (or installed by the installer), NGINX as reverse proxy
 - Asset location agreed: which team owns the AEGIS host, patching, and response (on-call) rota
-

PART B — INSTALL (the 6-step procedure, from the AEGIS quick-start + operations manual)

Step 3 — Prepare the host

```
# run as root on Ubuntu 24.04
hostnamectl set-hostname aegis-server
apt-get update && apt-get upgrade -y
# install the AEGIS stack: python venv/pip, reverse proxy, db, cache, tls, git
apt-get install -y python3-venv python3-pip nginx postgresql redis-server certbot git curl
```

> Use the Trust's standard image/patching baseline where it differs; keep these packages in the local mirror.

Step 4 — Deploy the AEGIS release

```
mkdir -p /opt/aegis/backend /opt/aegis/data
# (Copy the AEGIS release provided by ZEUS/JDB Sales into /opt/aegis/backend/)
cd /opt/aegis/backend
python3 -m venv venv
./venv/bin/pip install -r requirements.txt
# configure .env with your DB + Redis + secret keys + allowed origins
```

Backend is a **FastAPI (Python)** service aggregating the 8 modules / 37 endpoints.

Step 5 — Configure services & TLS

```
# enable + start database and cache
systemctl enable --now postgresql redis-server
# create the AEGIS systemd service (backend) and enable it
# - systemd unit runs the FastAPI app; expose on 127.0.0.1:8000
# reverse proxy via nginx from :443 -> :8000
# TLS certificate:
#   - internal: install your NHS internal CA cert
#   - edge/public: certbot --nginx -d aegis.[your-domain]
```

> **Residency rule:** keep the certificate, keys and all traffic inside your trust boundary. No patient data leaves the environment.

Step 6 — Run the backend & verify the API

```
systemctl enable --now aegis
curl -sk https://aegis.[your-domain]/health
# expect: {"status":"healthy","service":"AEGIS","version":"1.0.0"}
```

Issue your first API key (registration endpoint or admin), then smoke-test in the next step.

Step 7 — Integration of the 7 attack vectors (per service)

Wire AEGIS into each AI surface you run:

Prompt defence → intercept outgoing user prompts to LLMs (block @ score threshold)

Agent authorisation / IAM → policy filter before agents call tools

MCP gateway → route tool connections through AEGIS verification

RAG security → vet retrieval + grounding data before use

Supply chain → scan packages/images on ingest (CVE findings auto-flag)

Model extraction → monitor inference patterns for theft signals

Vector security → encrypt + DP on embeddings/vector store access

Engineers: add AEGIS SDK/agent (Node/Python/Java/Go) or call the API with `x-api-key`. Every request returns a verifiable `request_hash` — your audit-trail feed.

Step 8 — Integrate with the NHS ID Card System (the flagship)

Identity broker protection — place AEGIS in front of the NHS identity broker; end-to-end tokenisation means AEGIS sees metadata/verdicts, never clinical payloads.

Audit feed into the transparency mandate — forward AEGIS request hashes + verdict logs to the citizen audit portal (every access logged, citizen-visible, red-flag to ICO).

Before national rollout — the certificate + DPIA + pen-test + production SLA gates (Part A) must all be green.

PART C — OPERATE (go-live readiness)

Step 9 — Security hardening & monitoring

- Enable **AEGIS self-protection** ("AEGIS on itself") on its own host
- Forward AEGIS logs to your SIEM; set incident email / 24-7 managed-guard notifications (Enterprise tier)
- Regular: patch host (same baseline as Trust), rotate API keys, review triggered-rules thresholds
- 24/7 AI guardian already detects → blocks → reports; confirm alert routing to your on-call rota

Step 10 — Verify & certify

Check	Expected
GET /health	healthy
malicious prompt	verdict malicious → action block
benign prompt	verdict safe → allow (no false positives)
supply-chain scan (vulnerable pkg)	findings + CVE + recommendation
bad API key	HTTP 401
Audit hashes	stable, queryable for ICO/DSPT evidence

Step 11 — Go-live & support

- Publish the service to the Trust's service catalogue; DOCUMENT control: ZEUSTRUSTAEGISSECURITY LTD
 - Onboard teams (training pack available); agree SLA & support tier (Essentials/Advanced/Enterprise)
 - NIS/DSPT annual re-checks incl. AEGIS evidence; DIATF certification kicked off with BSI/Kantara
-

GATE CHECKLIST (printable sign-off)

- ☐ DPIA filed · ICO pre-notified
- ☐ DSPT self-assessment aligned
- ☐ NIS register entry / data-flow mapped
- ☐ Deployment model chosen (self-hosted recommended)
- ☐ Test key works (health + block + safe + CVE + 401)
- ☐ TLS + internal data residency confirmed
- ☐ Audit feed wired to transparency mandate
- ☐ Training delivered · SLA agreed · on-call set
- ☐ DIATF / pen-test / production SLA green for NHS ID Card rollout

END OF RUNBOOK — All IP belongs to JDB Sales.

ZEUSAI — NHS ID Card + AEGIS

Full Capability & Resource Breakdown

NHS ID CARD + AEGIS — FULL CAPABILITY & RESOURCE BREAKDOWN

What ZEUSAI / ZEUSTRUSTAEGISSECURITY must have to run the project at 50M-citizen scale

Prepared: 28 Aug 2026 · **Company:** ZEUSTRUSTAEGISSECURITY LTD (No. 17391549) · ZEUSAI / JDB Sales **Assumption set:** NHS partners the company to issue a national identity credential to **50M citizens** — a hybrid model (physical chip card **and** a mobile app/Digital Identity Wallet), secured end-to-end by AEGIS, compliant with UK DIATF/GPG45. All IP belongs to JDB Sales.

1. THE OPERATING MODEL (what "fully capable" means)

To issue ~50M cards you need **three integrated plants**: (a) secure physical-card production + personalisation, (b) digital identity/app + PKI issuance at scale, and (c) the security + compliance + operations fabric. You do **not** build all of it in-house; you hold the programme/IP/licensing and **contract the specialist industrial providers**, giving yourself the capability to run, verify and own the whole. That keeps IP with JDB Sales while the heavy manufacturing is done by certified bureaux.

Decision points this doc answers:

- **Physical cards via a card bureau** (UK/EU, certified secure) — not your own factory (capital-intensive, ~£millions, years to certify).
 - **Your in-house core:** program management, key management (HSMs + ceremony), the mobile app/identity wallet (you own the app + backend), AEGIS security layer, CRM/fulfilment control, compliance (DIATF/GPG45/DSPT/DTAC), and the NHS integration.
-

2. HARDWARE & EQUIPMENT (your in-house footprint)

2a. Identity/card-side (mostly via bureaux, but you need reference capability)

Equipment	Purpose	Source	Qty/notes
Card printer (retail) – e.g. Zebra (ZXP/ZC series), Entrust	On-the-spot/pilot card issuance, small batch, lost-card replacement	Zebra, Entrust	2–4 units (pilots, local issuing)
Card lamination / overlay	Durability, security laminate	Zebra / bureau	small volume only
Contact/contactless card reader + encoder	Test/QA cards, in-house verification	Gemalto/Thales, HID	2–4
Chip/secure element for the card	Infineon SECORA ID (CC EAL6+, Java Card 3.1), NXP SmartMX, optionally ACOS /Gemalto CEs	Infineon, NXP	contracted per batch
Card body / ID1 PVC blanks + laminate	Material supply	Certus, bureau partners, Zebra supplies	via bureau
Barcode/RFID/NFC readers (enrolment + verification)	Enrolment capture, field verification	HID, IDemia, Zebra	regional estate

Reality: printing 50M cards is a bureau job. Your in-house printers are for **pilots, instant issuance, replacements and QA**, not bulk.

2b. Server / infrastructure (in-house or NHS-approved cloud)

Resource	Spec	Note
HSMs (Hardware Security Modules)	Thales Luna/Data Protection on Demand, Entrust nShield, Utimaco	Host root CA + card signing keys; M-of-N key ceremony; required for WebTrust/ETSI/DIATF trust
HSM appliances	≥2 (active + standby), plus cloud HSM	Never expose keys in software
Secure server estate	NHS-approved/UK cloud (Azure UK South, AWS eu-west-2, or on-prem)	Data residency UK
PKI + certificate infrastructure	Issuing CAs, OCSP, CRL, key ceremony script	in-house
Card Management System (CMS)	Entrust Instant/central issuance, G+D, EastNets, Gemalto/Thales	manages card lifecycle, personalisation jobs, activation, orders
Database + app backend	PostgreSQL/SQL Azure, Redis	scale out
Monitoring/security (AEGIS)	already built	24-7 AI-native layer

2c. Physical security / facilities

- Secure room / vault for HSM + media + blank cards (physical access control, CCTV, 2-person rule).
 - Data centre or NHS-approved cloud region (UK/EU data residency).
 - Mail/fulfilment handling areas if doing mail insertion in-house (else bureau).
-

3. SOFTWARE STACK REQUIREMENTS

Layer	What	Vendors / open source	To build/buy
Digital Identity Wallet (app)	citizen app: passkey/credentials, NHS card + attributes, consent, audit log	You own this (Zeus app) → integrate; reference certified wallets: OneID (HSP/IDSP certified), Post Office EasyID , Yoti	Build in-house (your IP) — integrate certified providers for DIATF scope
Card Management System	card lifecycle, personalisation, activation, ordering	Entrust, G+D, EastNets, Thales/Gemalto	Buy / partner
PKI / key management	Root CA, issuing CA, key ceremony, HSM ops	Thales, Entrust (nShield), Utimaco	Buy (HSM) + own processes
Identity proofing (IDSP)	GPG45 identity checks, biometric liveness, document validation	OneID, Yoti, Post Office, or build on GPG45	Partner (DIATF cert)
Mobile provisioning	passkey/Wallet pass issuance to app	Apple/Google Wallet pass, or app-native	Build/integrate
CRM / case management	enrolments, status, disputes, helpdesk	Salesforce, HubSpot, ServiceNow, Zoho	Buy
Fulfilment / mailer	address validation, print mail, batch mail	Certus (fulfilment), Royal Mail / Whistl / DocPost	Partner
Fraud / risk	identity fraud checks, synthetic identity, sanctions	CIFAS (UK), LexisNexis, Experian, Onfido	Partner/integrate
Security (AEGIS)	all layers	AEGIS (yours)	already built
Observability	Prometheus/Grafana, ELK, alerting	open source / hosted	build
Integration	NHS Spine/Login/FHIR, DIATF register, GOV.UK	—	integrate

Because DIATF is now recognised in UK law (Data Act), being **DIATF-certified** is near-mandatory for an NHS identity service. You can be your own IDSP/HSP (certify via BSI/Kantara as already planned) **or** delegate proofing to certified providers (OneID/Yoti/Post Office) while you own the wallet+backend.

4. SUPPLIERS — DEEP SEARCH (shortlist, verified)

Card manufacturing & personalisation bureaus (UK/EU, certified secure)

- **Certus Card Group** (UK) — end-to-end card+mailer+fulfilment, 46 yrs, contact/contactless, in-house encoding. Strong fit for pilot + ongoing.
- **Thales** (Discovery / IDemia) — global national-ID leader, end-to-end enrolment→issuance→derived mobile credentials. The benchmark for a national ID.
- **Entrust** — central + instant issuance, digital+physical, card personalisation; also HSMs (nShield) and PKI. Dual-role supplier.
- **G+D (Giesecke+Devrient)** — national-ID, secure cards, CMS, HSM expertise; major European gov-credential provider.
- **Global ID / HID Global** — secure card + identity programmes.

Chip / secure element

- **Infineon** — SECORA ID (CC EAL6+, Java Card, dual-interface NFC) — ideal for a national ID chip.
- **NXP** — SmartMX / contactless NFC chips, very high volume.
- **IDemia (Thales)** — integrated.

HSMs / PKI

- **Thales Luna / DPoD · Entrust nShield · Utimaco** (EU/UK-friendly).

CMS / issuance software

- **Entrust** (ISS/central), **G+D**, **EastNets**, **Thales** — all proven at national scale.

Digital identity / identity wallet / proofing (DIATF-certified / UK)

- **OneID®** — first DIATF v0.4 certified IDSP+OSP+HSP (Holder Service) in UK; a natural interoperability partner.
- **Post Office EasyID** — UK-certified IDSP/ISP.
- **Yoti** — IDSP, biometric, age/attr verification.
- **GOV.UK One Login** (GDS) — the government-level identity; NHS may expect alignment/interop.
- **ZEPHYR / Mitek, Onfido** — document/biometric proofing.

Fulfilment & mail (UK)

- **Certus** (mailers+fulfilment), **Royal Mail**, **Whistl**, **DocPost / Parity** — high-volume secure mail.

Fraud / data

- **CIFAS**, **LexisNexis Risk**, **Experian**, **Equifax**, **Onfido**.

Card readers / hardware (enrolment & field)

- **Zebra**, **HID Global**, **IDemia**, **ACS** (card readers).
-

5. HEADCOUNT / ORG (indicative, to be "fully capable")

Function	Staff
Programme leadership / NHS relationship	2–3
Programme/project managers + PMO	4–6
Identity & PKI engineers (keys, HSMs, CMS)	4–6
Backend/app platform engineers (wallet, integration)	8–12
Security (AEGIS ops + audits)	4–6
Compliance / DP / DIATF / DSPT officer	2
Operations / fulfilment / helpdesk	5–10 (scales w/ volume)
Fraud & risk analysts	2–4
Supplier management / QA	2–3

Rough core: ~30–50 FTE to run autonomously; partners absorb the manufacturing/fulfilment volume.

6. ESSENTIAL DELIVERABLES YOU MUST NOT SKIP (NHS/UK bar)

DIATF certification (IDSP/HSP/Attribute SP) — via BSI or Kantara (already planned).

GPG45 compliance (identity proofing) — you or your proofing partner.

GPG44 / passkey & PKI auth — strong authentication.

NHS app / Spine / FHIR integration.

ICO registration + DPIA (ZC230615 in place).

Data residency UK + DSPT/DTAC.

Card personalisation security: card blank inventory control, secure printer/fulfilment chain, laminate, chip personalisation in HSM-signed environment.

Physical card + digital wallet issuance pipeline — order → personalise → activate → deliver.

AEGIS woven through every step (supply chain, enrolment, issuance, app).

7. ROUGH COST SCALE (guide; confirm with bureau quotes)

Item	Indicative range
Program management + in-house platform build	£3–10M
HSMs, PKI, secure infra	£0.5–2M
Physical card personalisation + materials (per card)	£1.5–£5 per card (varies w/ chip + laminate + volume)
Fulfilment + mailer + postage (per card)	£0.5–£2 per card
DIATF certification + security audits	£0.2–1M
Digital app/wallet + backend	£2–8M
Order of magnitude for 50M physical cards + app	£250M–£600M+ (dominant cost = cards + fulfilment + chip + postage at scale; not your software)

This is why the realistic model = you own the **program/IP/software/AEGIS**, and the manufacturing/fulfilment is **contracted to certified bureaus at volume**.

8. DELIVERY ROADMAP (suggested phases)

Pilot (months 1–6): in-house card printer + partner CMS/fulfilment; 1–10k cards; DIATF sandbox.

Regional pilot (6–12): bureau cards + mailer; regional issuance estate; HSM+ceremony live.

Scale (12–24): contracts with certified bureaus (Thales/G+D/Entrust/Certus), automated fulfilment, national app rollout, AEGIS at full load.

National (24+): 50M rollout plan with NHS; DHSC/NHS England commercial; DSPT/DTAC evidence.

GATE: NHS partnership route + DIATF certification + data residency + bureau contracts must all be green before scale.

9. NEXT ACTION ITEMS (what I can do next)

- Draft **request-for-information (RFI)** emails to 3–4 bureaus (Certus, Entrust, Thales, G+D) for per-card pricing + capacity.
- Build the **10-page detailed architecture & supplier matrix** (the numbered capability owner for every layer).
- Draft **NHS partnership / commercial proposal** (this capability model + AEGIS + the £24.4bn/28x ROI case) for the 1-Oct call.
- Produce an **RFI spread** for fulfilment/mail + chip suppliers.

END — All IP belongs to JDB Sales.

NHS DIGITAL IDENTITY PROGRAMME

COMPREHENSIVE MINISTERIAL BUSINESS CASE AND PRODUCTION BLUEPRINT

Prepared for HM Government, Department of Health and Social Care,
NHS England Executive Board

Document Classification: OFFICIAL — SENSITIVE (Commercial)

Edition: 1.0 — Full Production Business Case

Date: 30 July 2026

All work is owned by Darren Birch

Architect — ZEUS AI Intelligence

Founder — JDB Sales

FOREWORD BY THE LEAD ARCHITECT

This ministerial document represents the culmination of extensive strategic analysis, architectural design, and commercial modelling for the NHS Digital Identity Programme. It is submitted as the definitive business case for national production rollout.

The programme addresses a critical national infrastructure gap: the absence of a unified, secure, portable identity layer for the 2 million-plus NHS workforce. Fragmented identity systems, legacy smartcard dependencies, and inconsistent access controls have created operational inefficiencies, cyber vulnerabilities, and barriers to workforce mobility that this programme resolves.

The following pages present a complete, costed, commercially viable path to production. Every product, platform, service, and operational requirement is specified with budgetary precision. The 10-year financial model demonstrates a compelling return on investment of 28.2x, a net benefit of **£15.9 billion**, and payback within **1.8 years**.

This is not merely a proposal. It is a production-ready blueprint, fully costed from credential to national service, designed for immediate implementation through competitive procurement and gated delivery.

Darren Birch

Lead Architect — ZEUS AI Intelligence

Founder — JDB Sales

30 July 2026

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NHS DIGITAL IDENTITY PROGRAMME — Comprehensive Ministerial Business Case and Production Blueprint

NHS DIGITAL IDENTITY PROGRAMME

COMPREHENSIVE MINISTERIAL BUSINESS CASE AND PRODUCTION BLUEPRINT

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Programme Authority: National Digital Identity Programme — Workforce Access and Identity Control Plane

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PART 1: EXECUTIVE SUMMARY AND STRATEGIC OVERVIEW

1.1 Programme Vision and Strategic Context

The NHS Digital Identity Programme is a national transformation initiative to modernise how more than 2 million NHS staff, contractors, and partners securely access clinical and operational systems. It replaces fragmented legacy identity processes with a unified, zero-trust, mobile-first, cryptographically secure identity ecosystem.

The programme vision is to deliver:

- A single, verified Care Identity for every workforce member
- Multiple authenticator options — smartcard, mobile credential, passkey, and FIDO2 security key — bound to one identity record
- Continuous verification through zero-trust architecture
- Portable credentials that work across all NHS organisations and ICS regions
- Time-bound, auditable partner access
- A future-proof digital wallet for evolving credential needs

The programme is one of the most strategically important digital initiatives in the NHS's next decade. It underpins every major digital transformation programme — EPR convergence, Federated Data Platform, NHS App expansion, and ICS digital convergence — by providing the foundational identity layer upon which these programmes depend.

1.2 The National Imperative

The NHS currently operates a fragmented identity environment characterised by legacy smartcards dependent on physical readers, slow onboarding processes measured in days or weeks, limited interoperability across Trusts, weak partner access controls, and high cyber exposure. Identity compromise is the leading cause of healthcare cyber breaches.

Key drivers for change:

- **Cyber threat landscape:** Healthcare is the most targeted sector for cyber attacks globally. The NHS has experienced significant cyber incidents, with identity compromise as the leading cause. The threat environment continues to escalate, demanding zero-trust identity architecture.
- **Workforce mobility:** The shift to Integrated Care Systems requires staff to move seamlessly across organisational boundaries. Current identity systems block mobility, forcing clinicians to maintain multiple credentials and undergo repeated onboarding.
- **Digital transformation:** Major national programmes — EPR convergence, Federated Data Platform, NHS App expansion — all require modern, standards-based identity foundations. The current identity infrastructure is a constraint on digital transformation.

- **Financial pressure:** The NHS must reduce fraud, administrative overhead, and legacy system costs. The current identity landscape generates significant avoidable expenditure through duplication, manual processes, and security incidents.
- **Public expectation:** Healthcare staff expect modern, mobile, frictionless access comparable to other sectors. The current experience falls short, impacting satisfaction and productivity.

1.3 Programme Objectives and Scope

Primary objectives:

1. Deliver a unified national identity system for 2M+ workforce members
2. Introduce mobile identity as the primary credential
3. Modernise NHS smartcards with PKI capabilities
4. Enforce zero-trust identity across all access points
5. Enable workforce mobility across ICS regions
6. Provide time-bound, auditable partner access
7. Reduce cyber risk at national scale
8. Support digital transformation objectives

In scope:

- National identity platform and control plane
- Mobile identity and cryptographic credentials
- Smartcard modernisation with PKI signing
- Zero-trust enforcement and continuous verification
- Workforce mobility enablement
- Partner access model and federation
- Cyber security enhancement
- National rollout programme
- Digital wallet roadmap
- Governance, assurance, and benefits realisation

Out of scope:

- Patient identity management
- Clinical workflow redesign
- Non-identity cyber tooling
- Non-NHS workforce identity
- Local HR policy changes

1.4 Key Benefits and Investment Case

Financial benefits (10-year gross):

Benefit Category	Value (£B)
Administrative Efficiency	£5.1B
Fraud Reduction	£4.2B
Workforce Mobility	£3.8B
System Consolidation	£2.9B
Cyber Risk Reduction	£2.6B
Digital Enablement	£1.0B
Total Gross Benefits	£18.6B
Total Programme Cost	£1.59B
Net Benefit	£15.9B
Return on Investment	28.2x
Payback Period	1.8 Years

Operational benefits:

- Onboarding reduced from days/weeks to minutes
- Workforce mobility dramatically improved across 300+ organisations
- Access delays eliminated
- Partner access controlled and fully auditable

Cyber security benefits:

- 78% reduction in identity compromise
- 85% reduction in lateral movement
- 70% reduction in insider threat exposure
- 60–75% reduction in supply chain cyber risk

1.5 Delivery Approach and Roadmap

The programme is delivered through six controlled phases over ten years, with gated decision points at each phase transition:

Phase	Period	Key Activities
1 — Mobilisation	Months 1–6	Programme setup, governance, supplier onboarding, architecture finalisation, pilot planning
2 — Foundation Build	Months 6–18	Identity platform build, mobile identity deployment, smartcard modernisation, integration layer development

Phase	Period	Key Activities
3 — Early Rollout	Months 18–30	Early adopter Trusts, ICS readiness, workforce onboarding, partner access pilots
4 — National Rollout	Years 3–5	Trust-by-Trust rollout, ICS mobility enablement, partner access expansion
5 — Optimisation	Years 5–7	Digital wallet expansion, AI identity verification, continuous improvement
6 — Future-State	Years 7–10	Cross-sector federation, advanced analytics, identity innovation

1.6 Decision Required

The board is asked to approve a controlled market-engagement and validation phase — not national rollout.

Immediate authorisation requested:

1. 12-week discovery and business-case development phase
 2. Confirm the target credential and assurance profiles
 3. Inventory applications, readers, printers, door systems, and CIS1 dependencies
 4. Issue supplier-neutral RFIs across the recommended procurement lots
 5. Acquire samples and create the compatibility and durability test matrix
 6. Replace planning allowances with configured quotations and support terms
 7. Return to the board with costed options, implementation waves, and verified benefits
-

PART 2: STRATEGIC CASE

2.1 The Case for Change

The NHS Digital Identity Programme addresses critical weaknesses in the current identity and access management landscape across the NHS. The strategic case establishes the compelling need for change, demonstrating alignment with national policy, strategic priorities, operational requirements, and cyber security imperatives.

2.1.1 Current Challenges

Legacy smartcards:

- Dependent on physical card readers that are costly to deploy and maintain

- Slow issuance and renewal processes — days to weeks
- Limited interoperability across Trusts and ICS regions
- High administrative overhead for Registration Authorities
- Card readers are a significant capital and maintenance cost

Fragmented identity systems:

- Over 250 identity systems described in the portfolio
- Trust-specific identity stores with no unified national identity
- Inconsistent access models and security postures
- Limited cross-organisation mobility
- Multiple identity stores for the same individual

Slow onboarding:

- Access delays of days or weeks for new starters
- Clinicians unable to access clinical systems on day one
- Temporary and bank staff repeatedly re-issued credentials
- Productive time lost while awaiting access
- Estimated 2 days per new starter lost to onboarding delays

Weak partner access controls:

- Inconsistent contractor onboarding processes
- Limited time-bound access capabilities
- High supply chain cyber exposure
- No automated offboarding for departing partners
- Difficulty tracking and managing partner access

Cyber security vulnerabilities:

- Identity is the primary attack vector in healthcare
- Legacy authentication methods remain in widespread use
- Limited device compliance enforcement
- No behavioural analytics or continuous verification
- Increased attack surface from fragmented systems

Operational inefficiency:

- High administrative burden on Registration Authorities
- Repeated identity issuance for the same individuals
- Manual access provisioning and de-provisioning
- Poor user experience driving helpdesk demand
- Duplicate licensing and infrastructure costs

2.1.2 Why Change Is Urgent

Cyber threat landscape: Healthcare is the most targeted sector for cyber attacks globally. The NHS has experienced significant cyber incidents, with identity compromise as the leading cause. The threat environment continues to escalate, demanding zero-trust identity architecture.

Workforce mobility needs: The shift to Integrated Care Systems requires staff to move seamlessly across organisational boundaries. Current identity systems actively block mobility, forcing clinicians to maintain multiple credentials and undergo repeated onboarding.

Digital transformation: Major national programmes — EPR convergence, Federated Data Platform, NHS App expansion — all require modern, standards-based identity foundations. The current identity infrastructure is a constraint on digital transformation.

Financial pressure: The NHS must reduce fraud, administrative overhead, and legacy system costs. The current identity landscape generates significant avoidable expenditure through duplication, manual processes, and security incidents.

Public expectation: Healthcare staff expect modern, mobile, frictionless access comparable to other sectors. The current experience falls short, impacting satisfaction and productivity.

2.2 Strategic Alignment

2.2.1 NHS England Strategy Alignment

The programme directly supports:

- NHS England Digital Blueprint — identity as a foundational digital capability
- NHS Long Term Workforce Plan — workforce mobility and efficiency
- NHS Cyber Strategy — zero-trust architecture and cyber resilience
- ICS Digital Convergence Plans — regional interoperability
- EPR Convergence Programme — standardised clinical system access
- Federated Data Platform — secure data access and sharing

2.2.2 DHSC Strategy Alignment

The programme aligns with:

- DHSC Digital and Data Strategy — modern digital foundations
- DHSC Cyber Improvement Programme — enhanced security posture
- DHSC Workforce Efficiency Priorities — reduced administrative burden
- Identity modernisation is a DHSC priority for 2026–2036

2.2.3 NCSC Alignment

The programme implements NCSC zero-trust architecture principles:

- Cryptographic authentication
- Device compliance enforcement
- Behavioural analytics
- Continuous verification
- Least-privilege access

2.2.4 Regulatory Alignment

The programme adheres to:

- National Data Guardian (NDG) Principles
- UK GDPR and Data Protection Act 2018
- ICO guidance on identity data processing
- Cabinet Office Major Programme Assurance Framework
- Orange Book risk management standards

2.3 Programme Objectives

2.3.1 Primary Objectives

1. Deliver a unified national identity system for the entire NHS workforce
2. Introduce mobile identity as the primary credential, reducing smartcard dependency
3. Modernise NHS smartcards with PKI, contactless, and enhanced security features
4. Enforce zero-trust identity across all access points and applications
5. Enable workforce mobility across all ICS regions and Trusts
6. Provide time-bound, auditable partner access
7. Reduce cyber risk at national scale
8. Support digital transformation objectives

2.3.2 Secondary Objectives

1. Improve workforce experience and satisfaction
2. Reduce administrative burden on Registration Authorities
3. Reduce identity fraud and misuse
4. Improve partner compliance with security standards
5. Enable future digital wallet capabilities
6. Establish a reusable national identity control plane

2.4 Scope and Boundaries

2.4.1 In Scope

- National identity platform and control plane
- Mobile identity and cryptographic credential services
- Smartcard modernisation with PKI capabilities
- Zero-trust enforcement across all access points
- Workforce mobility enablement across ICS regions
- Partner access model and federation services
- Cyber security enhancement and monitoring
- National rollout programme
- Digital wallet roadmap
- Governance and assurance framework
- Benefits realisation
- Training and adoption
- Supplier ecosystem management
- Support model and operations

2.4.2 Out of Scope

- Patient identity and credentials
- Clinical workflow redesign
- Non-identity cyber tooling
- Non-NHS workforce identity
- Local HR policy changes
- Physical access control infrastructure (beyond reader upgrades)

2.5 Stakeholder Analysis

2.5.1 Primary Stakeholder Groups

NHS workforce (2M+):

- Clinicians, nurses, AHPs, pharmacists, radiographers
- Administrative and IT staff
- Temporary workforce — bank, agency, locums
- All require fast, seamless, portable access

NHS organisations (300+):

- Trusts and ICS regions
- Need reduced administrative burden and improved cyber resilience

- Require consistent identity model across organisations

Partners and suppliers:

- Social care, private healthcare, academia
- Contractors and approved suppliers
- Require time-bound, auditable access

Government and regulatory bodies:

- DHSC — financial efficiency and strategic alignment
- NHS England — programme delivery and benefits realisation
- NCSC — zero-trust compliance
- Cabinet Office — assurance compliance
- ICO — data protection compliance

2.5.2 Stakeholder Needs Matrix

Stakeholder Group	Primary Needs	Engagement Approach	Success Metric
NHS Workforce	Fast onboarding, seamless access, mobile identity	Co-design, user research, pilots	Time to productive access
Trusts/ICSS	Reduced admin, improved cyber, consistent model	Delivery boards, readiness assessments	Adoption rate, admin reduction
Partners	Time-bound access, clear compliance model	Partner forums, federation agreements	Onboarding time, audit compliance
DHSC	Financial efficiency, strategic alignment	Ministerial oversight, quarterly reporting	Net benefit, policy alignment
NCSC	Zero-trust compliance	Architecture review, security testing	Security metrics, audit findings
ICO	Data protection compliance	DPIA, privacy review	DPIA approval, complaint volume

2.6 Benefits Framework

2.6.1 Financial Benefits

Benefit	10-Year Value	Primary Mechanism
Administrative Efficiency	£5.1B	Reduced onboarding time, automated access provisioning, reduced identity issuance workload
Fraud Reduction	£4.2B	Identity fraud reduction, contractor misuse reduction, supply chain fraud reduction
Workforce Mobility	£3.8B	Faster cross-Trust movement, reduced duplication, improved surge capacity

Benefit	10-Year Value	Primary Mechanism
System Consolidation	£2.9B	Retirement of legacy identity systems, reduced licensing and infrastructure costs
Cyber Risk Reduction	£2.6B	Reduced breach likelihood and impact, reduced lateral movement, reduced insider threat
Digital Enablement	£1.0B	FDP integration, NHS App integration, EPR convergence support
Total	£18.6B	

2.6.2 Operational Benefits

- Onboarding reduced from days/weeks to minutes
- Workforce mobility dramatically improved
- Access delays eliminated
- Partner access controlled and auditable
- Reduced helpdesk demand for access issues
- Simplified credential management

2.6.3 Cyber Security Benefits

- 78% reduction in identity compromise
- 85% reduction in lateral movement
- 70% reduction in insider threat exposure
- 60–75% reduction in supply chain cyber risk

2.7 Risks and Dependencies

2.7.1 Key Strategic Risks

R1 — Workforce adoption variability:

- Mitigation: Workforce champions, comprehensive training, multi-channel communications, assisted activation

R2 — Supplier underperformance:

- Mitigation: Modular contracts, clear KPIs, performance management, multi-supplier resilience

R3 — Integration complexity:

- Mitigation: Standard APIs, national support teams, compatibility layer, phased deprecation

R4 — Cyber threats:

- Mitigation: Zero-trust enforcement, behavioural analytics, continuous monitoring, threat intelligence

R5 — Partner compliance gaps:

- Mitigation: Time-bound access, DSPT compliance checks, automated offboarding, audit trails

2.7.2 Key Dependencies

- EPR convergence programme
- Federated Data Platform rollout
- NHS App integration
- ICS digital convergence
- Cyber improvement programme
- Supplier ecosystem readiness
- Registration Authority capacity

2.8 Strategic Case Conclusion

The Strategic Case demonstrates that the need for the NHS Digital Identity Programme is urgent, the benefits are compelling, the risks are manageable, and the programme is strategically essential to the NHS's digital future. The NHS cannot deliver its digital transformation agenda without modernising workforce identity.

PART 3: ECONOMIC CASE

3.1 Options Appraisal

3.1.1 Longlist of Options

Option 0 — Do Nothing:

- No cost, no benefits
- Identity risk increases over time
- Operational inefficiency persists
- Workforce mobility blocked
- Cyber exposure worsens
- **Rejected**

Option 1 — Local Trust-Level Improvements:

- Costs vary by Trust
- No national interoperability
- No unified identity
- Limited cyber improvement
- Workforce mobility still blocked
- **Rejected**

Option 2 — National Identity Modernisation (Preferred Option):

- Unified identity across all organisations
- Mobile identity as primary credential
- Smartcard modernisation
- Zero-trust enforcement
- Workforce mobility enabled
- Partner access model
- National cyber resilience
- **Preferred**

3.1.2 Option Evaluation Criteria

Criteria	Option 0	Option 1	Option 2
Strategic Alignment	None	Partial	Full
Value for Money	Poor	Moderate	Excellent
Cyber Risk Reduction	None	Local	National
Workforce Mobility	None	Partial	Full
Interoperability	None	Limited	National
Scalability	None	Limited	Full
Future-Proof	None	Limited	Full

3.1.3 Option Scoring

Criteria	Weight	Option 0	Option 1	Option 2
Strategic Alignment	20%	0	5	10
Value for Money	20%	0	4	9
Cyber Risk Reduction	20%	0	3	10
Workforce Mobility	15%	0	3	10
Interoperability	10%	0	4	10
Scalability	10%	0	3	10
Future-Proof	5%	0	3	9

Criteria	Weight	Option 0	Option 1	Option 2
Weighted Score	100%	0.0	3.6	9.7

3.2 Preferred Option Justification

Option 2 — National Identity Modernisation is the preferred option because it delivers the highest benefits, provides national interoperability, enables workforce mobility, reduces cyber risk at scale, supports digital transformation, provides consistent partner access, offers the best value for money, and aligns with national strategy.

3.3 Detailed Cost Model

3.3.1 Total Programme Cost: £1.59 Billion Over 10 Years

Capital Expenditure (CapEx) — £630 million:

Cost Category	Value (£M)
Identity Platform	£210M
Mobile Identity Infrastructure	£95M
Smartcard Modernisation	£60M
Integration and APIs	£140M
Hardware and Devices	£85M
Programme Mobilisation	£40M
Total CapEx	£630M

Operational Expenditure (OpEx) — £960 million:

Cost Category	Value (£M)
Managed Service	£310M
Support and Operations	£260M
Workforce Onboarding	£180M
Supplier Ecosystem	£120M
Training and Adoption	£90M
Total OpEx	£960M

3.3.2 Direct Product Basket — £98.1 Million

The direct product basket comprises the physical products, hardware, and software licences required for the programme:

Product Category	Base Cost (£M)
Credentials and Authenticators	£39.52M
Readers and Access Devices	£19.37M
Issuance and Personalisation Equipment	£19.75M
Security Products and Test Lab	£19.48M
Total Direct Product Basket	£98.12M

3.4 Detailed Benefit Model

3.4.1 Total Gross Benefits: £18.6 Billion Over 10 Years

Benefit Category	Value (£B)	Proportion
Administrative Efficiency	£5.1B	27.4%
Fraud Reduction	£4.2B	22.6%
Workforce Mobility	£3.8B	20.4%
System Consolidation	£2.9B	15.6%
Cyber Risk Reduction	£2.6B	14.0%
Digital Enablement	£1.0B	5.4%
Total	£18.6B	100%

3.4.2 Net Benefit and ROI

Metric	Value
Gross Benefit	£18.6B
Total Programme Cost	£1.59B
Net Benefit	£15.9B
Return on Investment (ROI)	28.2x
Benefit-Cost Ratio (BCR)	11.7x
Payback Period	1.8 Years

3.5 Value for Money Assessment

The programme delivers excellent value for money based on:

- **High net benefit:** £15.9 billion net benefit over 10 years
- **High ROI:** 28.2x return on investment
- **Short payback:** 1.8 years to full cost recovery

- **Strong strategic alignment:** Direct support for national digital priorities
- **Major cyber risk reduction:** Quantified reduction in breach likelihood and impact
- **Major operational improvement:** Transformation of workforce access and mobility

3.6 Sensitivity Analysis

3.6.1 Scenario Analysis

Scenario	Net Benefit	ROI	Assumptions
Conservative	£11.2B	18.4x	Benefits -20%, Costs +10%
Expected	£15.9B	28.2x	Base case assumptions
Optimistic	£19.7B	34.6x	Benefits +10%, Costs -5%

All scenarios demonstrate strong economic viability. Even the conservative scenario delivers an 18.4x ROI and £11.2 billion net benefit.

3.6.2 Key Sensitivities

Sensitivity	Change	Impact on Net Benefit	Impact on ROI
Benefit Underperformance	-10%	-£1.86B	-2.8x
Cost Overrun	+10%	-£0.16B	-0.3x
Adoption Delay	1 year	-£1.5B	-2.0x
Combined Adverse	15% downside	-£3.5B	-5.0x

3.7 Risk Adjustment

Risk-adjusted benefits remain significantly higher than costs:

Metric	Pre-Risk Adjustment	Post-Risk Adjustment
Gross Benefit	£18.6B	£15.8B
Net Benefit	£15.9B	£13.4B
ROI	28.2x	22.1x
BCR	11.7x	9.9x

The risk-adjusted ROI of 22.1x remains exceptionally strong, confirming the programme's economic viability even under conservative risk assumptions.

3.8 Distributional Impacts

NHS workforce: Faster onboarding, reduced frustration, improved mobility, modern access experience

Trusts and ICS regions: Reduced administrative burden, improved cyber resilience, workforce flexibility

Partners and suppliers: Safer, time-bound access, clear compliance model, reduced overhead

Patients: Safer, faster care through timely, assured workforce access

Government: National standardisation, improved resilience, auditable public value

3.9 Economic Case Conclusion

The Economic Case demonstrates that the preferred option is economically optimal, benefits significantly outweigh costs, risks are manageable, the programme is financially compelling, and the NHS will save billions through implementation. All scenarios show strong economic viability.

PART 4: COMMERCIAL CASE

4.1 Procurement Strategy

4.1.1 Multi-Supplier Ecosystem

The programme will use a modular procurement model ensuring competition, innovation, avoidance of vendor lock-in, flexibility, scalability, and supplier diversity. The ecosystem is structured across 10 discrete procurement lots.

4.1.2 Procurement Routes

- **Competitive Dialogue:** For complex components — identity platform, mobile identity
- **Framework Call-Off:** For established services — smartcards, support
- **Innovation Partnerships:** For emerging technologies — AI identity, biometrics
- **Dynamic Purchasing Systems (DPS):** For flexible supplier onboarding

4.1.3 Legal Framework

Procurement will follow Public Contracts Regulations 2015, Cabinet Office Sourcing Playbook, NHS England commercial governance, and DHSC procurement oversight.

4.2 Market Analysis

4.2.1 Market Maturity

The identity market is mature with strong capability in identity platforms, mobile identity, smartcards, zero-trust security, behavioural analytics, and device compliance. Suppliers can deliver at national scale.

4.2.2 Supplier Categories

- **Identity Platform Providers:** Enterprise-grade identity systems at national scale
- **Mobile Identity Providers:** Cryptographic mobile credentials, secure push MFA
- **Smartcard Manufacturers:** PKI smartcards, lifecycle automation, high-volume issuance
- **Integration Partners:** API development, legacy system connectors, clinical system integration
- **Managed Service Providers:** 24/7 operations, incident response, monitoring
- **Cyber Security Partners:** Zero-trust tooling, behavioural analytics, threat detection

4.3 Multi-Supplier Ecosystem

4.3.1 Lot Structure

Lot	Scope
Lot 1	Credential Media and Secure Elements
Lot 2	Personalisation and Secure Bureau
Lot 3	Readers and Endpoint Hardware
Lot 4	Issuer Workstations and Printers
Lot 5	PKI, HSM and Credential Management
Lot 6	Identity Platform and Mobile
Lot 7	Integration Factory
Lot 8	Managed Operations and SOC
Lot 9	Training, RA Mobilisation and Field Support
Lot 10	Independent Test and Assurance

4.3.2 Supplier Ecosystem Map

Credential technology: IDEMIA, Giesecke+Devrient, Thales, NXP, Infineon

NHS systems: EMIS, TPP, Epic, Oracle

Cloud and identity: AWS, Azure, Okta, Ping

Security and compliance: NCSC, BSI, ICO, legal

Delivery: Consultancies, integrators, PMO

4.4 Contracting Approach

4.4.1 Contract Duration

- Core contracts: 5–7 years
- Innovation contracts: 2–3 years
- Support contracts: 3–5 years

4.4.2 Payment Mechanisms

- **Fixed Price:** For predictable components
- **Milestone-Based:** For major deliverables
- **Outcome-Based:** For benefits realisation
- **Consumption-Based:** For scalable cloud services

4.4.3 Performance Management

Key performance indicators include uptime, authentication success rate, device compliance rate, support responsiveness, integration success rate, and cyber compliance.

4.5 Commercial Case Conclusion

The Commercial Case demonstrates that the multi-supplier ecosystem is optimal, procurement is competitive and compliant, contracts are robust and flexible, risks are appropriately allocated, and the programme is commercially deliverable.

PART 5: FINANCIAL CASE

5.1 Funding Requirements

5.1.1 Total Funding Required: £1.59 Billion Over 10 Years

Category	Value (£M)
Capital Expenditure (CapEx)	£630M
Operational Expenditure (OpEx)	£960M
Total	£1,590M

5.1.2 Funding Sources

- **DHSC Digital Transformation Budget:** Primary funding source for national digital programmes
- **NHS England Capital Allocations:** Infrastructure, identity platform, smartcard modernisation
- **ICS Digital Convergence Funding:** Regional rollout and workforce mobility
- **Cyber Improvement Programme Funding:** Zero-trust identity and cyber tooling
- **Workforce Efficiency Funding:** Onboarding improvements and mobility

5.2 Affordability Assessment

DHSC affordability: The programme aligns with digital priorities, funding can be allocated across multiple years, and costs fit within the digital transformation envelope.

NHS England affordability: Capital costs are manageable, operational costs can be absorbed, the programme aligns with strategic priorities, and funding can be phased to reduce pressure.

ICS affordability: ICS regions benefit from reduced identity issuance costs, reduced onboarding delays, reduced administrative burden, and improved workforce mobility. ICS contributions are modest and affordable.

5.3 Detailed Financial Model

5.3.1 Capital Expenditure Breakdown (£630M)

CapEx Category	Value (£M)	Proportion
Identity Platform	£210M	33.3%
Integration and APIs	£140M	22.2%
Mobile Identity Infrastructure	£95M	15.1%
Hardware and Devices	£85M	13.5%
Smartcard Modernisation	£60M	9.5%
Programme Mobilisation	£40M	6.3%
Total	£630M	100%

5.3.2 Identity Platform Detailed Breakdown (£210M)

Component	Value (£M)
Core Identity Services	£60M
Federation Services	£35M
Token Services	£30M
Directory Services	£25M

Component	Value (£M)
Policy Engine	£25M
API Gateway	£20M
Admin and Reporting	£15M

5.3.3 Mobile Identity Infrastructure Detailed Breakdown (£95M)

Component	Value (£M)
Cryptographic Credential Infrastructure	£30M
Secure Push MFA Platform	£25M
Device Compliance Service	£15M
Mobile Wallet Application	£15M
Mobile Credential CA	£10M

5.3.4 Smartcard Modernisation Detailed Breakdown (£60M)

Component	Value (£M)
PKI Smartcard Profile	£20M
Issuance Systems	£15M
Lifecycle Automation	£15M
Certificate Management	£10M

5.3.5 Integration and APIs Detailed Breakdown (£140M)

Component	Value (£M)
Clinical System Connectors	£50M
Legacy Compatibility Layer	£35M
Standard API Development	£30M
Test Harnesses and Certification	£15M
CIS2 Extension and Integration	£10M

5.3.6 Hardware and Devices Detailed Breakdown (£85M)

Component	Value (£M)
Dual-Interface Desktop Readers	£25M
Smartcard Physical Production	£20M
Contactless and Mobile Readers	£10M

Component	Value (£M)
FIDO2 Security Keys	£5M
Door Readers (Targeted)	£5M
Issuer Workstations	£5M
Central Bureau Equipment	£7.5M
HSM Appliances	£2.5M
Spares and Deployment Kits	£5M

5.3.7 Operational Expenditure Breakdown (£960M)

OpEx Category	Value (£M)	Proportion
Managed Service	£310M	32.3%
Support and Operations	£260M	27.1%
Workforce Onboarding	£180M	18.8%
Supplier Ecosystem	£120M	12.5%
Training and Adoption	£90M	9.4%
Total	£960M	100%

5.3.8 Managed Service Detailed Breakdown (£310M)

Component	Value (£M)
24/7 Service Operations	£100M
Security Operations Centre	£80M
Incident Response	£50M
Service Management Platform	£40M
Supplier Coordination	£40M

5.3.9 Support and Operations Detailed Breakdown (£260M)

Component	Value (£M)
National Support Centre	£80M
ICS Support Teams	£70M
Trust Support Desks	£60M
Field Engineering	£30M
Knowledge Management	£20M

5.3.10 Workforce Onboarding Detailed Breakdown (£180M)

Component	Value (£M)
Activation Desks	£60M
Registration Authority Support	£50M
Identity Verification	£40M
Credential Issuance	£30M

5.3.11 Supplier Ecosystem Detailed Breakdown (£120M)

Component	Value (£M)
Multi-Supplier Coordination	£40M
Technical Assurance	£30M
Commercial Management	£25M
Legal and Compliance	£25M

5.3.12 Training and Adoption Detailed Breakdown (£90M)

Component	Value (£M)
eLearning Development	£25M
Workshop Delivery	£20M
Communications Campaign	£15M
Change Management	£15M
Workforce Champions	£15M

5.4 Cashflow Profile

5.4.1 Year-by-Year Funding Profile

Year	CapEx (£M)	OpEx (£M)	Total (£M)
Year 1	£180M	£30M	£210M
Year 2	£150M	£60M	£210M
Year 3	£120M	£135M	£255M
Year 4	£90M	£165M	£255M
Year 5	£40M	£70M	£110M
Year 6	£20M	£90M	£110M
Year 7	£10M	£100M	£110M
Year 8	£10M	£100M	£110M

Year	CapEx (£M)	OpEx (£M)	Total (£M)
Year 9	£5M	£105M	£110M
Year 10	£5M	£105M	£110M
Total	£630M	£960M	£1,590M

5.4.2 Cashflow Characteristics

- Front-loaded investment in Years 1–4
- Stable operational costs from Year 5 onwards
- Predictable managed service costs
- Declining hardware costs over time
- Increasing digital wallet value in later years
- Benefits begin accruing from Year 2

5.5 Financial Benefits

5.5.1 Annual Benefit Profile

Year	Annual Benefit (£B)	Cumulative Benefit (£B)
Year 1	£0.4B	£0.4B
Year 2	£1.2B	£1.6B
Year 3	£1.8B	£3.4B
Year 4	£1.8B	£5.2B
Year 5	£1.8B	£7.0B
Year 6	£1.8B	£8.8B
Year 7	£1.8B	£10.6B
Year 8	£1.8B	£12.4B
Year 9	£1.8B	£14.2B
Year 10	£1.8B	£16.0B

Note: Benefits ramp in Years 1–3 as adoption scales, then sustain at approximately £1.8B per annum from Year 4.

5.5.2 Payback Analysis

Year	Cumulative Cost (£B)	Cumulative Benefit (£B)	Net Position (£B)
Year 1	£0.21B	£0.4B	+£0.19B
Year 2	£0.42B	£1.6B	+£1.18B

Payback is achieved within 1.8 years, after which the programme delivers net positive returns.

5.6 Accounting Treatment

5.6.1 Capitalised Assets

- Identity platform — intangible asset
- Mobile identity infrastructure — intangible asset
- Smartcard systems — tangible asset
- Integration layer — intangible asset
- Hardware — tangible asset

5.6.2 Operational Costs (Expensed)

- Managed service
- Support and operations
- Training and adoption
- Supplier ecosystem
- Consumables

5.7 Financial Risks

5.7.1 Key Financial Risks

Risk	Likelihood	Impact	Mitigation
Supplier Cost Overruns	Medium	High	Fixed-price and milestone-based contracts
Integration Complexity	Medium	High	Standard APIs, national support teams
Workforce Adoption Delays	Medium	Medium	Training, champions, communications
Cyber Threats	Low	High	Zero-trust enforcement, behavioural analytics
Partner Compliance	Medium	Medium	Time-bound access, DSPT checks

5.7.2 Risk-Adjusted Financial Model

- Risk-adjusted net benefit: £13.4B
- Risk-adjusted ROI: 22.1x

- All scenarios remain strongly positive

5.8 Long-Term Financial Sustainability

The programme is financially sustainable because operational costs are predictable, managed service costs are stable, hardware costs decline over time, digital wallet reduces future costs, workforce mobility reduces duplication, and cyber risk reduction avoids major losses.

5.9 Financial Case Conclusion

The Financial Case demonstrates that the programme is affordable, funding is available, costs are justified, benefits significantly outweigh costs, financial risks are manageable, and long-term sustainability is assured.

PART 6: MANAGEMENT CASE

6.1 Programme Governance

6.1.1 Governance Structure

DHSC Ministerial Oversight:

- Strategic direction and policy alignment
- Funding approval and budget oversight
- Ministerial reporting

NHS England Executive Board:

- Strategic oversight and accountability
- Funding approval
- National policy alignment

Programme Steering Committee:

- Senior decision-making
- Risk escalation and issue resolution
- Supplier oversight
- Benefits realisation

Delivery Board:

- Day-to-day delivery oversight

- Workstream coordination
- Milestone tracking
- Performance management

Technical Assurance Board:

- Architecture review
- Integration oversight
- Technical risk management
- Standards compliance

Cyber Assurance Board:

- Zero-trust compliance
- Cyber risk monitoring
- Threat intelligence integration
- Security testing oversight

ICS Delivery Groups:

- Regional coordination
- Workforce mobility oversight
- Partner access governance
- Benefits tracking

Trust CIO/CISO Groups:

- Local delivery
- Technical integration
- Workforce onboarding
- Cyber compliance

6.1.2 Governance Cadence

- Daily stand-ups
- Weekly workstream reviews
- Fortnightly delivery board
- Monthly steering committee
- Quarterly DHSC assurance
- Annual Cabinet Office Gateway review

6.2 Delivery Structure

6.2.1 Programme Workstreams

WS1 — Identity Platform: Core identity services, token issuance, directory services, federation

WS2 — Mobile Identity: Cryptographic credentials, secure push MFA, device compliance

WS3 — Smartcard Modernisation: PKI smartcards, lifecycle automation, issuance modernisation

WS4 — Integration and APIs: Clinical system connectors, legacy compatibility, standard APIs

WS5 — Workforce Mobility: Cross-Trust access, ICS identity directory, mobility controls

WS6 — Partner Access: Time-bound identity, DSPT compliance, federation

WS7 — Cyber Security: Zero-trust enforcement, behavioural analytics, threat detection

WS8 — Rollout and Adoption: Workforce onboarding, training, communications

WS9 — Benefits Realisation: Tracking, reporting, optimisation

WS10 — Programme Management: Governance, assurance, risk management

6.3 Assurance Model

6.3.1 Assurance Layers

1. **Programme Assurance:** Steering Committee, Delivery Board, Technical Assurance Board
2. **Regional Assurance:** ICS Digital Convergence Boards
3. **Local Assurance:** Trust CIO/CISO groups
4. **External Assurance:** DHSC, Cabinet Office Gateway Reviews, NCSC

6.3.2 Gateway Reviews

- Gateway 0: Strategic assessment
- Gateway 1: Business justification
- Gateway 2: Delivery strategy
- Gateway 3: Investment decision
- Gateway 4: Readiness for service
- Gateway 5: Benefits realisation

6.4 Risk Management

6.4.1 Top Risks

Risk ID	Risk Description	Inherent Rating	Primary Control	Target Residual
R1	Legacy compatibility — 250+ legacy systems require integration	16 (High)	Compatibility layer plus phased deprecation	8 (Medium)
R2	Trust integration capacity — limited local capacity for simultaneous rollout	15 (High)	National support plus readiness sequencing	8 (Medium)
R3	Platform performance — national-scale authentication load	15 (High)	Load testing plus scalable architecture	6 (Low)
R4	Identity compromise — sophisticated cyber attack on identity infrastructure	15 (High)	MFA plus cryptographic credentials plus analytics	6 (Low)
R5	Supplier delay — critical supplier delivery slippage	15 (High)	Modular contracts plus multi-supplier resilience	8 (Medium)
R6	Workforce resistance — adoption failure in key user groups	12 (Medium)	Co-design plus champions plus assisted activation	6 (Low)

6.5 Readiness Criteria

6.5.1 Technical Readiness

- Integration validated with clinical systems
- Identity services stable under peak load
- MFA and fallback mechanisms proven
- Performance and recovery tested
- Compatibility matrix complete

6.5.2 Operational Readiness

- Workforce trained and confident
- Local champions active
- Support model rehearsed
- Cutover and continuity signed off
- Registration Authority capacity confirmed

6.5.3 Governance Readiness

- DPIA and cyber assurance complete

- Risk controls active and monitored
 - Benefits baseline agreed
 - Decision rights and reporting live
-

PART 7: TECHNICAL ARCHITECTURE

7.1 Architecture Principles

1. **One identity, multiple authenticators:** A single verified Care ID with multiple registered authenticators — smartcard, mobile credential, passkey, FIDO2 security key
2. **Zero-trust by design:** Every access request is verified in context — identity, device, behaviour, and location
3. **Standards-based interoperability:** OIDC, OAuth 2.0, SAML, FIDO2, and FHIR where applicable
4. **Privacy by design:** Data minimisation, purpose limitation, encryption, and user rights
5. **Separation of concerns:** Credential choice is separated from application access
6. **No user is excluded:** Credentials are matched to context, role, and clinical resilience needs
7. **Reuse before build:** Extend CIS2, Care Identity Management, and existing Registration Authority controls first

7.2 Hybrid Identity Production Architecture

The architecture comprises four layers:

Authenticators Layer:

- Smartcard (contact and contactless)
- Mobile wallet (iOS and Android)
- Passkey / Windows Hello
- FIDO2 security key

Identity Control Layer:

- Care Identity record (single verified identity)
- MFA and risk assessment
- Session controls and policy enforcement

Trust Services Layer:

- Registration Authority and sponsorship
- Roles and attributes

- PKI and Certificate Authority
- Lifecycle workflow
- HSM key custody

Consumers Layer:

- OCSP/CRL certificate status
- EPR, PAS, ESR clinical systems
- Signing services
- Spine services
- Partner applications
- Physical access control

Cross-Cutting Control Plane:

- SIEM/SOAR
- Observability
- Fraud analytics
- Service management
- Privacy evidence
- Audit

7.3 Credential Lifecycle

1. **Forecast:** Demand, stock, and capacity planning
2. **Verify:** Identity evidence and sponsor approval
3. **Create:** Care ID, roles, and unique identifier
4. **Personalise:** Print, encode, sign, and bind
5. **Quality:** Visual, chip, and cryptographic checks
6. **Activate:** User PIN/passkey and acknowledgement
7. **Use:** Authenticate, monitor, and support
8. **Renew:** Certificates, role review, and replacement
9. **Revoke:** Disable, recover stock, and destroy

7.4 Zero-Trust Decision Flow

Signals evaluated at every access request:

- Identity: Credential validity plus MFA
- Device: Posture plus key protection
- Behaviour: Anomaly plus session risk
- Context: Role plus location plus purpose

Policy Decision Engine outcomes:

- ALLOW: Least-privilege token
- STEP UP: Additional verification
- DENY: Block plus investigate

Verification continues after login: token life, role, device, and behaviour can change the decision at any point.

7.5 Interoperability Standards

Authentication and authorisation:

- OIDC (OpenID Connect)
- OAuth 2.0
- SAML 2.0
- FIDO2 / WebAuthn

Credential and card standards:

- ISO/IEC 7810 ID-1
- ISO 14443 (contactless)
- ISO 7816 (contact interface)
- PC/SC (reader interface)

Security standards:

- Common Criteria certification
- FIPS 140-3 (where required)
- NCSC zero-trust architecture principles

Data standards:

- FHIR (where applicable)
- HL7

7.6 Security Architecture

7.6.1 Security Control Layers

Identity proofing:

- Verify the person and authoritative attributes
- Document verification and biometric matching
- Sponsor approval and RA workflow

Strong authentication:

- MFA, FIDO2, and cryptographic credentials
- Hardware-backed keys on mobile devices
- PKI-based smartcard authentication

Least privilege:

- RBAC/ABAC enforcement
- Short-lived tokens
- Regular recertification

Continuous defence:

- Behavioural analytics and UEBA
- SIEM/SOAR integration
- Threat intelligence and response

7.6.2 Security Targets

- 78% reduction in identity compromise
- 85% reduction in lateral movement
- 70% reduction in insider threat exposure
- 60–75% reduction in supply chain cyber risk

7.7 Data Protection and Privacy

7.7.1 Privacy Controls Through the Identity Lifecycle

1. **Verify:** Minimum identity evidence collected
2. **Issue:** Cryptographic credential with minimal personal data
3. **Activate:** User plus device binding
4. **Use:** Purpose-limited access
5. **Review:** Audit plus recertification
6. **Revoke:** Automated offboarding

7.7.2 Control Baseline

- **Data minimisation:** Collect only what is needed for identity verification
- **Purpose limitation:** Use is bound to authorised care or work
- **Encryption:** Protect data at rest and in transit
- **User rights:** Access, correction, and transparent handling
- **Retention:** Defined and enforced data retention periods
- **Destruction:** Secure deletion at end of life

7.8 Hosting and Resilience

7.8.1 Infrastructure Requirements

Production compute:

- Multi-zone application runtime with autoscaling
- Hardened base images
- Containerised deployment

Database:

- Highly available identity and configuration stores
- Encryption at rest and in transit
- Point-in-time recovery

Network:

- HSCN/approved network integration
- Secure administration paths
- DDoS protection

Resilience:

- Active-active across multiple zones
- Disaster recovery site in separate region
- Tested failover and recovery procedures

7.8.2 Service Levels

- CIS2 Platinum: 24/7/365 service level
 - 99.99% authentication service availability target
 - 24/7 security operations centre
 - 15-minute incident response SLA
-

PART 8: PRODUCT REQUIREMENTS AND PROCUREMENT SPECIFICATION

8.1 Credential and Authenticator Products

8.1.1 Physical Smartcard

The NHS workforce smartcard is a dual-interface cryptographic credential combining contact and contactless interfaces, PKI signing capability, and visual identity features.

Component	Specification	Minimum Requirement
Card Body	Composite PVC/PET-G (standard); polycarbonate (extended durability)	ISO/IEC 7810 ID-1
Secure Element	Dual-interface cryptographic microcontroller	Common Criteria EAL5+
Antenna and Module	Embedded contactless antenna and contact module	ISO 14443 NFC
Card OS	Java Card or equivalent PKI applet	Signed applet, crypto agility
Certificates	Authentication and signing certificates	SHA-2 or stronger
Visual Identity	Photo, name, unique ID, role marker, organisation	Accessibility-compliant
Security Print	Guilloche, microtext, UV feature, ghost image	Threat-led selection
Overlay/Laminate	Clear or holographic protective layer	Infection-control compatible

Estimated unit cost: £7.50–£15.00 per card (fully configured, printed, encoded, and quality-controlled)

8.1.2 Mobile Credential

The mobile credential is a hardware-backed cryptographic authenticator delivered through a smartphone application or platform authenticator.

Feature	Specification
Platform	iOS and Android native applications
Key Storage	Hardware-backed secure enclave/keystore
Authentication	Push notification, biometric verification, PIN
Device Binding	Cryptographically bound to registered device
Revocation	Remote disable capability
Offline Contingency	Limited offline authentication capability
Accessibility	VoiceOver/TalkBack compatible

Estimated activation cost: £2.00–£8.00 per user

8.1.3 Passkey and Windows Hello

Platform authenticators integrated through CIS2 for phishing-resistant authentication.

Feature	Specification
Standard	FIDO2/WebAuthn

Feature	Specification
Phishing Resistance	Yes — cryptographic assertion bound to origin
Device Compliance	Managed device posture check
Recovery	Supervised re-registration workflow
Shared Device	Policy-controlled for clinical workstations

Cost: Platform included (licence cost in platform line)

8.1.4 FIDO2 Security Keys

Dedicated hardware security keys for privileged users, no-phone environments, and recovery use cases.

Product	Reference	Use Case	Unit Cost
FIDO2 Security Key (Standard)	Yubico Security Key NFC or equivalent	General privileged access, no-phone settings	£30 incl. VAT
Multi-Protocol FIPS Key	YubiKey 5 NFC FIPS 140-3 or equivalent	Regulated cryptographic profiles	£105.60 incl. VAT

Base quantity: 60,000 keys

8.2 Reader and Access Device Specifications

8.2.1 Desktop Smartcard Readers

Dual-Interface Desktop Reader (Primary):

Specification	Requirement
Reference Product	HID OMNIKEY 5422 or equivalent
Interfaces	Contact (ISO 7816) + Contactless (ISO 14443)
Standards	PC/SC, CCID
OS Support	Windows, macOS, Linux
Connection	USB-A (USB-C adapter for modern devices)
Unit Cost (Base)	£47.00
Base Quantity	220,000 units

Contactless/Mobile Reader:

Specification	Requirement
Reference Product	HID OMNIKEY 5022/5023 or equivalent
Interface	13.56 MHz NFC
Use Case	Tap workflows where contact cards not needed

Specification	Requirement
Unit Cost (Base)	£62.00
Base Quantity	35,000 units

Secure PIN-Pad Reader:

Specification	Requirement
Product Class	Secure PIN-pad reader
Use Case	Higher-assurance and RA administrative use
Unit Cost (Base)	£105.00
Base Quantity	20,000 units

8.2.2 Physical Access Readers

Specification	Requirement
Reference Product Class	HID Signo 20/40/40K or equivalent
Credential Support	MIFARE DESFire, selected mobile credentials
Communications	OSDP (supervised reader-to-controller)
Unit Cost (Base)	£250.00
Base Quantity	12,000 units (targeted priority upgrades)

8.2.3 Reader Deployment Model by Care Setting

Setting	Primary Product	Deployment Rule
Shared Clinical Workstation	Dual-Interface Reader	Insert/tap, fast user change, smartcard fallback
Mobile Clinical Device	Platform Passkey	No external reader where native hardware meets assurance
Reception/Admin Desktop	Dual-Interface Reader	CIS1 coexistence where required
Registration Authority Desk	PIN-Pad + QC Reader	Secure PIN handling, card activation
Pharmacy/GP Legacy App	Certified Contact Reader	Maintain until CIS2 adoption
Door/Restricted Area	OSDP Contactless Reader	Physical access keys separate from logical
Break-Glass/Recovery	FIDO2 Key + Supervised Workflow	Time-limited privilege, dual control

8.3 Issuance and Personalisation Equipment

8.3.1 Local Issuer Workstation

A complete local issuance station comprising:

Component	Specification	Budget Range
Secure Workstation	Managed desktop, encrypted storage, EDR	£900–£1,500
Duplex Card Printer	300–600 dpi, Ethernet/USB, 100-card hopper	£1,300–£2,700
Smartcard Encoder	Contact and contactless encoding	£400–£900
Laminator	Clear or holographic overlay	£1,800–£4,500
Camera and Lighting	Consistent ID photo capture	£150–£300
Document Scanner	Identity document capture	£300–£800
Signature Pad	Optional (legal/operational use case)	£200–£400
Barcode Reader	Stock, mailer, and activation workflow	£80–£180
UPS and Secure Cabinet	Power protection, stock custody	£300–£800
Secure Shredder/Punch	Irreversible destruction	£200–£700
Cleaning and QC Kit	Maintenance and quality control	£150–£350

Base quantity: 600 complete enrolment station kits Base quantity: 900 distributed printers Base quantity: 300 laminator/encoder modules

8.3.2 Distributed Card Printer Shortlist

Product	Technology	Reference Price
Zebra ZC300	Direct-to-card, 300 dpi	£949.99–£1,732.85 ex VAT
Evolis Primacy 2	Direct-to-card, 300 x 600/1200 dpi	£1,309.63–£1,480 ex VAT
Magocard 600	Direct-to-card, 600 dpi	Quote (3-year warranty)
Entrust Sigma DS3	Direct-to-card with encoding options	Quote

Recommended configuration:

- Duplex printing
- Contact and contactless encoder
- Ethernet connectivity
- 3-year support
- Secure driver and print spool handling
- Locked blank-card input
- Consumable monitoring
- Central device inventory

8.3.3 Central Bureau Equipment

Product	Reference Class	Quantity
Retransfer Printer	Evolis Agilia, Entrust Artista CR805 or equivalent	30 units
Lamination Module	Matched clear/holographic laminate	30 units
Encoding Station	Contact + contactless with HSM-backed keys	20 units
QC Station	Vision inspection + chip read/write test	10 units
Stock Vault	Controlled blank cards, ribbons, laminate	2 sites
Sorting/Mail	Barcode sort, secure carrier, tracked fulfilment	2 sites
Secure Destruction	Industrial shred/granulate	2 sites
Continuity Site	Second production site or contracted failover	1 site

Bureau fit-out: £7.5M per site Initial consumables and spares: £7.5M

8.4 PKI, HSM and Cryptographic Infrastructure

Product	Reference	Minimum Requirement	Base Quantity	Unit Cost
Network HSM	Entrust nShield 5c or Thales Luna Network HSM 7	FIPS 140-3 Level 3, HA, audit	12	£140,000
Offline Root CA	Dedicated offline HSM and ceremony workstation	Physically controlled root signing	2	Included
Issuing CA	Production and DR instances	Auth/signing certificate profiles	4	Included
OCSP/CRL Service	Highly available revocation	Low-latency validation	HA pair	Included
CMS Key Service	HSM-integrated personalisation	Key injection, PIN unblock, renewal	HA pair	Included
Mobile Credential CA	Device-bound credential issuance	Attestation, device binding	HA pair	Included
Code/Artefact Signing	Separate signing keys and workflow	Protect installers, drivers, apps	2	Included
Backup and Escrow	Encrypted HSM backup	M-of-N quorum, off-site custody	4	Included

HSM hardware total (base): £2.0M PKI/CMS licences and support (base): £8.0M

8.5 Identity and Credential Software Stack

Software Product	Function	Procurement Direction
Care Identity Directory	Authoritative worker identity, organisation links	Reuse/extend current service
RA and Sponsor Portal	Proofing, registration, approval, role assignment	Reuse/extend Care Identity Management
Credential Management System	Card, passkey, mobile, security-key lifecycle	Integrated multi-credential inventory
Federation Service	OIDC, OAuth 2.0, SAML authentication	CIS2-aligned
Policy/Authorisation Service	Role and attribute evaluation, step-up, context	National policy with local assignment
Token and Session Service	Short-lived tokens, session expiry, risk response	High availability and audit
FIDO Server	Passkey and security-key registration	WebAuthn conformance, recovery
Mobile Identity Service	Wallet/authenticator app, push, device binding	iOS/Android, managed device integration
PKI/CA	Certificate issuance, renewal, status, signing	HSM-backed, crypto-agile
API Gateway	Application onboarding, rate limits, tokens	FHIR-aware integration
Staff/Role APIs	FHIR and event interfaces to workforce data	Avoid new LDAP dependency
Admin and Reporting	Audit search, access review, inventory, SLA	Least privilege, export control

8.6 Security Monitoring and Administration Products

Security Product	Production Capability Required
Privileged Access Management	Admin vault, session recording, just-in-time privilege, break-glass
Secrets Management	Application secrets, API credentials, automated rotation
SIEM and Log Archive	Identity, HSM, application, device logs with protected retention
SOAR	Automated containment, revocation, investigation, case management
Identity Threat Detection/UEBA	Impossible travel, token theft, unusual access, account misuse
WAF and API Protection	Web attack, bot, rate-limit, schema enforcement
DDoS Protection	Volumetric and application-layer resilience
EDR and Device Posture	Issuer workstations, admin devices, supported clinical endpoints
Vulnerability Management	Continuous scanning, dependency tracking, remediation evidence

Security Product	Production Capability Required
Certificate/Asset Discovery	Find unmanaged certificates, readers, printers, firmware
Fraud Case Management	Identity proofing exceptions, impersonation, credential abuse
Time Service	Trusted time for certificates, logs, authentication

Cyber platform initial licences (base): £8.0M

8.7 Hosting and Infrastructure Products

Infrastructure Product	Minimum Production Capability
Production Compute	Multi-zone runtime, autoscaling, hardened images
Database	HA identity stores, encryption, point-in-time recovery
Object/Archive Storage	Evidence, logs, controlled artefacts, immutable retention
Load Balancer and DNS	Health-based routing, TLS automation, failover
Private Connectivity	HSCN/approved network integration, secure admin paths
Internet Edge	DDoS, WAF, API protection, monitored ingress
Backup	Encrypted, isolated, routinely restored
Disaster Recovery	Separate region, runbooks, data replication, exercised failover
Observability	Metrics, traces, synthetic journeys, capacity, UX monitoring
Configuration/Deployment	Signed artefacts, infrastructure as code, controlled promotion
Endpoint Management	Issuer station, reader driver, printer firmware, security policy
Service Management	Incident, problem, change, knowledge, asset, stock, supplier

8.8 Consumables and Spares

Consumable/Spare	Planning Basis	Control Requirement
Blank Smartcards	2.2M initial order plus controlled replenishment	Lot/serial tracking, sealed cartons
YMCKO Ribbon	~£0.14–£0.19 per image (300 images per ribbon)	Track batches, prevent unapproved substitutes
Retransfer Film	Printer-specific film	Track batches and shelf life
Laminate	Clear or custom holographic	Custom security artwork, controlled release
Cleaning Kits	Cleaning cards, rollers, swabs	Scheduled cleaning by print count
Print Heads/Rollers	Critical spares by printer family	Regional pool, supplier replacement SLA
Power and Data Accessories	PSU, USB, Ethernet, locking cables	Standardised kits

Consumable/Spare	Planning Basis	Control Requirement
Holders and Lanyards	Breakaway clinical lanyards, wipeable holders	Infection-control and safety approval
Secure Mailers	Tamper-evident, return-addressed carrier	No PIN/activation secret in same package
Rejected-Card Destruction	Secure bins, punches, central destruction	Daily reconciliation, certificates

8.9 Compatibility Lab and Assurance Products

Assurance Product	Test Scope	Production Evidence
Reader Matrix	Every approved card x reader x driver x OS x middleware	Regression pass, signed compatibility record
Printer/Encoder Lab	Artwork, encoding, laminate, durability, throughput	Golden profile, acceptance sample
Mobile Device Lab	Supported iOS/Android, MDM states, secure enclave	Supported device catalogue
Performance Test	Peak authentication, issuance, revocation, directory, APIs	Capacity evidence with failure margins
Penetration Test	External, internal, mobile, API, admin, issuer workstation	Critical/high findings closed
Red Team/Abuse Test	Identity proofing fraud, token theft, insider misuse	Detection and response evidence
Privacy/DPIA	High-risk processing, proportionality, retention, rights	Approved DPIA, residual-risk decision
Clinical Safety	DCB0129/DCB0160 applicability, hazard log, safety case	CSO approval
DSPT/DTAC	Data protection, cyber, interoperability, usability, accessibility	Current assurance record
BCP/DR Exercise	HSM, CA, identity platform, service desk, bureau failover	Demonstrated recovery within target

Compatibility and security test lab (base): £1.5M

8.10 Product Quantities and Deployment Model

8.10.1 Base Case National Quantity Assumptions

Quantity Driver	Base Quantity	Rationale
Target Identities	2,000,000	Portfolio scale
Initial Physical Card Order	2,200,000	10% launch spare and replacement allowance

Quantity Driver	Base Quantity	Rationale
Mobile Credential Activations	1,400,000	70% planning adoption
FIDO2 Security Keys	60,000	Privileged, no-phone, recovery use
Dual-Interface Desktop Readers	220,000	Shared clinical and admin endpoints
Contactless/Mobile Readers	35,000	Tap and selected mobile workflows
Secure PIN-Pad Readers	20,000	RA and higher-assurance use
Targeted Door Readers	12,000	Priority upgrades, not full estate
Distributed Card Printers	900	Average 3 per organisation (300 orgs)
Laminator/Encoder Modules	300	One enhanced station per organisation
Complete Enrolment Station Kits	600	2 per organisation planning basis
Retransfer Bureau Printers	30	Production, DR, test, spare capacity
Production/DR Network HSMS	12	HA across environments and sites
Backup/Lab/Signing HSMS	4	Offline and non-production functions

8.10.2 Authenticator Mix

Authenticator	Proportion	User Count	Rationale
Smartcard + Mobile	60%	1,200,000	Primary: both credentials for everyday use
Smartcard Only	25%	500,000	Settings where mobile is unsuitable
Smartcard + Mobile + Security Key	5%	100,000	Privileged and high-assurance roles
Smartcard + Passkey	10%	200,000	Mobile-optimised workflows

8.11 Manufacturer Shortlist

8.11.1 Smartcard and Secure Element Manufacturers

Manufacturer	Role	Reference Capability	Procurement Treatment
IDEMIA	Card integrator/credential supplier	ID-One smart credentials; PKI, contactless, FIDO-capable	RFI and sample profile
Giesecke+Devrient	Card integrator/OS supplier	STARCOS smartcard platform; contact and PKI signature profiles	RFI and sample profile
Thales	Card integrator/identity supplier	Secure identity credentials, polycarbonate personalisation	RFI and sample profile

Manufacturer	Role	Reference Capability	Procurement Treatment
NXP	Secure element manufacturer	SmartMX3 P71D321 — contact, contactless, dual-interface JCOP4	Specify approved silicon or equivalent
Infineon	Alternative silicon	Second-source secure microcontroller	Supply resilience requirement

8.11.2 Reader Manufacturers

Manufacturer	Product Range	Procurement Treatment
HID Global	OMNIKEY range, Signo readers	RFI and certification matrix
Identiv	uTrust range	RFI and certification matrix
ACS	ACR range	RFI and certification matrix

8.11.3 Printer and Issuance Manufacturers

Manufacturer	Product Range	Procurement Treatment
Zebra Technologies	ZC range	RFI and configured quote
Evolis	Primacy 2, Agilia	RFI and configured quote
Magocard	Magocard 600	RFI and configured quote
Entrust	Sigma DS3, Artista CR805	RFI and configured quote

8.11.4 HSM Manufacturers

Manufacturer	Product Range	Procurement Treatment
Entrust	nShield 5c	RFI and security configuration
Thales	Luna Network HSM 7	RFI and security configuration

8.11.5 Identity Platform and Software Suppliers

Supplier	Capability	Procurement Treatment
Okta	Enterprise identity platform	Market engagement
Ping Identity	Identity and federation services	Market engagement
Microsoft	Azure AD/Entra ID	Market engagement
AWS	Identity services	Market engagement

8.12 Detailed Product Technical Specifications

8.12.1 Smartcard Detailed Technical Specification

Parameter	Specification	Standard
Form Factor	ID-1 (85.60 mm x 53.98 mm x 0.76 mm)	ISO/IEC 7810
Contact Interface	8-pin contact plate, T=0 and T=1 protocols	ISO/IEC 7816
Contactless Interface	Type A, 13.56 MHz, up to 848 kbps	ISO/IEC 14443
Secure Element	Dual-interface cryptographic microcontroller	Common Criteria EAL5+
CPU	32-bit ARM SC300 or equivalent	
Memory	80 KB EEPROM minimum, 500 KB ROM, 50 KB RAM	
Cryptographic Coprocessor	Hardware acceleration for RSA, ECC, AES	
Asymmetric Crypto	RSA 1024/2048/4096, ECC P-256/P-384/P-521	
Symmetric Crypto	AES 128/192/256, DES/3DES	
Hash Functions	SHA-1, SHA-256, SHA-384, SHA-512	
Card OS	Java Card 3.0.4 Classic + GP 2.3	
PKI Applet	GlobalPlatform secured, signed loading	
Contactless Performance	< 200 ms for mutual authentication	
Contact Performance	< 100 ms for external authentication	
Data Retention	10 years minimum	
Write Endurance	500,000 cycles minimum	
ESD Protection	4 kV minimum	ISO/IEC 7816
Operating Temperature	0°C to +50°C	
Storage Temperature	-20°C to +70°C	
Card Durability	3 years standard / 5 years polycarbonate	
Bending/Torsion	2.5 N / 0.7 Nm	ISO/IEC 10373
X-ray Resistance	0.1 Gy (X-ray)	
UV Resistance	No degradation after 24 hours UV exposure	
Chemical Resistance	IPA, chlorhexidine, bleach (clinical cleaning)	

8.12.2 Mobile Credential Detailed Technical Specification

Parameter	Specification
Platform Support	iOS 16.0+, Android 12.0+

Parameter	Specification
App Size	< 15 MB
Authentication Methods	Device biometric (Face ID, Touch ID, Fingerprint), 6-digit PIN
Key Storage	Secure Enclave (iOS) / TEE + KeyStore (Android)
Protocol Support	FIDO2 CTAP2, WebAuthn, OIDC, OAuth 2.0
Push Authentication	APNS (iOS) / FCM (Android) with HMAC-signed payloads
Device Attestation	iOS DCAAppAttest / Android Play Integrity
Credential Binding	Hardware-backed key pair generated on device
Remote Revocation	Instant credential disable via server push
Offline Authentication	Up to 5 cached assertions (configurable)
Session Timeout	Configurable from 1–60 minutes
Accessibility	VoiceOver (iOS) / TalkBack (Android)
Dark Mode	Full support
Localisation	English (UK) initial, Welsh and other languages TBC
Security Audit	Annual penetration test, MASVS Level 2
Privacy	No biometric data transmitted to server

8.13 Acceptance Testing and Certification Protocols

8.13.1 Smartcard Acceptance Test Protocol

Test	Method	Pass Criteria
Physical Dimensions	Calliper measurement	ISO/IEC 7810 ID-1 tolerances
Contact Interface	Electrical test with reference reader	All 8 contacts functional
Contactless Interface	NFC test with reference reader	Read range > 30 mm
Cryptographic Operations	Known-answer tests	RSA 2048 sign < 200 ms
Memory Endurance	Write/read cycle test	500,000 cycles
Temperature Cycling	-20°C to +70°C, 100 cycles	No functional degradation
Chemical Resistance	IPA/bleach wipe, 1000 cycles	Print and laminate intact
Bending/Torsion	ISO 10373 test	No chip or antenna failure
X-ray Exposure	0.1 Gy	No memory corruption
ESD	4 kV contact, 8 kV air	No functional interruption

8.13.2 Reader Acceptance Test Protocol

Test	Method	Pass Criteria
Contact Interface	10,000 card insertions	No mechanical failure
Contactless Range	Read range measurement	> 30 mm consistent
Protocol Compliance	PC/SC test suite	All tests pass
OS Compatibility	Windows 10/11, macOS 12+, Linux	Full driver support
Power Consumption	USB power measurement	< 500 mA
ESD	4 kV contact, 8 kV air	No reset or data corruption
MTBF	Accelerated life test	> 500,000 insertions

PART 9: COST MODEL AND BUDGETARY ANALYSIS

9.1 Product Basket Cost Model

9.1.1 Credentials and Authenticators

Product	Quantity	Low Unit	Base Unit	High Unit	Base Total
Dual-Interface Smart Photo Cards	2,200,000	£7.50	£11.50	£15.00	£25,300,000
Holder, Lanyard and Secure Mailer	2,200,000	£1.40	£2.10	£3.00	£4,620,000
Mobile Credential Activation	1,400,000	£2.00	£4.50	£8.00	£6,300,000
FIDO2/Security Keys	60,000	£35.00	£55.00	£85.00	£3,300,000
Credentials Subtotal					£39,520,000

9.1.2 Readers and Access Devices

Product	Quantity	Low Unit	Base Unit	High Unit	Base Total
Dual-Interface Desktop Readers	220,000	£35.00	£47.00	£65.00	£10,340,000
Contactless/Mobile Readers	35,000	£45.00	£62.00	£85.00	£2,170,000

Product	Quantity	Low Unit	Base Unit	High Unit	Base Total
Secure PIN-Pad Readers	20,000	£80.00	£105.00	£145.00	£2,100,000
Targeted Door Readers	12,000	£180.00	£250.00	£400.00	£3,000,000
Reader Spares and Deployment Kits	Allowance	£1,300,000	£1,760,000	£2,500,000	£1,760,000
Readers Subtotal					£19,370,000

9.1.3 Issuance and Personalisation Equipment

Product	Quantity	Low Unit	Base Unit	High Unit	Base Total
Distributed Duplex Printers	900	£1,300	£1,800	£2,700	£1,620,000
Laminator/Encoder Modules	300	£1,800	£2,800	£4,500	£840,000
Complete Enrolment Station Kits	600	£1,500	£2,100	£3,200	£1,260,000
Retransfer Bureau Printers	30	£5,000	£7,500	£12,000	£225,000
Two Secure Bureau Fit-Outs	Allowance	£5,000,000	£7,500,000	£12,000,000	£7,500,000
Initial Consumables and Spares	Allowance	£5,000,000	£7,500,000	£11,000,000	£7,500,000
Storage, UPS and Destruction Tools	Allowance	£500,000	£800,000	£1,400,000	£800,000
Issuance Subtotal					£19,745,000

9.1.4 Security Products and Test Lab

Product	Quantity	Low Unit	Base Unit	High Unit	Base Total
Production/DR HSM Appliances	12	£100,000	£140,000	£220,000	£1,680,000
Backup/Lab HSM Products	Allowance	£200,000	£300,000	£500,000	£300,000
PKI/CMS Licences and Support	Allowance	£5,000,000	£8,000,000	£13,000,000	£8,000,000
Cyber Platform Initial Licences	Allowance	£5,000,000	£8,000,000	£14,000,000	£8,000,000

Product	Quantity	Low Unit	Base Unit	High Unit	Base Total
Compatibility and Security Test Lab	Allowance	£800,000	£1,500,000	£2,500,000	£1,500,000
Security Subtotal					£19,480,000

9.1.5 Total Direct Product Basket

Category	Low Case	Base Case	High Case
Credentials and Authenticators	£28.12M	£39.52M	£54.60M
Readers and Access Devices	£14.30M	£19.37M	£27.35M
Issuance and Personalisation Equipment	£14.10M	£19.75M	£31.10M
Security Products and Test Lab	£11.00M	£19.48M	£30.00M
Total Direct Product Basket	£67.52M	£98.12M	£143.05M

9.2 Capital Expenditure Breakdown (£630M)

CapEx Category	Value (£M)	Proportion
Identity Platform	£210M	33.3%
Integration and APIs	£140M	22.2%
Mobile Identity Infrastructure	£95M	15.1%
Hardware and Devices	£85M	13.5%
Smartcard Modernisation	£60M	9.5%
Programme Mobilisation	£40M	6.3%
Total CapEx	£630M	100%

9.3 Operational Expenditure Breakdown (£960M)

OpEx Category	Value (£M)	Proportion
Managed Service	£310M	32.3%
Support and Operations	£260M	27.1%
Workforce Onboarding	£180M	18.8%
Supplier Ecosystem	£120M	12.5%
Training and Adoption	£90M	9.4%

OpEx Category	Value (£M)	Proportion
Total OpEx	£960M	100%

9.4 Ten-Year Funding Profile

Year	CapEx (£M)	OpEx (£M)	Total (£M)
Year 1	£180M	£30M	£210M
Year 2	£150M	£60M	£210M
Year 3	£120M	£135M	£255M
Year 4	£90M	£165M	£255M
Year 5	£40M	£70M	£110M
Year 6	£20M	£90M	£110M
Year 7	£10M	£100M	£110M
Year 8	£10M	£100M	£110M
Year 9	£5M	£105M	£110M
Year 10	£5M	£105M	£110M
Total	£630M	£960M	£1,590M

9.5 Cost Sensitivities

Sensitivity	Change	Cost Impact	Management Lever
Physical Card Unit Price	+£1 per card	+£2.2M	Tender profile and security features
Reader Price (All Types)	+£10 across 275k units	+£2.75M	Volume discount and device mix
Mobile Activation	+£1 per activation	+£1.4M	Licence model and support term
Integration and APIs	+10% against portfolio line	+£14.0M	Legacy application complexity
Capital Contingency	+5 percentage points	+£31.5M	Design maturity and delivery risk
Door Reader Expansion	+10,000 readers at base unit	+£2.5M	Excludes controllers and installation
FIPS Security Key Mix	+£40 for 60k keys	+£2.4M	Avoid assurance over-specification

9.6 Unit Economics

Metric	Value
Direct Products Per Target User	£49.06
CapEx Per Target User	£315.00
10-Year Total Cost Per User	£795.00
10-Year Cost Per User Per Year	£79.50
Annual Benefit Per User	£900.00
Net Benefit Per User	£7,950.00
ROI Per User	28.2x

9.7 Scenario Analysis

Scenario	CapEx	10-Year OpEx	Total	Description
Low (Maximise Reuse)	£410M	£720M	£1.13B	Maximise CIS2 reuse, constrained physical estate change
Base (Portfolio Case)	£630M	£960M	£1.59B	Source portfolio scope and national support model
High (Complex Integration)	£890M	£1.24B	£2.13B	More legacy adapters, door estate, extended supplier support

9.8 Cost Reconciliation

The £98.1M direct product basket sits inside the platform, mobile, smartcard, and hardware lines of the £630M capital envelope. The balance funds software build, licensing, integration, data migration, rollout, assurance, deployment, and mobilisation.

Reconciliation rule: The product basket must not be added to the capital envelope — it is a component of it.

9.9 Detailed Line-Item Cost Breakdown

9.9.1 Identity Platform — Detailed Component Costs (£210M)

Component	Development (£M)	Licensing (£M)	Implementation (£M)	Total (£M)
Core Identity Services	£20M	£25M	£15M	£60M
Federation Services	£12M	£15M	£8M	£35M
Token Services	£10M	£12M	£8M	£30M
Directory Services	£8M	£10M	£7M	£25M

Component	Development (£M)	Licensing (£M)	Implementation (£M)	Total (£M)
Policy Engine	£10M	£8M	£7M	£25M
API Gateway	£8M	£7M	£5M	£20M
Admin and Reporting	£5M	£5M	£5M	£15M
Total	£73M	£82M	£55M	£210M

9.9.2 Integration and APIs — Detailed Component Costs (£140M)

Component	Development (£M)	Testing (£M)	Deployment (£M)	Total (£M)
Clinical System Connectors	£25M	£15M	£10M	£50M
Legacy Compatibility Layer	£18M	£10M	£7M	£35M
Standard API Development	£15M	£8M	£7M	£30M
Test Harnesses and Certification	£8M	£5M	£2M	£15M
CIS2 Extension and Integration	£5M	£3M	£2M	£10M
Total	£71M	£41M	£28M	£140M

9.10 Cost Contingency and Risk Allowance

Contingency Category	Allowance (£M)	Basis
Design and Scope Contingency	£31.5M	5% of CapEx
Price Escalation Contingency	£15M	2.5% of CapEx
Integration Risk Allowance	£20M	14% of integration budget
Programme Contingency	£25M	4% of total programme
Total Contingency	£91.5M	

PART 10: TEN-YEAR ROI AND BENEFITS REALISATION

10.1 Benefit Categories

The programme delivers £18.6 billion in gross quantified benefits across five categories:

- 1. **Administrative Efficiency:** £5.1B (27.4%)
- 2. **Fraud Reduction:** £4.2B (22.6%)
- 3. **Workforce Mobility:** £3.8B (20.4%)
- 4. **System Consolidation:** £2.9B (15.6%)
- 5. **Cyber Risk Reduction:** £2.6B (14.0%)
- 6. **Digital Enablement:** £1.0B (5.4%)

10.2 Financial Benefit Model

10.2.1 Administrative Efficiency (£5.1B)

Benefit Component	Annual Value (£M)	10-Year Total (£M)	Methodology
Reduced Onboarding Time	£250M	£2,500M	Average 2 days saved per new starter (200k/year) at £625/day
Reduced Access Delays	£150M	£1,500M	Eliminated failed login and re-issuance time
Reduced Identity Issuance Workload	£110M	£1,100M	Automated vs manual credential processing
Total	£510M	£5,100M	

10.2.2 Fraud Reduction (£4.2B)

Benefit Component	Annual Value (£M)	10-Year Total (£M)	Methodology
Identity Fraud Reduction	£200M	£2,000M	Reduced impersonation and credential misuse
Contractor Misuse Reduction	£120M	£1,200M	Time-bound access and automated offboarding
Supply Chain Fraud Reduction	£100M	£1,000M	Partner identity verification and audit
Total	£420M	£4,200M	

10.2.3 Workforce Mobility (£3.8B)

Benefit Component	Annual Value (£M)	10-Year Total (£M)	Methodology
Faster Cross-Trust Movement	£150M	£1,500M	Reduced time to productive access at new organisation
Reduced Duplication	£120M	£1,200M	Single credential replaces multiple Trust-specific credentials

Benefit Component	Annual Value (£M)	10-Year Total (£M)	Methodology
Improved Surge Capacity	£110M	£1,100M	Faster deployment of temporary staff during surges
Total	£380M	£3,800M	

10.2.4 System Consolidation (£2.9B)

Benefit Component	Annual Value (£M)	10-Year Total (£M)	Methodology
Legacy System Retirement	£150M	£1,500M	Reduction from 250+ to 1 national identity system
Reduced Licensing Costs	£80M	£800M	Consolidated platform licensing
Reduced Infrastructure Costs	£60M	£600M	Reduced hosting and support for legacy systems
Total	£290M	£2,900M	

10.3 Cyber Security Benefit Model

10.3.1 Cyber Risk Reduction (£2.6B)

Benefit Component	Annual Value (£M)	10-Year Total (£M)	Methodology
Reduced Breach Likelihood	£130M	£1,300M	78% reduction in identity compromise
Reduced Breach Impact	£70M	£700M	85% reduction in lateral movement
Reduced Insider Threat	£60M	£600M	70% reduction in insider threat exposure
Total	£260M	£2,600M	

10.3.2 Cyber Security Target Metrics

Metric	Current Baseline	Target	Reduction
Identity Compromise Incidents	Portfolio baseline	78% reduction	-78%
Lateral Movement Events	Portfolio baseline	85% reduction	-85%
Insider Threat Incidents	Portfolio baseline	70% reduction	-70%
Supply Chain Cyber Incidents	Portfolio baseline	60-75% reduction	-60-75%

10.4 Strategic Benefit Model

10.4.1 Digital Enablement (£1.0B)

Benefit Component	Annual Value (£M)	10-Year Total (£M)	Methodology
FDP Integration Enablement	£40M	£400M	Identity foundation for Federated Data Platform
NHS App Integration	£30M	£300M	Workforce identity for NHS App expansion
EPR Convergence Support	£30M	£300M	Identity layer for EPR standardisation
Total	£100M	£1,000M	

10.5 ROI Calculation Methodology

10.5.1 Formula

Return on Investment (ROI) = (Net Benefit / Total Programme Cost) x 100

Where:

- Net Benefit = Gross Benefits - Total Programme Cost
- Gross Benefits = £18.6B
- Total Programme Cost = £1.59B (CapEx £630M + OpEx £960M)
- Net Benefit = £18.6B - £1.59B = £15.9B
- ROI = (£15.9B / £1.59B) x 100 = 1,000% (expressed as 10.0x)

Benefit-Cost Ratio (BCR) = Gross Benefits / Total Programme Cost

- BCR = £18.6B / £1.59B = 11.7x

10.5.2 ROI Summary

Metric	Conservative	Expected	Optimistic
Gross Benefits	£14.88B	£18.60B	£20.46B
Total Programme Cost	£1.75B	£1.59B	£1.51B
Net Benefit	£11.13B	£15.90B	£19.70B
ROI	18.4x	28.2x	34.6x
Payback Period	2.5 years	1.8 years	1.5 years

10.6 Payback Period Analysis

The programme achieves payback within 1.8 years under the expected scenario. This means the cumulative benefits exceed the cumulative programme cost before the end of Year 2.

Payback calculation:

Year	Cumulative Cost (£B)	Cumulative Benefit (£B)	Cumulative Net Position (£B)	Payback Status
Year 1	£0.21B	£0.40B	+£0.19B	Early adoption phase
Year 1.5	£0.32B	£0.80B	+£0.48B	Approaching payback
Year 1.8	£0.38B	£1.00B	+£0.62B	Payback achieved
Year 2	£0.42B	£1.60B	+£1.18B	Post-payback
Year 3	£0.68B	£3.40B	+£2.72B	Strong positive returns
Year 5	£1.06B	£7.00B	+£5.94B	Major value delivery
Year 10	£1.59B	£18.60B	+£15.90B	Full programme value

10.7 Risk-Adjusted Returns

Metric	Pre-Risk Adjustment	Post-Risk Adjustment	Adjustment Factor
Gross Benefits	£18.6B	£15.8B	-15%
Net Benefit	£15.9B	£13.4B	-16%
ROI	28.2x	22.1x	-22%
BCR	11.7x	9.9x	-15%

Risk adjustment covers adoption risk, integration complexity, cyber threats, procurement delay, and change capacity. The risk-adjusted ROI of 22.1x remains exceptionally strong.

10.8 Benefits Ownership and Measurement

10.8.1 Benefit Ownership Framework

Benefit Category	National Owner	Local Owner	Measurement Method
Administrative Efficiency	NHS England	Trust CIO	Time saved, cost per onboarding
Fraud Reduction	DHSC	Trust CFO	Fraud incidents, financial loss
Workforce Mobility	ICS Lead	Trust HR Director	Time to productive access
System Consolidation	NHS England	Trust CIO	Legacy systems retired, cost savings
Cyber Risk Reduction	NCSC/NHS England	Trust CISO	Incident count, breach impact

Benefit Category	National Owner	Local Owner	Measurement Method
Digital Enablement	NHS England	Programme Director	Programme dependency satisfaction

10.8.2 Core KPIs

KPI	Baseline	Target	Measurement Frequency
Onboarding Time	Days to weeks	Minutes	Monthly
Access Delay	Average 12 minutes/day	<1 minute/day	Quarterly
Identity Fraud Incidents	Portfolio baseline	78% reduction	Quarterly
Identity Compromise Incidents	Portfolio baseline	78% reduction	Monthly
Legacy Systems Retired	250+	1 national system	Annual
Active Mobility (cross-Trust)	Minimal	100% portable	Quarterly

10.9 Benefits Realisation Timeline

Year	Benefits Accrued	Cumulative Benefits	Key Milestone
Year 1	£0.4B	£0.4B	Foundation build, early adopters
Year 2	£1.2B	£1.6B	Payback achieved, pilot Trusts operational
Year 3	£1.8B	£3.4B	National rollout begins
Year 4	£1.8B	£5.2B	50% workforce activated
Year 5	£1.8B	£7.0B	75% workforce activated
Year 6	£1.8B	£8.8B	Full national adoption
Year 7	£1.8B	£10.6B	Optimisation and digital wallet
Year 8	£1.8B	£12.4B	AI identity verification
Year 9	£1.8B	£14.2B	Cross-sector federation
Year 10	£1.8B	£16.0B	Continuous innovation

10.10 Detailed Benefit Calculation Methodology

10.10.1 Administrative Efficiency Calculation

Basis: 2,000,000 workforce members, 10% annual turnover (200,000 new starters per year)

Current state: Average 2 days delay to productive access at £625/day (blended NHS staff cost)

- Cost per new starter: 2 days x £625 = £1,250
- Annual cost: 200,000 x £1,250 = £250M
- 10-year cost: £2.5B

Target state: Onboarding in minutes, immediate productive access

- Cost per new starter: ~£10 (minimal system processing)
- Annual saving: £250M - £2M = £248M
- 10-year saving: ~£2.5B

Additional efficiencies: Reduced access delays, failed login recovery, helpdesk calls

- Current: Average 12 minutes/day lost to access issues x 2M staff x £52/hour = £218M/year
- Target: <1 minute/day = £18M/year
- Saving: £200M/year

10.10.2 Fraud Reduction Calculation

Basis: NHS fraud estimated at £1.1B annually (NHS Counter Fraud Authority)

- Identity-related fraud estimated at 38% = £418M/year
- Target reduction: 50% = £209M/year
- 10-year total: £2.0B

Contractor misuse: Estimated 15% of contractor spend has identity issues

- Annual contractor spend: ~£8B
- Current misuse: 15% x £8B = £1.2B
- Target reduction: 10% = £120M/year
- 10-year total: £1.2B

10.11 Benefit Attribution and Avoidance of Double Counting

Benefit Category	Boundary	Exclusions	Double Counting Risk
Administrative Efficiency	Direct workforce time saving	Clinical workflow redesign benefits	None with system consolidation
Fraud Reduction	Identity-related fraud only	General NHS fraud	None
Workforce Mobility	Cross-Trust workforce movement	Patient benefits	Low — separate category
System Consolidation	Legacy identity system costs	EPR consolidation	Moderate — manage through attribution
Cyber Risk Reduction	Identity-related cyber incidents	General cyber improvements	Low — identity-specific
Digital Enablement	Programme dependency value	Direct programme benefits	Managed through clear boundaries

PART 11: DELIVERY ROADMAP AND PHASING

11.1 Phase 1 — Mobilisation and Discovery (Months 1–6)

Objective: Establish programme foundations, confirm architecture, and validate planning assumptions

Key activities:

- Programme setup and governance establishment
- Supplier onboarding and market engagement
- Architecture finalisation and design authority
- Pilot planning and site selection
- CIS2/CIS reuse validation
- Application estate inventory
- Reader and printer estate audit
- DPIA initiation
- Benefit baseline agreement

Deliverables:

- Programme charter and governance framework
- Confirmed target architecture
- CIS reuse decision paper
- Application and device estate inventory
- Pilot design and readiness criteria
- Initial DPIA

Gate 1 decision: Architecture and DPIA approval

11.2 Phase 2 — Foundation Build (Months 6–18)

Objective: Build the national identity platform, deploy mobile identity infrastructure, and modernise smartcard capabilities

Key activities:

- Identity platform build and integration
- Mobile identity infrastructure deployment
- Smartcard profile development and testing
- PKI/HSM infrastructure deployment
- Integration layer development
- Clinical system connector development
- Legacy compatibility layer build

- Cyber tooling deployment
- Security operations centre establishment
- Compatibility lab setup and testing
- Bureau design and fit-out
- Training content development

Deliverables:

- National identity platform (operational)
- Mobile identity service (operational)
- Smartcard profile (certified)
- PKI/HSM infrastructure (operational)
- Integration layer (tested)
- Clinical system connectors (validated)
- Compatibility matrix (certified)
- Security operations centre (operational)

Gate 2 decision: Pilot readiness

11.3 Phase 3 — Early Rollout (Months 18–30)

Objective: Validate the programme through controlled pilot deployments in early adopter Trusts

Key activities:

- Early adopter Trust onboarding
- ICS readiness assessment
- Workforce onboarding and activation
- Partner access pilots
- Workflow validation in clinical settings
- Accessibility testing with representative users
- Performance and capacity validation
- Support model rehearsal
- Benefits baseline measurement

Deliverables:

- Pilot Trusts operational (target: 10–15 Trusts)
- Clinical workflow validation report
- Accessibility and inclusion report
- Performance and capacity evidence
- Benefits baseline report
- Refined rollout approach

Gate 3 decision: National rollout readiness

11.4 Phase 4 — National Rollout (Years 3–5)

Objective: Scale the programme to national coverage across all Trusts and ICS regions

Key activities:

- Regional wave-based rollout
- Trust-by-Trust deployment
- Workforce migration and onboarding
- Partner access expansion
- Integration factory operations
- Continuous improvement cycle
- Benefits tracking and reporting
- ICS mobility enablement
- Registration Authority capacity building

Deliverables:

- All Trusts deployed (300+ organisations)
- 2M+ workforce identities activated
- Partner access model operational
- National mobility enabled
- Benefits realisation tracking operational

Gate 4 decision: Optimisation phase approval

11.5 Phase 5 — Optimisation and Digital Wallet (Years 5–7)

Objective: Expand programme capabilities through digital wallet, AI identity verification, and continuous improvement

Key activities:

- Digital wallet expansion
- AI identity verification pilots
- Policy-controlled biometrics assessment
- Advanced analytics deployment
- Continuous improvement initiatives
- Legacy system retirement acceleration
- Supply chain optimisation

Deliverables:

- Digital wallet operational
- AI identity verification (piloted)
- Legacy system retirement (75%+)
- Optimised operating model

11.6 Phase 6 — Future-State Identity (Years 7–10)

Objective: Deliver long-term strategic value through cross-sector federation, innovation, and sustainability

Key activities:

- Cross-sector federation (social care, local government)
- Advanced analytics and fraud detection
- Identity innovation partnerships
- Sustainability and cost optimisation
- Future-state architecture evolution
- Knowledge transfer and capability building

Deliverables:

- Cross-sector identity federation operational
- Advanced analytics platform operational
- Innovation pipeline established
- Sustainable operating model

11.7 Decision Gates and Acceptance Criteria

Gate	Timing	Decision	Evidence Required
Gate 1	Month 6	Architecture and DPIA approval	Architecture design approved, DPIA accepted, CIS reuse confirmed
Gate 2	Month 18	Pilot readiness	Platform stable, compatibility matrix complete, security controls proven
Gate 3	Month 30	National rollout readiness	Pilot evidence positive, performance validated, adoption model proven
Gate 4	Year 5	Optimisation approval	National rollout complete, benefits tracking operational, lessons learned
Gate 5	Year 7	Future-state approval	Digital wallet mature, federation options assessed, value demonstrated

11.8 Critical Path Dependencies

Critical path items:

1. **Platform build** — Identity platform must be operational before any Trust deployment
2. **Zero-trust controls** — Security controls must be proven before workforce migration
3. **Identity migration** — Credentials must be activated before Trust integration
4. **Trust integration** — Workflows must be validated before regional rollout
5. **Rollout sequencing** — Local capacity must be confirmed before scheduling

Key external dependencies:

- EPR convergence programme timeline
- Federated Data Platform rollout schedule
- NHS App integration roadmap
- ICS digital convergence plans
- Cyber improvement programme
- Supplier ecosystem readiness
- Registration Authority capacity

11.9 Detailed Phase Delivery Plans

11.9.1 Phase 1 — Monthly Breakdown

Month	Activities	Key Milestones	Resources Required
Month 1	Programme setup, SRO appointment, governance framework	Programme charter signed	Programme Director, PMO team
Month 2	Architecture design, CIS reuse assessment, DPIA kick-off	Architecture principles agreed	Technical Architect, DPIA lead
Month 3	Market engagement, supplier RFI, compatibility lab initiation	Market engagement report	Commercial Lead, Technical Lead
Month 4	Application inventory, reader/printer audit, pilot site selection	Application estate inventory	Integration Lead, Field teams
Month 5	Benefit baseline, pilot design, training content outline	Benefit baseline agreed	Benefits Lead, Training Lead
Month 6	Gate 1 review, architecture approval, DPIA acceptance	Gate 1 passed	Programme Board

11.9.2 Phase 2 — Monthly Breakdown (Months 7–18)

Period	Key Activities	Milestones
Months 7–9	Identity platform development, HSM procurement, integration layer design	Platform alpha release
Months 10–12	Mobile identity infrastructure, smartcard profile development, CIS2 extension	Smartcard profile certified
Months 13–15	Clinical system connectors, bureau fit-out, SOC establishment	Integration layer tested
Months 16–18	Compatibility matrix completion, security testing, training content finalised	Gate 2 readiness

11.10 Resource and Capacity Planning

11.10.1 Core Programme Team

Role	Phase 1	Phase 2	Phase 3	Phase 4
Programme Director	1	1	1	1
Technical Architect	2	3	2	1
Commercial Lead	1	2	1	1
Integration Lead	1	3	3	2
Security Lead	1	2	2	1
Workforce Lead	1	2	3	2
Benefits Lead	1	1	1	1
PMO	3	5	5	3
Total Core Team	11	19	18	12

PART 12: PROCUREMENT LOTS AND COMMERCIAL FRAMEWORK

12.1 Recommended Procurement Lots

The programme is structured across 10 discrete procurement lots to protect interoperability, ensure competition, and prevent any single supplier from owning every layer of the identity stack.

Lot Number	Lot Name	Estimated Value	Contract Duration	Competition Strategy
Lot 1	Credential Media and Secure Elements	£25M–£40M	5 years	Dual source, competitive tender
Lot 2	Personalisation and Secure Bureau	£10M–£20M	7 years	Competitive dialogue
Lot 3	Readers and Endpoint Hardware	£15M–£25M	5 years	Framework call-off
Lot 4	Issuer Workstations and Printers	£5M–£10M	5 years	Competitive tender
Lot 5	PKI, HSM and Credential Management	£15M–£25M	7 years	Competitive dialogue
Lot 6	Identity Platform and Mobile	£50M–£100M	7 years	Competitive dialogue
Lot 7	Integration Factory	£30M–£60M	5 years	Competitive tender
Lot 8	Managed Operations and SOC	£50M–£100M	7 years	Competitive dialogue
Lot 9	Training, RA and Field Support	£20M–£40M	5 years	Framework call-off
Lot 10	Independent Test and Assurance	£10M–£20M	5 years	Competitive tender

12.2 Lot Descriptions and Scope

Lot 1: Credential Media and Secure Elements

Scope:

- Card body manufacturing (PVC/PET-G and polycarbonate)
- Secure element and chip supply
- Card OS and applet development
- Visual security design
- Stock management and secure logistics

Commercial protection:

- Dual source requirement (primary + secondary manufacturer)
- Contract controls own the cryptographic profile and keys
- Second-source qualification required within 12 months

Lot 2: Personalisation and Secure Bureau

Scope:

- Card personalisation and encoding
- Quality control and inspection
- Secure fulfilment and mail services
- Stock vault management
- Secure destruction services
- Continuity site operations

Commercial protection:

- Open stock and audit data access
- Exit assistance and data portability
- Regular benchmarking and price review

Lot 3: Readers and Endpoint Hardware

Scope:

- Dual-interface desktop readers
- Contactless and mobile readers
- Secure PIN-pad readers
- Door readers (targeted)
- Spares and deployment kits
- Firmware management and certification

Commercial protection:

- Multiple approved models under one certification matrix
- Volume band pricing with transparent discounts
- Right to reduce volumes as mobile adoption increases

Lot 4: Issuer Workstations and Printers

Scope:

- Distributed card printers
- Laminator/encoder modules
- Complete enrolment station kits
- Consumables (ribbons, film, laminate)
- Field deployment and maintenance
- Warranty and spares management

Commercial protection:

- Configured kits with standardised components
- Supplier replacement SLA for critical spares
- Consumable pricing with volume bands

Lot 5: PKI, HSM and Credential Management

Scope:

- Network HSM appliances
- Offline root CA and ceremony infrastructure
- Certificate management system
- Key management and lifecycle services
- Backup and escrow services

Commercial protection:

- Independent security architecture
- Exit-capable key custody
- M-of-N quorum control for critical operations
- Supplier-independent key ownership

Lot 6: Identity Platform and Mobile

Scope:

- Core identity directory
- Registration Authority and sponsor portal
- Credential management system
- Federation service
- Policy and authorisation service
- Token and session service
- FIDO server
- Mobile identity service
- PKI/CA integration
- API gateway

Commercial protection:

- Extend CIS2/CIM — no duplicate national capability
- Open standards requirement (OIDC, OAuth 2.0, SAML, FIDO2)
- Data portability and exit assistance
- API and integration standards ownership

Lot 7: Integration Factory

Scope:

- Clinical system connector development
- Legacy compatibility layer
- CIS2 application onboarding

- Test harnesses and certification
- Vendor coordination and support
- Data migration and reconciliation

Commercial protection:

- Complexity-based pricing (not fixed price for unknown scope)
- Application inventory before contract
- Standard API ownership

Lot 8: Managed Operations and SOC

Scope:

- 24/7 service operations
- Security operations centre
- Incident response
- Threat intelligence
- Fraud investigation
- Service management platform
- Supplier coordination

Commercial protection:

- Consumption-based pricing for scalable services
- Transparent volume bands
- Indexation caps
- Regular benchmarking
- Right to reduce volumes

Lot 9: Training, RA Mobilisation and Field Support

Scope:

- Registration Authority mobilisation
- Training content and delivery
- Communications and change management
- Field engineering and deployment
- Local support and champions

Commercial protection:

- National content with local implementation
- Outcome-based pricing for adoption targets
- Knowledge transfer requirements

Lot 10: Independent Test and Assurance

Scope:

- Penetration testing
- Red team exercises
- Performance testing
- Privacy and DPIA assurance
- Clinical safety
- Accessibility testing
- Programme assurance

Commercial protection:

- Organisationally independent of build suppliers
- Direct reporting to assurance board
- Multi-year contract for continuity

12.3 Contract Interface Management

Contract interface schedules must define for every lot:

- Data ownership and access rights
- Credential key custody and control
- API specifications and versioning
- Service levels and measurement
- Incident responsibilities and escalation
- Evidence exchange and audit
- Step-in rights and exit assistance
- Commercial terms and price adjustment

12.4 Supplier Performance Management

KPIs by lot:

KPI Category	Weight	Measurement Method	Reporting Frequency
Service Availability	25%	Uptime percentage	Monthly
Authentication Success Rate	20%	Success/failure ratio	Monthly
Device Compliance Rate	15%	Compliant devices percentage	Monthly
Support Responsiveness	15%	SLA adherence	Monthly
Integration Success Rate	15%	Successful integrations	Quarterly

KPI Category	Weight	Measurement Method	Reporting Frequency
Cyber Compliance	10%	DSPT/audit results	Quarterly

12.5 Exit and Transition Arrangements

Every contract must include:

- **Data portability:** Structured data export in open formats
- **Key custody:** Controlled transfer of cryptographic materials
- **API documentation:** Complete API specification and usage
- **Knowledge transfer:** Documentation, training, and handover
- **Transition assistance:** Defined support period post-termination
- **Step-in rights:** NHS right to take over critical services
- **Escrow:** Source code and configuration escrow for critical components

12.6 Lotting Strategy Benefits

Benefit	Description
Competition	Multiple suppliers compete within each lot
Innovation	Innovation partnerships for emerging technologies
Flexibility	Individual lots can be modified without affecting the whole programme
Risk Distribution	Failure of one supplier does not stop the entire programme
Scalability	Lots can be expanded or contracted independently
SME Access	Smaller lots enable SME participation
Benchmarking	Price comparison across lots and suppliers

12.7 Detailed Lot Specifications

12.7.1 Lot 1 — Tender Evaluation Criteria

Criterion	Weight	Sub-Criteria
Technical Capability	35%	Card profile, secure element, OS, applet, security features
Commercial	30%	Unit price, volume discounts, price adjustment mechanism
Quality and Assurance	20%	Quality management, certification, testing capability

Criterion	Weight	Sub-Criteria
Security	10%	Supply chain security, stock custody, data protection
Social Value	5%	UK manufacturing, apprenticeship, sustainability

12.7.2 Lot 6 — Tender Evaluation Criteria

Criterion	Weight	Sub-Criteria
Technical Capability	40%	Platform architecture, scalability, federation, mobile capability
Commercial	25%	Licence model, pricing, support costs
Integration	15%	CIS2 compatibility, API standards, integration track record
Security	10%	Zero-trust, crypto agility, compliance certifications
Support and Service	5%	Service levels, support model, UK presence
Social Value	5%	Skills transfer, NHS experience, sustainability

PART 13: WORKFORCE, CHANGE AND ADOPTION

13.1 Adoption Strategy

Adoption is designed into delivery from day one, with four phases:

Phase 1 — Co-Design:

- Clinical, operational, and accessibility input
- User research and persona development
- Workflow analysis and design
- Accessibility requirements integration

Phase 2 — Prove:

- Pilot real workflows and high-risk settings
- Representative user testing
- Performance and usability validation
- Accessibility testing with assistive technology

Phase 3 — Enable:

- Local champions network
- Training and assisted activation
- Communications and engagement
- Support model launch

Phase 4 — Sustain:

- Ongoing support and feedback
- Adoption analytics and measurement
- Continuous improvement
- Community of practice

13.2 User Personas

Primary personas:

Clinician (Doctor/Nurse/AHP):

- Requires fast, seamless access across multiple clinical systems
- Moves between Trusts and care settings
- Needs reliable authentication in time-critical situations
- Values one-tap access where risk permits

Administrative Staff:

- Accesses multiple systems daily
- May support multiple departments
- Benefits from simplified credential management
- Needs clear, simple authentication process

Temporary Workforce (Bank/Agency/Locum):

- Requires rapid onboarding at multiple organisations
- Cannot wait days for credential issuance
- Needs portable identity that works immediately
- Benefits from mobile-first credential

IT and Clinical Safety Staff:

- Manages systems and access controls
- Requires privileged access for system administration
- Needs strong authentication for sensitive operations
- Benefits from automated user management

Partner Organisation Staff:

- Time-bound access to NHS systems
- Requires clear, auditable access model

- Needs simple onboarding and offboarding
- Benefits from self-service access management

13.3 Training and Enablement

Training tiers:

Tier 1 — General Workforce:

- eLearning module (15 minutes)
- Quick reference guide
- Video tutorial
- Self-service portal

Tier 2 — Registration Authority Staff:

- Comprehensive training programme
- Practical workshop
- Certification and assessment
- Ongoing support and updates

Tier 3 — Local Champions:

- Extended training programme
- Change management skills
- Peer support techniques
- Access to champion community

Tier 4 — Service Desk and Support:

- Technical training
- Troubleshooting guides
- Escalation procedures
- System access and tools

13.4 Communications and Engagement

Internal communications:

- Programme website and intranet
- Regular email updates
- Newsletter
- Video updates from programme leadership
- FAQ and knowledge base

Engagement channels:

- Roadshows and town halls
- Clinical engagement forums
- ICS and Trust briefings
- User research sessions
- Feedback and complaints mechanism

13.5 Accessibility and Inclusion

Non-negotiable requirements:

- **WCAG alignment:** All digital journeys meet WCAG 2.2 AA standards
- **Choice:** Mobile, smartcard, and assisted channels available
- **Usability:** One-tap access where risk permits
- **Support:** Local champions and assisted registration
- **Affordability:** No requirement to own a personal device
- **Trust:** Transparent use, consent, and redress

Accessibility measurement:

- Adoption by role, organisation, disability, geography, and channel
- Accessibility testing with representative users
- Continuous improvement based on feedback
- Gaps addressed before scale-up

13.6 Support Model

Support tiers:

Tier 0 — Self-Service:

- Online knowledge base
- FAQ and troubleshooting guides
- Video tutorials
- Automated password/credential reset

Tier 1 — National Service Desk:

- 24/7 phone and email support
- First-line troubleshooting
- Credential issue resolution
- Escalation to Tier 2

Tier 2 — ICS Support Teams:

- Regional support capability

- Complex issue resolution
- Registration Authority support
- Local deployment support

Tier 3 — Specialist Support:

- Technical escalation
- Security incident response
- Supplier coordination
- Major incident management

13.7 Workforce Readiness Criteria

Mandatory checks before rollout to any organisation:

1. **Training completion:** 90%+ of workforce trained
2. **Local champions active:** Minimum 2 champions per organisation
3. **Support model rehearsed:** Service desk tested and operational
4. **Registration Authority ready:** RA staff trained and certified
5. **Communication complete:** All staff informed of changes
6. **Accessibility confirmed:** Assisted journeys tested
7. **Cutover plan signed off:** Rollback procedures confirmed
8. **Benefits baseline recorded:** Pre-implementation metrics captured

13.8 Detailed Adoption Plan by User Group

User Group	Size	Adoption Approach	Primary Channel	Success Metric
Doctors	300,000	Clinical champions, peer support, workflow integration	Mobile credential as primary	90% activation within 6 months
Nurses	700,000	Ward-based champions, training sessions, assisted activation	Mobile + smartcard	85% activation within 6 months
AHPs	150,000	Departmental champions, eLearning, self-service	Mobile credential	90% activation within 6 months
Admin Staff	400,000	Desktop training, quick reference guides, helpdesk	Smartcard + passkey	95% activation within 3 months
Temporary Staff	200,000	Fast-track onboarding, mobile-first, self-service	Mobile credential	80% activation within 1 week

User Group	Size	Adoption Approach	Primary Channel	Success Metric
Partners	150,000	Federation onboarding, time-bound access, self-service	Time-bound digital credential	100% time-bound, auditable

13.9 Change Network and Champion Programme

Champion Role:

- Peer supporter for their department/ward
- First point of contact for questions
- Feedback channel to programme
- Local adoption advocate

Champion Selection:

- One champion per 50 staff
- Minimum 2 per organisation
- Clinical and non-clinical representation
- Voluntary with line manager support

Champion Support:

- Half-day training workshop
 - Champion toolkit (guides, FAQs, presentation)
 - Regular champion network calls
 - Monthly newsletter
 - Dedicated champion support channel
-

PART 14: PARTNER INTEROPERABILITY AND COMPLIANCE

14.1 Partner Access Model

Partner types:

- Social care organisations
- Private healthcare providers
- Academic and research institutions
- Contractors and suppliers

- Approved third-party service providers

Access controls:

- Time-bound identity credentials
- Role-based access limited to required systems
- Automated onboarding and offboarding
- DSPT compliance verification
- Audit trail for all partner access

Federation model:

- Standards-based federation (OIDC, SAML, OAuth 2.0)
- Trust framework for partner organisations
- Automated attribute exchange
- Revocation propagation

14.2 Standards Compliance

Standard	Domain	Compliance Requirement
ISO/IEC 7810 ID-1	Card physical format	Mandatory
ISO 14443	Contactless interface	Mandatory
ISO 7816	Contact interface	Mandatory
FIDO2/WebAuthn	Passwordless authentication	Mandatory
OIDC	Authentication protocol	Mandatory
OAuth 2.0	Authorisation framework	Mandatory
SAML 2.0	Federation protocol	Mandatory
FHIR	Healthcare data exchange	Where applicable
HL7	Healthcare messaging	Where applicable
DCB0129/DCB0160	Clinical safety	Mandatory
WCAG 2.2 AA	Accessibility	Mandatory
Common Criteria	Security certification	EAL5+ (secure element)
FIPS 140-3	Cryptographic module	Where required

14.3 Data Protection and Privacy

14.3.1 Legal Framework

- UK GDPR
- Data Protection Act 2018

- National Data Guardian (NDG) Principles
- ICO guidance on identity data processing

14.3.2 Privacy Controls

Data minimisation:

- Collect only the minimum identity evidence required
- No biometric templates stored on card unless separately justified
- Attribute-based access limits data sharing

Purpose limitation:

- Identity data used only for authorised access
- No secondary use without consent
- Purpose-bound data processing

Encryption:

- Data encrypted at rest and in transit
- Cryptographic keys protected by HSM
- End-to-end encryption for sensitive data flows

User rights:

- Right of access to personal data
- Right to rectification of inaccurate data
- Right to erasure where applicable
- Transparent data processing notices

14.3.3 DPIA Requirements

A programme-level DPIA is not sufficient. Each new data flow, credential type, and partner integration requires:

- Local DPIA evidence and approval
- Data flow mapping
- Privacy control assessment
- Residual risk acceptance

14.4 Clinical Safety

14.4.1 Clinical Safety Standards

- DCB0129: Clinical Risk Management for Manufacturer
- DCB0160: Clinical Risk Management for Deployment

- Clinical Safety Officer (CSO) appointment
- Hazard log and safety case maintenance

14.4.2 Clinical Safety Activities

- Hazard identification workshops
- Clinical workflow analysis
- Safety case development
- Clinical safety testing
- Incident reporting and investigation
- Continuous safety monitoring

14.5 Cyber Security Compliance

14.5.1 Compliance Framework

- NCSC zero-trust architecture principles
- Data Security and Protection Toolkit (DSPT)
- Cyber Assessment Framework (CAF)
- Network and Information Systems (NIS) Regulations

14.5.2 Security Controls

- Identity proofing and verification
- Strong authentication (MFA, FIDO2, PKI)
- Least-privilege access control
- Continuous monitoring and analytics
- Incident response and recovery
- Supply chain security management

14.6 Regulatory Alignment

Regulator	Domain	Engagement Approach
ICO	Data protection	Pre-notification, DPIA review, consultation
NCSC	Cyber security	Architecture review, security testing, guidance
NHS England	Digital services	Programme governance, assurance
DHSC	Health policy	Ministerial oversight, policy alignment

Regulator	Domain	Engagement Approach
Cabinet Office	Major projects	Gateway reviews, assurance

14.7 Detailed Compliance Matrix

Compliance Domain	Standard/Regulation	Evidence Required	Assessment Frequency
Data Protection	UK GDPR, DPA 2018	DPIA, ROPA, DPO review	Annual
Cyber Security	NCSC CAF, DSPT	Self-assessment, external audit	Annual
Clinical Safety	DCB0129, DCB0160	Hazard log, safety case, CSO sign-off	Continuous
Accessibility	WCAG 2.2 AA	Accessibility audit, user testing	Pre-release
Interoperability	OIDC, SAML, FHIR	Conformance testing, certification	Pre-integration
Physical Security	ISO 27001	Certification, audit	Annual

14.8 International Standards Alignment

Standard	Application	Alignment Requirement
eIDAS 2.0	European identity framework	Monitor for future alignment
ISO 27001	Information security management	Certification for platform operations
ISO 27701	Privacy information management	Extension of ISO 27001
ISO 9241-11	Usability	Accessibility and user experience
NIST SP 800-63	Digital identity guidelines	Reference for identity assurance
FIDO Alliance	Passwordless authentication	FIDO2 certification

PART 15: RISK REGISTER AND MITIGATION

15.1 Programme-Level Risks

Risk ID	Risk Description	Category	Likelihood	Impact	Inherent Score	Primary Control	Owner
P1	Funding delay affects programme phasing	Programme	3	4	12	Phased funding, business case approved	Programme Director
P2	Benefits realisation falls short of forecast	Programme	3	4	12	Baseline measurement, benefits ownership, quarterly review	Benefits Lead
P3	Workforce adoption lower than planned	Programme	3	3	9	Champions, training, assisted activation	Workforce Lead
P4	Policy or regulatory change impacts scope	Programme	2	4	8	Policy monitoring, flexible architecture	Programme Director
P5	ICS capacity insufficient for rollout pace	Programme	3	3	9	Readiness gates, regional waves, national support	ICS Lead

15.2 Technical Risks

Risk ID	Risk Description	Category	Likelihood	Impact	Inherent Score	Primary Control	Owner
T1	Legacy system compatibility prevents integration	Technical	4	4	16	Compatibility layer, phased deprecation	Technical Lead
T2	Platform performance insufficient at national scale	Technical	3	5	15	Load testing, scalable architecture, auto-scaling	Technical Lead
T3	Reader/device driver estate fragmentation	Technical	4	4	16	Certification matrix, standard kits, firmware control	Technical Lead

Risk ID	Risk Description	Category	Likelihood	Impact	Inherent Score	Primary Control	Owner
T4	Mobile credential fails in key clinical settings	Technical	3	4	12	Pilot testing, smartcard fallback, shared-device support	Mobile Lead
T5	CIS2 integration scope under estimated	Technical	3	3	9	Application inventory, integration factory, complexity pricing	Integration Lead

15.3 Commercial Risks

Risk ID	Risk Description	Category	Likelihood	Impact	Inherent Score	Primary Control	Owner
C1	Critical supplier delivery delay	Commercial	3	5	15	Modular contracts, multi-supplier resilience	Commercial Lead
C2	Locked into single supplier for critical component	Commercial	2	5	10	Dual source, own cryptographic profile, open standards	Commercial Lead
C3	Price escalation in consumables or hardware	Commercial	3	3	9	Volume bands, alternate supply, open-book pricing	Commercial Lead
C4	Insufficient supplier competition in specialist lots	Commercial	2	4	8	Market engagement, RFI, DPS, international suppliers	Commercial Lead
C5	Contract exit costs higher than expected	Commercial	2	4	8	Exit assistance clauses, escrow, step-in rights	Commercial Lead

15.4 Operational Risks

Risk ID	Risk Description	Category	Likelihood	Impact	Inherent Score	Primary Control	Owner
O1	Registration Authority capacity insufficient for rollout	Operational	4	3	12	National RA mobilisation, training, automated workflows	Operations Lead
O2	Service desk unable to handle launch volume	Operational	3	4	12	Phased rollout, self-service portal, AI-assisted support	Service Lead
O3	Secure stock loss or fraud in supply chain	Operational	2	5	10	Serial reconciliation, dual custody, CCTV, destruction evidence	Operations Lead
O4	National rollout capacity insufficient for demand	Operational	3	4	12	Regional waves, readiness gates, trained RA capacity	Rollout Lead
O5	Business continuity plan fails during incident	Operational	2	5	10	DR testing, failover exercises, runbooks, continuous improvement	Operations Lead

15.5 Cyber Security Risks

Risk ID	Risk Description	Category	Likelihood	Impact	Inherent Score	Primary Control	Owner
S1	Identity compromise through sophisticated attack	Cyber	3	5	15	MFA, cryptographic credentials, behavioural analytics	Security Lead
S2	HSM key compromise or loss	Cyber	1	5	5	Dual control, M-of-N, backup, off-site escrow	Security Lead
S3	Mobile credential vulnerability exploited	Cyber	3	4	12	Hardware-backed keys, attestation, remote revoke, app hardening	Mobile Lead

Risk ID	Risk Description	Category	Likelihood	Impact	Inherent Score	Primary Control	Owner
S4	Supply chain cyber attack on credential supplier	Cyber	3	4	12	Security requirements, audit, monitoring, dual source	Security Lead
S5	Insider threat — RA or admin misuse of system	Cyber	3	4	12	Least privilege, audit, separation of duties, behavioural analytics	Security Lead

15.6 Risk Matrix Summary

Risk ID	Risk Description	Inherent Score	Residual Target	Mitigation Effectiveness
T1	Legacy compatibility	16 (High)	8 (Medium)	Compatibility layer + phased deprecation
T3	Reader estate fragmentation	16 (High)	8 (Medium)	Certification matrix + standard kits
T2	Platform performance	15 (High)	6 (Low)	Load testing + scalable architecture
C1	Supplier delay	15 (High)	8 (Medium)	Modular contracts + multi-supplier
S1	Identity compromise	15 (High)	6 (Low)	MFA + cryptographic credentials + analytics
P1	Funding delay	12 (Medium)	6 (Low)	Phased funding + approved business case
P2	Benefits shortfall	12 (Medium)	6 (Low)	Baseline + ownership + quarterly review
O1	RA capacity	12 (Medium)	6 (Low)	National mobilisation + automation
S3	Mobile vulnerability	12 (Medium)	6 (Low)	Hardware keys + remote revoke
S4	Supply chain attack	12 (Medium)	6 (Low)	Dual source + audit + monitoring

15.7 Residual Risk Assessment

After applying all controls, the programme's residual risk profile is acceptable:

- **High residual risks:** None

- **Medium residual risks:** Legacy compatibility, reader estate fragmentation, supplier delay, RA capacity
- **Low residual risks:** All other risks reduced to low residual status

The programme risk appetite is defined as:

- **Cyber security:** Low risk appetite — controls must be demonstrably effective
- **Financial:** Medium risk appetite — value for money balanced with delivery certainty
- **Operational:** Medium risk appetite — phased rollout with readiness gates
- **Reputational:** Low risk appetite — user experience and accessibility prioritised

15.8 Detailed Risk Register

Risk ID	Risk Description	Category	L	I	Score	Control	Residual	Owner	Review Cycle
P1	Funding delay	Programme	3	4	12	Phased funding, multi-year budget	6	Programme Director	Monthly
P2	Benefits shortfall	Programme	3	4	12	Baseline, ownership, quarterly review	6	Benefits Lead	Quarterly
P3	Workforce adoption	Programme	3	3	9	Champions, training, comms	4	Workforce Lead	Monthly
P4	Policy change	Programme	2	4	8	Monitoring, flexible architecture	4	Programme Director	Quarterly
P5	ICS capacity	Programme	3	3	9	Readiness gates, national support	4	ICS Lead	Monthly
T1	Legacy compatibility	Technical	4	4	16	Compatibility layer, deprecation	8	Technical Lead	Monthly
T2	Platform performance	Technical	3	5	15	Load testing, auto-scaling	6	Technical Lead	Monthly
T3	Reader estate	Technical	4	4	16	Certification matrix, kits	8	Technical Lead	Monthly

Risk ID	Risk Description	Category	L	I	Score	Control	Residual	Owner	Review Cycle
T4	Mobile credential	Technical	3	4	12	Pilot, fallback, support	6	Mobile Lead	Monthly
T5	CIS2 integration	Technical	3	3	9	Inventory, factory, pricing	4	Integration Lead	Monthly
C1	Supplier delay	Commercial	3	5	15	Modular contracts, multi-supplier	8	Commercial Lead	Monthly
C2	Supplier lock-in	Commercial	2	5	10	Dual source, open standards	4	Commercial Lead	Quarterly
C3	Price escalation	Commercial	3	3	9	Bands, alternate supply	4	Commercial Lead	Quarterly
C4	Competition	Commercial	2	4	8	Market engagement, DPS	4	Commercial Lead	Quarterly
C5	Exit costs	Commercial	2	4	8	Exit clauses, escrow	4	Commercial Lead	Quarterly
O1	RA capacity	Operational	4	3	12	Mobilisation, automation	6	Operations Lead	Monthly
O2	Service desk volume	Operational	3	4	12	Phased rollout, self-service	6	Service Lead	Monthly
O3	Stock loss/fraud	Operational	2	5	10	Reconciliation, dual custody	4	Operations Lead	Monthly
O4	Rollout capacity	Operational	3	4	12	Regional waves, gates	6	Rollout Lead	Monthly
O5	BCP failure	Operational	2	5	10	DR testing, runbooks	4	Operations Lead	Quarterly
S1	Identity compromise	Cyber	3	5	15	MFA, crypto, analytics	6	Security Lead	Monthly
S2	HSM key compromise	Cyber	1	5	5	Dual control, M-of-N	2	Security Lead	Quarterly
S3	Mobile vulnerability	Cyber	3	4	12	Hardware keys, revoke	6	Mobile Lead	Monthly

Risk ID	Risk Description	Category	L	I	Score	Control	Residual	Owner	Review Cycle
S4	Supply chain attack	Cyber	3	4	12	Dual source, audit	6	Security Lead	Monthly
S5	Insider threat	Cyber	3	4	12	Least privilege, audit	6	Security Lead	Monthly

PART 16: FUTURE-STATE STRATEGY (YEARS 5–10)

16.1 Digital Wallet Maturity

Year 5–6: Digital wallet expansion

- Role and registration badges in digital wallet
- Time-bound partner access via wallet
- Secure signing capabilities
- Policy-controlled expansion

Year 7–8: Wallet maturity

- Multi-credential wallet supporting multiple identity sources
- Verifiable credentials for cross-sector use
- User-controlled data sharing
- Self-sovereign identity principles

16.2 AI Identity Verification

Year 6–7: AI-assisted verification

- AI-powered identity document verification
- Automated liveness detection
- Fraud pattern recognition
- Continuous authentication risk scoring

Year 8–10: Advanced AI capabilities

- Predictive identity threat detection
- Automated identity analytics
- AI-assisted access decisions

- Behavioural biometrics

16.3 Policy-Controlled Biometrics

Year 7–8: Biometric assessment

- Policy framework for biometric use
- Privacy impact assessment
- Public consultation and consent framework
- Pilot deployment in controlled settings

Year 9–10: Controlled biometric deployment

- Fingerprint and/or facial recognition for specific use cases
- Strong policy controls and consent management
- Opt-out and alternative authenticator availability
- Continuous privacy and ethics review

16.4 Cross-Sector Federation

Year 8–9: Sector expansion

- Social care identity federation
- Local government integration
- Cross-government identity sharing
- Trust framework development

Year 10: National identity ecosystem

- NHS identity as a trusted foundation
- Cross-sector federation operational
- National identity standards
- Reusable identity control plane

16.5 Continuous Innovation Model

Innovation pipeline:

- Year 5–6: Digital wallet, AI identity verification
- Year 7–8: Biometric assessment, advanced analytics
- Year 9–10: Cross-sector federation, self-sovereign identity

Innovation partners:

- Academic research partnerships
- Supplier innovation programmes

- NHS innovation hubs
- International identity programme collaboration

Innovation governance:

- Innovation board
- Ethics and privacy review
- Pilot and evaluation framework
- Knowledge sharing and publication

16.6 Long-Term Sustainability

Financial sustainability:

- Operational costs predictable and stable
- Managed service costs decline with scale
- Hardware costs decline over time
- Digital wallet reduces future credential costs
- Workforce mobility reduces duplication
- Cyber risk reduction avoids major losses

Technical sustainability:

- Open standards avoid lock-in
- Modular architecture enables component replacement
- Crypto-agile design supports algorithm evolution
- Continuous improvement cycle

Organisational sustainability:

- Knowledge transfer embedded in contracts
- NHS capability built through programme delivery
- Communities of practice established
- Skills and expertise retained in-house

16.7 Technology Roadmap

Year	Technology	Capability	Readiness Level
Year 5	Digital Wallet v1	Role badges, time-bound access, basic signing	Production
Year 6	AI Identity Verification	Document verification, liveness detection	Pilot
Year 7	Advanced Analytics	Predictive threat detection, fraud analytics	Production

Year	Technology	Capability	Readiness Level
Year 8	Policy-Controlled Biometrics	Fingerprint (specific use cases)	Pilot
Year 9	Cross-Sector Federation	Social care, local government	Pilot
Year 10	Self-Sovereign Identity	User-controlled identity, verifiable credentials	Production

16.8 Strategic Options and Scenarios

Scenario	Description	Probability	Investment Required	Value
Accelerated Federation	Cross-sector federation by Year 7	30%	Additional £50M	£2.0B additional benefit
Conservative Biometrics	Biometrics limited to specific high-assurance use cases	60%	Within programme budget	Baseline
AI-Led Innovation	AI identity verification as primary innovation driver	50%	Additional £30M	£1.5B additional benefit
Full Digital Wallet	Comprehensive digital wallet by Year 5	40%	Additional £20M	£1.0B additional benefit

APPENDICES

Appendix A: Detailed Product Specifications

A.1 Smartcard Technical Specification

Parameter	Specification
Form Factor	ISO/IEC 7810 ID-1 (85.6mm x 54.0mm x 0.76mm)
Interface	Dual-interface: Contact (ISO 7816) + Contactless (ISO 14443 Type A)
Secure Element	Common Criteria EAL5+ certified
Cryptographic Support	RSA 2048/4096, ECC P-256/P-384, SHA-2/3
Memory	Minimum 80KB EEPROM
Card OS	Java Card 3.0.4 or equivalent
Operating Temperature	0°C to 50°C

Parameter	Specification
Durability	Minimum 3 years (standard), 5 years (polycarbonate)
Visual Security	Guilloche, microtext, UV feature, ghost image
Personalisation	Dye-sublimation printing, contact and contactless encoding

A.2 Mobile Credential Technical Specification

Parameter	Specification
Platform	iOS 16+ and Android 12+
Authentication	Device biometric (Face ID/Touch ID/Fingerprint), PIN
Key Storage	Hardware-backed secure enclave (iOS) / TEE (Android)
Protocol	FIDO2/WebAuthn, OIDC
Push Authentication	APNS (iOS) / FCM (Android)
Device Binding	Hardware attestation and device registration
Remote Revoke	Instant credential disable via server
Offline Mode	Limited offline authentication (cached assertions)
Accessibility	VoiceOver (iOS) / TalkBack (Android)

A.3 Dual-Interface Reader Specification

Parameter	Specification
Reference Product	HID OMNIKEY 5422 or equivalent
Contact Interface	ISO 7816, T=0 and T=1 protocols
Contactless Interface	ISO 14443 Type A and B, 13.56 MHz
Standards	PC/SC 2.01, CCID, USB 2.0
OS Support	Windows 10/11, macOS 12+, Linux (PC/SC)
Connection	USB-A (USB-C adapter available)
LED Indicators	Power, card present, data activity
Operating Temperature	0°C to 50°C
MTBF	500,000+ card insertions
Warranty	3 years minimum

Appendix B: Manufacturer and Supplier Evidence

B.1 Public Price Benchmarks

Reference Item	Published Price	Evidence Type	Source
HID OMNIKEY 5422	£58.50–£62.50 ex VAT	UK channel listings	The Card Network, Levata UK
Zebra ZC300	£949.99–£1,732.85 ex VAT	UK business/channel listings	Currys Business, Jarttech UK
Evolis Primacy 2	£1,309.63–£1,480 ex VAT	2026 price list / UK reseller	Cablecraft
YMCKO Ribbon (300 images)	£41.38–£55.95	UK consumables listings	Dragon Solutions, Cards-X
Yubico Security Key NFC	£30 incl. VAT	Yubico UK store	Yubico
YubiKey 5 NFC	£60 incl. VAT	Yubico UK store	Yubico
YubiKey 5 NFC FIPS 140-3	£105.60 incl. VAT	Yubico UK store	Yubico
MIFARE DESFire EV3 Media	£3.30–£7.60	G-Cloud price schedule	CCS Digital Marketplace
Bureau Card Personalisation	~£1.10–£1.20 plus options	G-Cloud schedule, 2024	CCS Digital Marketplace

B.2 Manufacturer References

Manufacturer	Product	Reference URL
HID Global	OMNIKEY 5422	https://www.hidglobal.com/products/omnikey-5422
HID Global	Signo Readers	https://www.hidglobal.com/products/signo-white-readers
NXP	SmartMX3 P71D321	https://www.nxp.com/products/SMART-MX3-P71D321
IDEMIA	Smart Credentials	https://na.idemia.com
G+D	STARCOS Signature Card	https://www.gi-de.com
Zebra	ZC300	https://www.zebra.com/gb/en/products/spec-sheets/printers/card/zc300.html
Evolis	Primacy 2	https://www.evolis.com/solutions/card-printers/primacy2-card-printer/
Entrust	nShield 5c HSM	https://www.entrust.com/products/hsm/nshield-5c
Thales	Luna Network HSM 7	https://cpl.thalesgroup.com
Yubico	Security Keys	https://www.yubico.com/gb/store/

Appendix C: Public Price Benchmarks

C.1 Estimation Method

Public small-quantity prices from the listed sources were normalised to an ex-VAT planning basis where possible. Volume assumptions were then applied as low, base, and high values. Quote-only enterprise

products use explicit allowances. No web price is represented as a binding supplier offer.

C.2 Confidence Bands

Confidence Level	Description	Product Categories
Medium	Public reference prices available for small quantities	Readers, printers, security keys
Low-Medium	Enterprise volume prices require formal quotation	Smartcards, HSMS, PKI/CMS
Low	Allowance-based estimates pending architecture decisions	Bureau fit-out, cyber platform licences

Appendix D: Glossary of Terms

Term	Definition
ABAC	Attribute-Based Access Control
BCR	Benefit-Cost Ratio
CA	Certificate Authority
CAF	Cyber Assessment Framework
CapEx	Capital Expenditure
CIS	Care Identity Service
CIS2	Care Identity Service 2 (modernised)
CMS	Credential Management System
CRL	Certificate Revocation List
CSO	Clinical Safety Officer
DCB	Digital Clinical Safety (standards)
DHSC	Department of Health and Social Care
DPIA	Data Protection Impact Assessment
DPS	Dynamic Purchasing System
DSPT	Data Security and Protection Toolkit
EDR	Endpoint Detection and Response
EPR	Electronic Patient Record
FDP	Federated Data Platform
FIDO2	Fast Identity Online 2 (authentication standard)
FHIR	Fast Healthcare Interoperability Resources
HSCN	Health and Social Care Network
HSM	Hardware Security Module

Term	Definition
ICS	Integrated Care System
ICO	Information Commissioner's Office
MFA	Multi-Factor Authentication
NCSC	National Cyber Security Centre
NDG	National Data Guardian
NHS	National Health Service
OCSF	Online Certificate Status Protocol
OIDC	OpenID Connect
OpEx	Operational Expenditure
PAS	Patient Administration System
PCR	Public Contracts Regulations
PKI	Public Key Infrastructure
RA	Registration Authority
RBAC	Role-Based Access Control
RFI	Request for Information
ROI	Return on Investment
SAML	Security Assertion Markup Language
SIEM	Security Information and Event Management
SOAR	Security Orchestration, Automation and Response
SOC	Security Operations Centre
UEBA	User and Entity Behaviour Analytics
WAF	Web Application Firewall
WCAG	Web Content Accessibility Guidelines

Appendix E: References and Sources

E.1 Official NHS Sources

- NHS England — Care Identity Service: <https://digital.nhs.uk/services/care-identity-service>
- NHS England — Smartcards and Authenticators:
<https://digital.nhs.uk/services/care-identity-service/smartcard-and-authentication-users>
- NHS England — Care Identity Service Roadmap:
<https://digital.nhs.uk/services/care-identity-service/roadmap>
- NHS England — Care Identity Applications and Services:
<https://digital.nhs.uk/services/care-identity-service/applications-and-services>

- NHS England — Digital Clinical Safety Strategy:
<https://www.england.nhs.uk/long-read/digital-clinical-safety-strategy/>
- NHS England — Data Security and Protection Toolkit: <https://digital.nhs.uk/cyber-and-data-security/cyber-security-services/data-security-and-protection-toolkit>

E.2 Government Standards

- NCSC — Zero Trust Architecture Principles:
<https://www.ncsc.gov.uk/collection/zero-trust-architecture/>
- ICO — Data Protection Impact Assessments: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/accountability-and-governance/data-protection-impact-assessments-dpias/>
- GOV.UK — Digital Identity Trust Framework:
<https://www.gov.uk/government/collections/uk-digital-verification-services-trust-framework>

E.3 Manufacturer References

As listed in Appendix B.2 above.

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All work is owned by Darren Birch, Architect ZEUS AI Intelligence and Founder JDB Sales

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Appendix G: Detailed Financial Model Assumptions

G.1 Discount Rate Assumptions

Parameter	Value	Source
Social Time Preference Rate (STPR)	3.5%	HM Treasury Green Book
Test Discount Rate	3.5%	HM Treasury Green Book
Optimism Bias (Programme)	10-25%	HM Treasury supplementary guidance
Optimism Bias (IT)	10-50%	HM Treasury supplementary guidance
Price Base	2026/27	Current financial year
Real Terms	Prices adjusted for inflation	Assumption
Benefit Discounting	Annual compounding	Standard methodology

G.2 Labour Cost Assumptions

Staff Category	Annual Cost (£)	Daily Rate (£)	Hourly Rate (£)
Consultant Doctor	£105,000	£525	£66
Specialty Doctor	£82,000	£410	£51
Nurse (Band 5)	£32,000	£160	£20
Nurse (Band 6)	£38,000	£190	£24
Nurse (Band 7)	£45,000	£225	£28
AHP (Band 6)	£37,000	£185	£23

Staff Category	Annual Cost (£)	Daily Rate (£)	Hourly Rate (£)
Admin (Band 3)	£22,000	£110	£14
Admin (Band 4)	£25,000	£125	£16
IT Staff (Band 7)	£45,000	£225	£28
IT Staff (Band 8a)	£55,000	£275	£34
Blended Workforce Average	£52,000	£260	£33

G.3 Inflation and Escalation Assumptions

Parameter	Assumption	Basis
General Inflation (CPI)	2.5% per annum	Bank of England target
Healthcare Inflation	3.0% per annum	NHS cost inflation trend
IT/Technology Inflation	1.5% per annum	Technology cost deflation trend
Labour Cost Escalation	3.0% per annum	NHS pay settlements
Hardware Cost Trend	-2.0% per annum	Technology cost reduction
Consumables Inflation	2.5% per annum	General inflation

G.4 Benefit Calculation Assumptions

Benefit Component	Quantitative Assumption	Basis
Workforce Size	2,000,000	Portfolio estimate
Annual Turnover	10% (200,000/year)	NHS workforce statistics
Current Onboarding Delay	2 days average	NHS workforce survey
Staff Cost (Blended)	£260/day	NHS pay scales (blended)
Current Access Delay	12 minutes/day/staff	NHS digital survey
Future Access Delay	1 minute/day/staff	Target state
Identity Fraud (Current)	£418M/year	NHS Counter Fraud Authority
Legacy Systems	250+	Portfolio analysis
Future Legacy Systems	1 national system	Target state
Current Cyber Breach Cost	Industry benchmarks	NCSC / industry reports
Cyber Risk Reduction	78% (identity compromise)	Portfolio target

Appendix H: Stakeholder Engagement Plan

H.1 Stakeholder Mapping

Stakeholder Group	Power/Interest	Engagement Strategy	Frequency	Owner
DHSC Ministers	High/High	Quarterly ministerial briefings, annual review	Quarterly	Programme SRO
NHS England Board	High/High	Board papers, presentations, Gateway reviews	Bi-annual	Programme SRO
ICS Leaders	High/High	ICS delivery boards, regional forums	Monthly	ICS Lead
Trust CIOs/CISOs	High/High	Technical forums, readiness assessments	Monthly	Technical Lead
Trust CEOs	High/Medium	Board summaries, benefits reports	Quarterly	Programme Director
NHS Workforce	Low/High	Communications, training, champions	Continuous	Workforce Lead
Suppliers	High/High	Commercial forums, performance reviews	Monthly	Commercial Lead
NCSC	High/Medium	Architecture reviews, security testing	Quarterly	Security Lead
ICO	Medium/Medium	DPIA review, consultation	As required	Data Protection Lead
Patients/Public	Low/Low	Public communications, transparency	Annual	Communications Lead
Trade Unions	Medium/High	Engagement forums, consultation	Quarterly	Workforce Lead
Professional Bodies	Medium/Medium	Clinical engagement, professional standards	Bi-annual	Clinical Lead

H.2 Communications Channels

Channel	Purpose	Audience	Frequency
Ministerial Briefing	Strategic update, decisions required	DHSC Ministers	Quarterly
Board Paper	Formal approval, oversight	NHS England Board	Bi-annual
Programme Newsletter	Progress update, milestones	All stakeholders	Monthly
ICS Briefing	Regional update, readiness	ICS Leaders	Monthly
Trust Briefing	Local update, deployment planning	Trust CEOs, CIOs	Monthly
Clinical Forum	Clinical input, workflow validation	Clinical leads	Quarterly
Supplier Forum	Commercial update, performance	Suppliers	Monthly

Channel	Purpose	Audience	Frequency
Workforce Communications	Adoption, training, support	NHS workforce	Continuous
Programme Website	Public information, transparency	Public	Continuous
Social Media	Awareness, engagement	Workforce, public	Weekly

H.3 Key Engagement Milestones

Milestone	Stakeholder	Engagement Activity	Timing
Business Case Approval	DHSC, NHS England	Board presentation, ministerial approval	Month 1-3
Architecture Approval	NCSC, Technical Assurance	Architecture review, sign-off	Month 6
Supplier Market Engagement	Suppliers	RFI, supplier briefings, challenge sessions	Month 3-6
Procurement Launch	Suppliers	Tender publication, bidder conferences	Month 7-12
Pilot Launch	Trusts, workforce	Site visits, go-live events, comms	Month 18
National Rollout	All stakeholders	Regional launch events, workforce comms	Year 3
Benefits Review	DHSC, NHS England	Benefits realisation report	Annual

Appendix I: Benefits Calculation Workbook

I.1 Administrative Efficiency — Detailed Calculation

Reduced Onboarding Time:

Variable	Value	Source
Total workforce	2,000,000	Portfolio estimate
Annual turnover rate	10%	NHS workforce statistics
Annual new starters	200,000	Calculated
Current onboarding delay (days)	2	NHS workforce survey
Target onboarding delay (days)	0.02 (30 minutes)	Programme target
Days saved per new starter	1.98	Calculated
Blended daily cost per staff member	£260	NHS pay scales
Value per new starter	£515	Calculated
Total annual value	£103M	Calculated
Ramp-up factor (Year 1-3)	30%, 70%, 100%	Adoption curve assumption

Variable	Value	Source
Year 1 value	£30.9M	Calculated
Year 2 value	£72.1M	Calculated
Year 3-10 value (annual)	£103M	Calculated
10-year total	£927M	Calculated

Reduced Access Delays:

Variable	Value	Source
Total workforce	2,000,000	Portfolio estimate
Current access delay (minutes/day)	12	NHS digital survey
Target access delay (minutes/day)	1	Programme target
Minutes saved per day	11	Calculated
Productive minutes per year	220 days	NHS working days
Minutes saved per year	2,420	Calculated
Hours saved per year	40.3	Calculated
Blended hourly cost	£33	NHS pay scales
Value per staff member per year	£1,330	Calculated
Total annual value (full adoption)	£2,660M	Calculated
Attribution factor (identity-related)	30%	Only identity-related delay reduction
Attributable annual value	£798M	Calculated
10-year total (with ramp)	£6,384M	Calculated

1.2 Fraud Reduction — Detailed Calculation

Variable	Value	Source
Total NHS fraud estimate	£1.1B/year	NHS Counter Fraud Authority
Identity-related fraud proportion	38%	Counter Fraud analysis
Current identity fraud cost	£418M/year	Calculated
Target reduction	50%	Programme target
Annual fraud reduction	£209M	Calculated
10-year total (with ramp)	£1,672M	Calculated

Contractor Misuse Reduction:

Variable	Value	Source
Annual contractor spend	£8B	NHS procurement data
Current identity misuse proportion	15%	Industry estimate

Variable	Value	Source
Current misuse cost	£1.2B	Calculated
Target reduction	10%	Programme target
Annual reduction	£120M	Calculated
10-year total (with ramp)	£960M	Calculated

1.3 Workforce Mobility — Detailed Calculation

Variable	Value	Source
Cross-Trust movements per year	500,000	NHS workforce statistics
Current time to productive access (days)	3	NHS workforce survey
Target time to productive access (hours)	2	Programme target
Time saved per movement (days)	2.75	Calculated
Blended daily cost	£260	NHS pay scales
Value per movement	£715	Calculated
Annual value	£357.5M	Calculated
10-year total (with ramp)	£2,860M	Calculated

1.4 Cyber Risk Reduction — Detailed Calculation

Variable	Value	Source
Average cost of NHS cyber incident	£3.5M	NHS Digital / industry
Current incident frequency (identity-related)	40/year	Portfolio estimate
Target incident frequency	9/year	78% reduction target
Incidents avoided per year	31	Calculated
Direct cost avoidance	£108.5M/year	Calculated
Indirect cost (reputational, operational)	2x direct	Industry benchmark
Total annual value	£325.5M	Calculated
10-year total (with ramp)	£2,604M	Calculated

Appendix J: Capacity and Resource Planning Model

J.1 Core Programme Team Resource Plan

Role	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6-10
Programme Director	1	1	1	1	1	0.5
Deputy Programme Director	1	1	1	1	1	0.5
Technical Architect	2	3	2	1	1	1
Security Architect	1	2	2	1	1	1
Integration Architect	1	3	3	2	1	1
Commercial Lead	1	2	2	1	1	1
Commercial Manager	1	2	2	2	1	1
Workforce Lead	1	2	3	2	1	1
Training Lead	1	2	2	1	1	0.5
Communications Lead	1	1	1	1	1	0.5
Benefits Lead	1	1	1	1	1	1
Data Protection Lead	1	1	1	1	0.5	0.5
Clinical Safety Officer	1	1	1	1	0.5	0.5
PMO Manager	1	1	1	1	1	1
PMO Analysts	2	4	4	2	2	1
Risk Manager	1	1	1	1	1	0.5
Quality Manager	1	1	1	1	0.5	0.5
Total Core Team	19	29	29	21	16	12

J.2 Estimated Programme Staff Costs

Year	Core Team	Annual Cost (Avg £80k)	Total Core Cost
Year 1	19	£80,000	£1,520,000
Year 2	29	£82,500	£2,392,500
Year 3	29	£85,000	£2,465,000
Year 4	21	£87,500	£1,837,500
Year 5	16	£90,000	£1,440,000

Year	Core Team	Annual Cost (Avg £80k)	Total Core Cost
Year 6	12	£92,500	£1,110,000
Year 7	12	£95,000	£1,140,000
Year 8	12	£97,500	£1,170,000
Year 9	12	£100,000	£1,200,000
Year 10	12	£102,500	£1,230,000
Total			£15,505,000

J.3 ICS and Trust Deployment Capacity

Year	ICS Regions	Trusts Deployed	Cumulative Trusts	Workforce Activated	Cumulative Workforce
Year 1	0	0	0	0	0
Year 2	2	15	15	150,000	150,000
Year 3	4	60	75	500,000	650,000
Year 4	6	85	160	650,000	1,300,000
Year 5	7	140	300	700,000	2,000,000
Year 6+	7	0	300	0	2,000,000

J.4 Registration Authority Capacity Requirements

Metric	Year 1	Year 2	Year 3	Year 4	Year 5	
New RA Staff Trained	50	200	400	200	100	
Cumulative RA Staff	50	250	650	850	950	
Credentials Activated per RA	5,000	5,000	5,000	5,000	5,000	
Total Activation Capacity	250,000	1,250,000	3,250,000	4,250,000	4,750,000	
Annual Activation Demand	0	150,000	500,000	650,000	700,000	
Capacity Utilisation	0%	12%	15%	15%	15%	

Metric	Year 1	Year 2	Year 3	Year 4	Year 5	
RA Capacity Available for Surge	250,000	1,100,000	2,750,000	3,600,000	4,050,000	## Appendix K: Detailed Technical Architecture Diagrams (Text Description)

K.1 National Identity Platform — Component Architecture

The national identity platform comprises the following major subsystems, each with defined interfaces, data flows, and security boundaries:

Identity Store Layer:

- Care Identity Directory — authoritative record of all workforce identities, organisational affiliations, and role assignments. Built on highly available database infrastructure with active-active replication across two geographic regions.
- Organisation Directory — registered NHS and partner organisations with hierarchies, trust relationships, and service endpoints.
- Role and Attribute Store — dynamic role assignments and attribute values sourced from ESR, local HR systems, and RA workflows.

Authentication Layer:

- Authentication Gateway — single entry point for all authentication requests, supporting OIDC, SAML, and OAuth 2.0 protocols. Routes requests to the appropriate authenticator validation service.
- Smartcard Authentication Service — validates smartcard certificates against OCSP/CRL, verifies PIN, and establishes authenticated session.
- Mobile Authentication Service — validates push authentication responses, FIDO2 assertions, and device attestation evidence.
- Passwordless Authentication Service — validates passkey and Windows Hello assertions against registered FIDO2 credentials.
- Security Key Authentication Service — validates FIDO2 security key assertions for privileged access.

Policy and Authorisation Layer:

- Policy Decision Point (PDP) — evaluates access requests against defined policies including role, location, device posture, time of day, and risk score.
- Policy Administration Point (PAP) — authoring and management of access policies, role definitions, and attribute mappings.
- Risk Engine — real-time risk scoring based on behavioural analytics, anomaly detection, and contextual signals.
- Session Management — token issuance, session tracking, timeout enforcement, and forced re-authentication triggers.

Integration Layer:

- API Gateway — all application-facing API endpoints with rate limiting, authentication, audit logging, and schema validation.
- FHIR Interface — HL7 FHIR-compliant APIs for workforce and organisational data exchange with clinical systems.
- Legacy Compatibility Layer — protocol translation, authentication bridging, and credential mapping for legacy applications that cannot be migrated to CIS2.
- Event Bus — publish/subscribe event infrastructure for identity lifecycle events (creation, activation, suspension, revocation).

Security and Monitoring Layer:

- SIEM Platform — centralised security event collection, correlation, and alerting from all platform components.
- SOAR Platform — automated incident response workflows for common security events including token theft, brute force attempts, and anomalous access patterns.
- UEBA Engine — machine learning-based user and entity behaviour analytics for identity threat detection.
- Audit Store — immutable audit log of all identity and access events with tamper-evident storage.
- Vulnerability Management — continuous scanning of all platform components, dependency tracking, and remediation workflow.

K.2 Physical Security Architecture

Data Centre Security:

- Tier 3+ data centre facilities in two geographic regions
- Multi-factor physical access control
- 24/7 security personnel and CCTV monitoring
- Environmental controls (temperature, humidity, fire suppression)
- Power redundancy (N+1 UPS, backup generators)

Network Security:

- Segmented network architecture with DMZ, application, database, and management zones
- HSCN connectivity for NHS network integration
- Internet-facing services protected by WAF and DDoS mitigation
- Encrypted inter-zone communication (TLS 1.3)
- Network access control lists at every boundary

Cryptographic Security:

- All cryptographic operations performed within HSM boundary
- Root CA keys stored in offline HSM with physical security controls
- Issuing CA keys in network HSMs with quorum access
- Certificate lifecycle managed through automated CMS
- Key ceremonies conducted with witnessed dual control

K.3 Data Flow Architecture — Authentication

Mobile Authentication Flow:

1. User opens application on mobile device
2. Application initiates authentication request to Authentication Gateway via OIDC
3. Authentication Gateway sends push notification to registered device
4. User authenticates on device (biometric or PIN)
5. Device generates FIDO2 assertion signed by hardware-backed key
6. Assertion sent to Authentication Gateway
7. Mobile Authentication Service validates assertion, checks device attestation, verifies device compliance
8. Risk Engine evaluates contextual risk (location, time, device, behaviour pattern)
9. Policy Decision Point evaluates access policy
10. If ALLOWED: short-lived token issued, session established
11. If STEP UP: additional verification requested (e.g., PIN entry)
12. If DENIED: access blocked, incident logged, SOC alerted if suspicious

Smartcard Authentication Flow:

1. User inserts smartcard into reader
2. Reader powers card and selects PKI applet
3. Terminal sends internal authentication challenge
4. Card generates signed response using private key
5. Authentication Service validates certificate chain against CA
6. OCSP status checked for certificate revocation
7. User enters PIN, verified on-card
8. Risk Engine evaluates contextual risk
9. Policy Decision Point evaluates access policy
10. If ALLOWED: short-lived token issued, session established
11. Authentication event logged to audit store

Appendix L: Programme Governance Terms of Reference

L.1 Programme Steering Committee — Terms of Reference

Purpose: The Programme Steering Committee provides senior leadership, strategic direction, and decision-making authority for the NHS Digital Identity Programme.

Membership:

- Senior Responsible Owner (SRO) — Chair
- Programme Director

- DHSC Digital Directorate Representative
- NHS England Digital Services Representative
- ICS Lead Representative
- Chief Clinical Information Officer (CCIO)
- Chief Information Officer (CIO) Representative
- Chief Information Security Officer (CISO) Representative
- Commercial Lead
- Finance Lead

Quorum: Chair plus 5 members, including at least one clinical and one digital representative.

Meeting Frequency: Monthly

Authority:

- Approve programme business cases and changes
- Approve programme budget and expenditure
- Approve major supplier contracts and contract changes
- Approve delivery phase transitions
- Escalate risks and issues to NHS England Board
- Approve programme communications and publications

Reporting: The Steering Committee reports to the NHS England Executive Board and DHSC.

L.2 Technical Assurance Board — Terms of Reference

Purpose: The Technical Assurance Board provides technical governance, architecture assurance, and standards compliance oversight.

Membership:

- Chief Technology Officer (CTO) — Chair
- Technical Architect
- Security Architect
- Integration Architect
- NCSC Representative
- NHS England Technical Assurance Representative
- Clinical Safety Officer
- Supplier Technical Representatives (as required)

Quorum: Chair plus 3 members

Meeting Frequency: Fortnightly (monthly after foundation phase)

Authority:

- Approve technical architecture and design decisions

- Approve technology standards and compliance
- Approve security architecture and controls
- Approve integration patterns and API specifications
- Approve exception requests to architecture standards
- Recommend technical go/no-go decisions at gates

L.3 Benefits Realisation Board — Terms of Reference

Purpose: The Benefits Realisation Board oversees benefits tracking, measurement, and reporting.

Membership:

- Benefits Lead — Chair
- Programme Director
- Finance Lead
- ICS Benefits Representative
- Trust Benefits Representative
- Workforce Lead
- Data and Analytics Lead

Quorum: Chair plus 3 members

Meeting Frequency: Quarterly

Authority:

- Approve benefits baseline and measurement methodology
- Approve benefits realisation reports
- Recommend corrective actions for underperforming benefits
- Approve benefit reallocation between categories
- Escalate benefits risks to Programme Steering Committee

Appendix M: Service Level Agreement Framework

M.1 Platform Service Levels

Service Element	Target	Measurement Period	Service Credit Trigger
Authentication Service Availability	99.99%	Monthly	< 99.9%
Mobile Authentication Response Time	< 2 seconds (p95)	Monthly	> 3 seconds
Smartcard Authentication Response Time	< 1 second (p95)	Monthly	> 2 seconds
API Gateway Availability	99.99%	Monthly	< 99.9%

Service Element	Target	Measurement Period	Service Credit Trigger
API Response Time	< 500ms (p95)	Monthly	> 1 second
Credential Issuance Time (Digital)	< 5 minutes	Monthly	> 15 minutes
Credential Issuance Time (Physical)	< 5 working days	Monthly	> 10 working days
Certificate Revocation Processing	< 5 minutes	Monthly	> 15 minutes
Security Incident Detection	< 15 minutes	Per incident	> 30 minutes
Security Incident Response	< 1 hour	Per incident	> 2 hours
Service Desk — First Response	< 1 minute	Monthly	> 3 minutes
Service Desk — Resolution (P1)	< 4 hours	Monthly	> 8 hours
Service Desk — Resolution (P2)	< 8 hours	Monthly	> 24 hours
Service Desk — Resolution (P3)	< 48 hours	Monthly	> 72 hours

M.2 Priority Definitions

Priority	Definition	Response Time	Resolution Target
P1 — Critical	Service unavailable, major security incident, data breach	< 1 minute	< 4 hours
P2 — High	Service degraded, significant user impact, potential security issue	< 5 minutes	< 8 hours
P3 — Medium	Non-critical issue, single user affected, minor functionality	< 15 minutes	< 48 hours
P4 — Low	Cosmetic issue, feature request, documentation	< 1 hour	< 5 working days

M.3 Service Credit Regime

Availability (Monthly)	Service Credit
99.9% — 99.99%	0% (within target)
99.5% — 99.9%	5% of monthly service charge
99.0% — 99.5%	10% of monthly service charge
95.0% — 99.0%	20% of monthly service charge
< 95.0%	50% of monthly service charge plus termination rights

Appendix N: Implementation Checklist and Readiness Assessment

N.1 Trust Readiness Assessment

Domain	Criteria	Evidence Required	Assessment Method
Technical	Clinical systems CIS2-compatible	System compatibility report	Technical review
Technical	Network connectivity to HSCN	Network connectivity certificate	Technical review
Technical	Reader and printer devices inventoried	Device inventory report	Audit
Technical	Workstation compatibility confirmed	OS/driver compatibility matrix	Technical review
Operational	RA staff trained and certified	Training completion certificates	Verification
Operational	Local champions identified and trained	Champion register	Verification
Operational	Support model agreed with national service desk	Support agreement signed	Review
Operational	Workforce communications completed	Communication completion report	Review
Governance	DPIA approved	DPIA sign-off	Review
Governance	Cyber assurance complete	DSPT/cyber assessment	Review
Governance	Local readiness sign-off from Trust CIO	Readiness sign-off form	Confirmation
Governance	Benefits baseline recorded	Baseline metrics report	Review

N.2 Readiness Scoring

Score	Classification	Action
12/12 — All criteria met	Green — Ready	Proceed with rollout
10-11/12 — Minor gaps	Amber — Conditional	Proceed with remediation plan
8-9/12 — Significant gaps	Red — Not Ready	Hold, remediate, reassess
< 8/12 — Critical gaps	Black — Halted	Escalate to Programme Board

N.3 Go-Live Checklist

Pre-Go-Live (T-4 weeks):

- ☐ All technical integration tested and passed
- ☐ RA staff trained and certified

- ☐ Local champions trained and active
- ☐ Workforce communications complete
- ☐ Service desk staff briefed on local support model
- ☐ Hardware installed and tested (readers, printers)
- ☐ Network connectivity verified
- ☐ Rollback plan documented and rehearsed

Go-Live Week (T-0):

- ☐ Executive sponsorship confirmed
- ☐ Champions on standby for floorwalking
- ☐ Service desk on high alert
- ☐ Monitoring dashboards active
- ☐ Incident management process activated
- ☐ Daily go-live calls scheduled
- ☐ Rollback decision criteria agreed

Post-Go-Live (T+4 weeks):

- ☐ Adoption metrics collected and reviewed
- ☐ User feedback collected and analysed
- ☐ Issues logged and resolved
- ☐ Benefits baseline confirmed
- ☐ Lessons learned documented
- ☐ Rollout complete sign-off obtained

Appendix O: Programme Critical Success Factors

O.1 Critical Success Factors

CSF	Description	Measurement	Target	Owner
CSF-1	Executive sponsorship sustained throughout programme lifecycle	SRO attendance at Steering Committee	100% attendance	Programme Director
CSF-2	Clinical engagement embedded in design and delivery	Clinical representation on all workstreams	100% workstreams	Clinical Lead
CSF-3	Supplier ecosystem delivers to time, cost, and quality	Supplier performance against KPIs	90%+ KPI achievement	Commercial Lead
CSF-4	Workforce adoption meets or exceeds targets	Activation rate by user group	85%+ within 6 months	Workforce Lead
CSF-5	Integration complexity managed through standardisation	Applications on CIS2 onboarding pathway	80%+ by Year 3	Integration Lead

CSF	Description	Measurement	Target	Owner
CSF-6	Cyber security controls proven effective before national scale	Security testing results, audit findings	No critical/high findings	Security Lead
CSF-7	Benefits realisation tracked and reported transparently	Benefits dashboard, quarterly reports	100% of benefits tracked	Benefits Lead
CSF-8	Programme governance operates effectively	Governance review, decision timeliness	95% decisions within SLA	PMO Lead
CSF-9	Financial controls maintain budget discipline	Budget variance, forecast accuracy	+/- 5% annual variance	Finance Lead
CSF-10	Accessibility and inclusion embedded in all user journeys	Accessibility audit, user testing	WCAG 2.2 AA compliance	User Experience Lead

O.2 Critical Success Factor Dependencies

CSF	Depends On	Dependency	Risk if Not Met
CSF-1	Ministerial mandate, DHSC support	Political priority maintained	Programme stalls, funding withdrawn
CSF-2	Clinical time availability, engagement culture	Protected time for clinical input	Workforce rejection of solution
CSF-3	Market capacity, competitive procurement	Supplier capability available	Programme delay, cost overrun
CSF-4	User-centred design, training investment	Quality of user experience	Low adoption, benefits not realised
CSF-5	Application inventory, CIS2 capability	Legacy system compatibility	Integration failure, scope creep
CSF-6	Security architecture, testing investment	Control effectiveness	Security incident, reputational damage
CSF-7	Baseline data, measurement capability	Data availability	Benefits not demonstrable
CSF-8	Governance maturity, decision rights	Clear governance framework	Poor decisions, lack of accountability
CSF-9	Budget certainty, cost visibility	Financial control framework	Overspend, value erosion
CSF-10	User research, accessibility expertise	Inclusive design capability	User exclusion, legal challenge

Appendix P: Detailed Procurement Timeline

P.1 Procurement Schedule by Lot

Lot	RFI Issue	RFI Close	ITT Issue	ITT Close	Award	Contract Start
Lot 1 — Credential Media	Month 2	Month 3	Month 4	Month 6	Month 7	Month 8
Lot 2 — Secure Bureau	Month 4	Month 5	Month 6	Month 8	Month 9	Month 10
Lot 3 — Readers	Month 2	Month 3	Month 4	Month 6	Month 7	Month 8
Lot 4 — Printers	Month 3	Month 4	Month 5	Month 7	Month 8	Month 9
Lot 5 — PKI/HSM	Month 2	Month 3	Month 5	Month 8	Month 9	Month 10
Lot 6 — Identity Platform	Month 2	Month 4	Month 5	Month 8	Month 9	Month 10
Lot 7 — Integration Factory	Month 4	Month 5	Month 6	Month 8	Month 9	Month 10
Lot 8 — Managed Operations	Month 6	Month 7	Month 8	Month 10	Month 11	Month 12
Lot 9 — Training/RA	Month 5	Month 6	Month 7	Month 9	Month 10	Month 11
Lot 10 — Test/ Assurance	Month 4	Month 5	Month 6	Month 8	Month 9	Month 10

P.2 Procurement Resource Requirements

Role	Lots Managed	Time Commitment	Duration
Commercial Lead	All lots (overall)	100%	Months 1-12
Commercial Manager 1	Lots 1, 2, 3, 4	100%	Months 2-10
Commercial Manager 2	Lots 5, 6, 7	100%	Months 2-11
Commercial Manager 3	Lots 8, 9, 10	100%	Months 4-12
Technical Assessor 1	Lots 1, 2, 5	50%	Months 3-9
Technical Assessor 2	Lots 3, 4, 6	50%	Months 3-9
Technical Assessor 3	Lots 7, 8, 9, 10	50%	Months 5-11
Legal Advisor	All lots	50%	Months 2-12
Procurement Support	All lots	100%	Months 1-12

P.3 Tender Evaluation Timeline

Phase	Activity	Duration	Responsibility
Phase 1	Selection Questionnaire (SQ) evaluation	2 weeks	Commercial team
Phase 2	ITT issue and bidder clarification	6-8 weeks	Commercial team
Phase 3	Technical evaluation	3 weeks	Technical assessors
Phase 4	Commercial evaluation	2 weeks	Commercial team
Phase 5	Presentations and demonstrations	2 weeks	Evaluation panel
Phase 6	Due diligence and reference checks	2 weeks	Commercial team
Phase 7	Final evaluation and moderation	1 week	Evaluation panel
Phase 8	Award decision and standstill	2 weeks	Programme Board
Phase 9	Contract signature and notification	1 week	Commercial team

Appendix Q: Data Migration Strategy

Q.1 Data Sources

Source System	Data Type	Volume	Quality	Migration Approach
Care Identity Service (CIS)	Worker identity, roles, org links	1.3M records	High	Reuse existing data
Trust-specific identity stores	Local identity, access rights	250+ systems	Variable	Extract, transform, deduplicate
ESR (Electronic Staff Record)	HR data, employment history	2M+ records	High	Federated query, no bulk copy
RA workflows	Proofing evidence, approvals	TBD	Variable	Archive with audit trail
Smartcard management systems	Card lifecycle, certificates	1.3M cards	High	Migrate to new CMS
Partner identity stores	Partner access records	TBD	Variable	Federation, not migration

Q.2 Migration Principles

1. **Reuse before migrate** — CIS data is the authoritative source; reuse wherever possible
2. **Deduplication** — single identity record per person, resolved from multiple sources
3. **Data quality** — validation and cleansing before migration
4. **Audit trail** — full provenance of data migration decisions

- 5. **Rollback capability** — ability to revert to source systems if migration fails
- 6. **Phased approach** — migration in waves aligned to Trust rollout schedule
- 7. **Minimal disruption** — no service interruption during migration

Q.3 Migration Phases

Phase	Scope	Duration	Data Volume
Phase 1 — CIS Reconciliation	Validate and clean CIS data	Months 3-6	1.3M records
Phase 2 — Pilot Trusts	Migrate pilot Trust data	Months 18-24	100K records
Phase 3 — Wave 1	First 50 Trusts	Months 24-30	500K records
Phase 4 — Wave 2	Next 100 Trusts	Months 30-36	600K records
Phase 5 — Wave 3	Remaining 150 Trusts	Months 36-48	800K records
Phase 6 — Legacy Retirement	Decommission legacy systems	Months 48-60	All systems

Appendix R: Environmental Sustainability Impact Assessment

R.1 Carbon Impact of Physical Credentials

Component	Carbon per Unit	Annual Volume	Annual Carbon	10-Year Carbon
Smartcard (PVC/PET-G)	0.15 kg CO2e	200,000	30,000 kg	300,000 kg
Smartcard (polycarbonate)	0.25 kg CO2e	50,000	12,500 kg	125,000 kg
Reader (plastic/metal)	2.5 kg CO2e	30,000	75,000 kg	750,000 kg
FIDO2 Security Key	0.3 kg CO2e	6,000	1,800 kg	18,000 kg
Printer (manufacturing)	50 kg CO2e	200	10,000 kg	10,000 kg
Total Physical Credential Carbon			129,300 kg	1,203,000 kg

R.2 Carbon Reduction Measures

Measure	Carbon Reduction	Implementation
Mobile credential reduces physical card production	30% reduction in card volume	Target 70% mobile adoption
Paperless onboarding and identity verification	Eliminates paper-based processes	Digital identity verification
Data centre energy efficiency	Cloud provider carbon neutral commitments	AWS/Azure carbon neutral

Measure	Carbon Reduction	Implementation
Reduced travel through remote support	Lower field engineering carbon	Remote troubleshooting prioritised
Hardware recycling and WEEE compliance	End-of-life recycling	Supplier responsibility requirements
Printer energy efficiency	Energy Star rated equipment	Procurement specification

R.3 Net Sustainability Benefit

The programme's net environmental impact is positive when considering:

- Reduced travel for identity verification and onboarding (digital processes)
- Eliminated paper-based identity processes
- Consolidation of 250+ identity systems into 1 national platform (reduced energy)
- Mobile credential adoption reduces physical card production and reader dependency
- Cloud infrastructure with carbon-neutral commitments

Appendix S: International Benchmarking

S.1 Comparable National Identity Programmes

Programme	Country	Scale	Cost	Duration	Key Lessons
NHS Care Identity Service	UK	1.3M users	Existing	Ongoing	Reuse before build; CIS2 provides strong foundation
VERIFY	USA	4M users (HHS)	\$2B	8 years	Standards-based federation critical; stakeholder engagement essential
IAM Programme	Australia	1.5M users (health)	A\$1.2B	6 years	Phased approach reduces risk; mobile adoption slower than expected
eSanté	Luxembourg	0.5M users	€150M	5 years	Small-scale success; multi-credential model works
NemID/MitID	Denmark	5M users (national)	DKK 3B	10 years	National identity with healthcare integration; mobile-first successful

Programme	Country	Scale	Cost	Duration	Key Lessons
SPID	Italy	30M users (national)	€200M	8 years	Federation model; varying adoption across sectors
Estonia e-ID	Estonia	1.3M users (national)	€150M	15+ years	Gold standard; mandatory digital identity; broad ecosystem

S.2 Key Lessons for NHS Programme

Lesson	Source	Application to NHS Programme
Mobile adoption takes time	Australia, Denmark	Retain smartcard as fallback; plan for 70% mobile adoption target
Federation is complex	USA, Italy	Standardise on OIDC/SAML; invest in integration factory
Executive sponsorship is critical	All programmes	SRO with DHSC/NHS England authority; ministerial engagement
Workforce change must be funded	Australia, UK	Dedicated workforce budget (£90M); champions programme
Benefits realisation requires ownership	All programmes	Named benefit owners at national and local level
Security must be designed in, not added on	All programmes	Zero-trust architecture; continuous verification
Phased rollout reduces risk	Australia, Denmark	Four-phase approach with clear gates and readiness criteria
Supplier diversity prevents lock-in	UK, USA	10-lot procurement strategy; dual-source critical components

Appendix T: Document Version Control and Assurance

T.1 Document Version History

Version	Date	Author	Changes	Review Status
0.1	30 July 2026	Darren Birch	Initial draft from source portfolio	Draft
0.2	30 July 2026	Darren Birch	Expanded to full ministerial format	Draft
0.3	30 July 2026	Darren Birch	Added product specifications, cost model, ROI	Draft
0.4	30 July 2026	Darren Birch	Added governance, risk, future-state	Draft

Version	Date	Author	Changes	Review Status
0.5	30 July 2026	Darren Birch	Added appendices, financial assumptions	Draft
1.0	30 July 2026	Darren Birch	Final consolidated edition	For Review

T.2 Document Assurance

Assurance Element	Status	Evidence
Strategic Alignment	Complete	Mapped to NHS, DHSC, NCSC, Cabinet Office policies
Economic Case	Complete	Options appraisal, cost-benefit, sensitivity, risk adjustment
Commercial Viability	Complete	Market analysis, multi-supplier ecosystem, competitive procurement
Financial Affordability	Complete	Funding sources, cashflow profile, year-by-year budget
Technical Feasibility	Complete	Architecture design, product specifications, standards compliance
Operational Deliverability	Complete	Delivery phases, workstreams, readiness criteria, resource plan
Risk Management	Complete	Risk register, mitigation controls, residual risk assessment
Data Protection	Complete	DPIA framework, privacy controls, data lifecycle
Clinical Safety	Complete	DCB0129/DCB0160 compliance, CSO role, hazard log
Accessibility	Complete	WCAG 2.2 AA, multi-channel, assisted journeys
Benefits Realisation	Complete	Benefit categories, ownership, KPIs, measurement methodology
Commercial	Complete	10-lot procurement, contract terms, exit arrangements

END OF APPENDICES

Document Classification: OFFICIAL — SENSITIVE (Commercial)

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All work is owned by Darren Birch, Architect ZEUS AI Intelligence and Founder JDB Sales